

SHARP HYPOCOERCIVE BOUNDS FOR PERTURBATIVE NONEQUILIBRIUM LIFTS

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ABSTRACT. For centered slow–fast lifts whose effective macroscopic dynamics need not be reversible, we establish quantitative hypocoercive estimates with explicit dependence on the macroscopic spectral gap. The generators take the form

$$\mathcal{L} = \gamma \mathcal{R} + \mathcal{V}_{\text{skew}} + \eta \mathcal{V}_{\text{pert}},$$

with fast relaxation rate γ and nonequilibrium amplitude $\eta \geq 0$. Centering ensures that the slow–slow component of the transport vanishes, so the leading slow dynamics is encoded in the second-order Schur complement, and the reference self-adjoint slow operator

$$Q_O := -P_S \mathcal{V}_{\text{skew}} \mathcal{R}^+ \mathcal{V}_{\text{skew}} P_S$$

sets the macroscopic time scale through its gap $\lambda(Q_O)$.

Our central observation is that the Dolbeault–Mouhot–Schmeiser (DMS) corrector recovers the kinetic acceleration scale only when its resolvent shift is calibrated to the gap, $\varpi = \lambda(Q_O)$. A shift fixed independently of $\lambda(Q_O)$ suppresses the weakest macroscopic mode by an unnecessary factor; the gap-scale shift restores uniform visibility of every slow mode. Under uniform structural constants, including a gap-rescaled flux bound, this calibration yields

$$\gamma_{\text{opt}} \asymp \sqrt{\lambda(Q_O)}, \quad \max_{\gamma > 0} \Lambda_{\text{hypo}}(\gamma) \asymp \sqrt{\lambda(Q_O)},$$

together with a perturbative forcing window of the same order whenever the loss is non-degenerate. A complementary singular-value argument shows that for fixed slow scale, any decay exponent recoverable from a fixed-norm estimate is bounded above by a quantity of order $\gamma^{-1} + \eta$ in the overdamped regime, with an improved prefactor under contractivity. Together, the two bounds single out an intermediate-friction window $\gamma \asymp \sqrt{\lambda(Q_O)}$ as the only regime in which the present method delivers the square-root profile. Four models of independent interest serve as test cases: a confining nonequilibrium Langevin diffusion on $\mathbb{R}^{2d}$, a gravitactic active Brownian particle without translational diffusion on the torus, a biased circular telegraph (run-and-tumble) process on $\mathbb{S}^1$, and a boundary-driven two-qubit XXZ Lindblad model in the quantum Zeno regime.

1. INTRODUCTION

1.1. Problem and main idea. Many models of physical and computational interest produce stationary states that violate detailed balance, and quantitative convergence to such nonequilibrium steady states (NESS) is a central question across kinetic theory, sampling, soft matter, and open quantum systems. Four classes of models motivate the present work and serve, respectively, as test cases for our framework. Driven underdamped Langevin dynamics, in which a non-gradient force breaks reversibility and the invariant density is no longer Gibbsian [22, 30, 29], yield steady states that are only implicitly defined and for which quantitative convergence rates remain scarce. Active-matter dynamics in soft matter [2, 39] replace a Gibbsian equilibrium by a current-carrying steady state generated by self-propulsion; gravitactic and bottom-heavy active particles [44, 40, 42] are a paradigmatic instance. Velocity-jump processes such as the classical Goldstein–Kac telegraph

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[18, 23] and its asymmetric and circular variants [27, 8], equivalently the one-dimensional run-and-tumble particles of active-matter theory [41, 6], support nontrivial steady-state currents whenever the switching rates are unequal. Boundary-driven open quantum dynamics, in which reservoirs act on only part of a spin system, produce NESS and Zeno-limit effective dynamics [25, 34, 35, 5].

A common feature of these four models is that the underlying microscopic dynamics is a *lift* of an effective slow process: an auxiliary fast variable (momentum, swim direction, internal velocity, boundary-coupled spin) is appended to a sluggish observable of physical interest (configuration, position, angle, protected spin), and the slow variable evolves only through a slow-fast-slow excursion. In the reversible regime, kinetic lifting accelerates relaxation: by augmenting an overdamped diffusion with auxiliary momentum-like degrees of freedom, one trades a diffusive γ^{-1} rate for a ballistic $\sqrt{m}$ rate, where m is the gap of the underlying overdamped operator. Discrete-time precursors of this acceleration mechanism include the work of Chen–Lovász–Pak [7] and the non-reversible samplers of Diaconis–Holmes–Neal [9]; in continuous time, related results appear in the optimal-drift analysis of Lelièvre–Nier–Pavliotis [26], the underdamped Langevin estimates of Cao–Lu–Wang [4], and the operator-theoretic lift framework developed by Eberle–Lörler and collaborators [14, 15, 13], in which the $\sqrt{m}$ scale is identified as the natural ceiling for second-order reversible lifts. Brigati–Lörler–Wang [3] have recently combined the hypocoercivity and lift viewpoints. All these works share a single reference point: the macroscopic effective operator is self-adjoint.

The present paper extends the kinetic acceleration mechanism, in a perturbative nonequilibrium regime, to *centered* second-order lifts whose effective macroscopic dynamics need not be reversible. Throughout, by a centered lift we mean one whose slow-slow component of the transport vanishes (Section 3.3), so that the macroscopic variables move only through a slow-fast-slow detour. This is the natural class for which adiabatic elimination produces a genuine second-order Schur complement on the slow space; the four model families above all fall in this class once the perturbation is centered correctly.

The technical obstacle in extending kinetic acceleration past detailed balance is that the invariant density is in general not explicit, and the macroscopic dissipation reaches the slow variables only through the slow-to-fast-to-slow detour. The classical modified- L^2 method of Villani [43] and Dolbeault–Mouhot–Schmeiser [11, 12] (henceforth *DMS*) measures this detour through a shifted resolvent on the macroscopic subspace, and the choice of resolvent shift is conventionally treated as an order-one structural constant. Our central observation is that this conventional choice systematically suppresses the weakest macroscopic mode in centered second-order lifts in the small-gap regime, and consequently loses the kinetic acceleration scale entirely: the optimized lower-bound profile degrades from $\sqrt{\lambda(Q_O)}$ to $\lambda(Q_O)^{5/2}$ as $\lambda(Q_O) \downarrow 0$. Calibrating the shift to the macroscopic gap restores the correct $\sqrt{\lambda(Q_O)}$ scaling and, as shown in [16, 28], also controls the perturbative nonequilibrium loss in the same units.

Setting. We work with generators on a Hilbert space $\mathcal{H}$ of the form

$$(1) \quad \mathcal{L} = \gamma \mathcal{R} + \mathcal{V}_{\text{skew}} + \eta \mathcal{V}_{\text{pert}},$$

where $\gamma > 0$ is the fast relaxation scale, $\mathcal{R} \leq 0$ is the self-adjoint dissipative part, $\mathcal{V}_{\text{skew}}$ is conservative and skew-adjoint, and $\eta \mathcal{V}_{\text{pert}}$ records a nonequilibrium forcing of amplitude $\eta \geq 0$. The macroscopic subspace is the kernel of the dissipator,

$$\mathcal{H}_S := \ker(\mathcal{R}), \quad \mathcal{H}_F := \mathcal{H}_S^\perp,$$

with orthogonal projections P_S and $P_F := \text{id} - P_S$. The componentwise centering condition

$$P_S \mathcal{V}_{\text{skew}} P_S = 0, \quad P_S \mathcal{V}_{\text{pert}} P_S = 0,$$

together with $\mathcal{V} := \mathcal{V}_{\text{skew}} + \eta \mathcal{V}_{\text{pert}}$, kills the slow-slow direct drift, so that adiabatic elimination produces the effective macroscopic generator

$$\mathcal{L}_O = -P_S \mathcal{V} \mathcal{R}^+ \mathcal{V} P_S,$$

where $\mathcal{R}^+$ is the Moore–Penrose pseudoinverse of $\mathcal{R}$ on $\mathcal{H}_F$ and zero on $\mathcal{H}_S$ (Definition 3.4). At $\eta = 0$ this reduces to the self-adjoint reference operator

$$(2) \quad Q_O := -P_S \mathcal{V}_{\text{skew}} \mathcal{R}^+ \mathcal{V}_{\text{skew}} P_S,$$

whose spectral gap $\lambda(Q_O)$ measures the macroscopic time scale. Since $\mathcal{L}_O$ is generally non-self-adjoint when $\eta > 0$, we anchor our calibration to the self-adjoint reference Q_O rather than to $\mathcal{L}_O$ itself.

The gap-scale shift. The mechanism is already visible on a single macroscopic mode. In the DMS modified- L^2 method, a corrector with shift $\varpi > 0$ enters the Lyapunov functional, and a spectral mode of $-Q_O$ at eigenvalue $\lambda > 0$ contributes to the macroscopic coercivity through the multiplier

$$\frac{\lambda}{\varpi + \lambda}.$$

With ϖ fixed independently of $\lambda(Q_O)$, this multiplier behaves like $\lambda(Q_O)/\varpi$ on the weakest mode. The macroscopic coercivity is therefore suppressed inside the DMS dissipation matrix by an avoidable factor of order $\lambda(Q_O)$, and the kinetic lower-bound scale is lost. Recalibrating to the gap,

$$\varpi_* := \lambda(Q_O),$$

restores uniform visibility, since every mode then satisfies $\lambda/(\varpi_* + \lambda) \geq 1/2$. Section 2 carries out this calculation in the equilibrium underdamped Langevin model, where the algebra is fully explicit; Section 4 abstracts the part of the argument that survives perturbative nonequilibrium forcing.

Contribution. The contributions of the paper are threefold. First, we isolate and prove the gap-scale calibration $\varpi_* = \lambda(Q_O)$ of the DMS corrector as the mechanism by which the kinetic square-root scale is recovered for centered small-gap lifts; the choice removes the artificial loss caused by a fixed shift inside the DMS coercivity mechanism and is the canonical gap-scale choice for the determinant–trace estimate used here. We do not claim global optimality of ϖ_* over all admissible shifts. Second, we quantify the linear-in- η nonequilibrium penalty in the perturbative regime through an explicit cross-term constant C_{cross} and a uniform window $\eta^*(\gamma_{\text{opt}}) \asymp \sqrt{\lambda(Q_O)}$, providing the first DMS-type rate in this generality with a forcing window matched to the kinetic scale. Third, we pair the lower bound with a matching overdamped upper bound in the same reference norm via a singular-value-gap argument, which pins down the intermediate-friction window $\gamma \asymp \sqrt{\lambda(Q_O)}$ as the only regime in which a square-root profile can be obtained. Because the framework is model-agnostic within the centered second-order class, the same constants are checked in the four models above: the confining nonequilibrium Langevin diffusion, the gravitactic active Brownian particle, the biased circular telegraph process, and the boundary-driven two-qubit XXZ chain. None of these is a black-box specialization of a single existing theorem; each requires a model-specific verification of the structural inputs.

1.2. Main results. The principal convergence estimate is Theorem 5.2, stated below in informal form. Fix a distinguished invariant vector $\mathbf{1} \in \mathcal{H}$ (the constant function in L^2 examples) and write

$$\mathcal{H}_0 := \{g \in \mathcal{H} : \langle g, \mathbf{1} \rangle = 0\}$$

for the codimension-one mass-zero subspace on which all coercive estimates take place. We assume the existence of a stationary state $h_{\gamma, \eta} \in \mathbf{1} + \mathcal{H}_0$, so that convergence to stationarity reduces, via the centering step (Lemma 4.2), to a semigroup estimate on $\mathcal{H}_0$; this is the role of Assumption 4.1 below.

Theorem A (informal). Under the structural inputs of Section 4—fast microscopic coercivity, centered coupling, a self-adjoint Schur complement Q_O with positive macroscopic gap $\lambda(Q_O) > 0$, two-term flux regularity, and relative perturbation bounds—the centered adjoint semigroup satisfies

$$\|e^{t\mathcal{L}^\dagger} g\| \leq C e^{-c(\Lambda_{\text{hypo}}(\gamma) - \eta B_{\text{pert}}(\gamma))t} \|g\|, \quad g \in \mathcal{H}_0,$$

with $C \leq 1 + \sqrt{2}$ and $c = 2 - \sqrt{2}$. Here $\Lambda_{\text{hypo}}(\gamma)$ is the lower-rate profile produced by the unperturbed 2×2 DMS dissipation matrix at the gap-scale shift $\varpi = \lambda(Q_O)$, and $B_{\text{pert}}(\gamma)$ is the structural penalty due to the nonequilibrium forcing. Optimization in γ under uniform structural constants yields

$$\gamma_{\text{opt}} \asymp \sqrt{\lambda(Q_O)}, \quad \sup_{\gamma > 0} \Lambda_{\text{hypo}}(\gamma) \asymp \sqrt{\lambda(Q_O)},$$

matching the classical kinetic acceleration law. When the perturbative loss is non-degenerate, the admissible forcing window at $\gamma = \gamma_{\text{opt}}$ is also of this order: $\eta^*(\gamma_{\text{opt}}) \asymp \sqrt{\lambda(Q_O)}$; see Corollary 5.3.

Theorem B (informal). Under the same structural assumptions, any decay exponent in an exponential reference-norm estimate of the centered semigroup is bounded above by a quantity of order $\gamma^{-1} + \eta$ when the slow scale and the structural constants are held fixed. Contractivity is not required for this scale; when it holds, the reference prefactor improves from C to $1 + \log C$. In particular, when $\eta \lesssim \gamma^{-1}$ the overdamped regime is genuinely diffusive, and no square-root rate can be obtained from a fixed-norm estimate.

Together, Theorems A and B single out the intermediate-friction window $\gamma \asymp \sqrt{\lambda(Q_O)}$ as the only regime in which the present method delivers a square-root DMS lower-bound profile, and identify the gap-scale shift $\varpi_* = \lambda(Q_O)$ as the calibration that exposes it. Section 6 verifies the mechanism in four models—a nonequilibrium underdamped Langevin family on $\mathbb{R}^{2d}$, a gravitactic active Brownian particle on the torus, a biased circular telegraph process on the circle, and a boundary-driven two-qubit XXZ Lindblad model—and recovers the predicted $\sqrt{\lambda(Q_O)}$ rate profile uniformly along a small-gap scaling.

1.3. Related work. *Hypocoercivity and modified L^2 methods.* The modified- L^2 hypocoercive framework originates with Villani [43] and Dolbeault–Mouhot–Schmeiser [11, 12]. For confining kinetic Fokker–Planck and Langevin dynamics we also draw on the regularity viewpoint of Hérau [20] and Helffer–Nier [19]. Explicit, optimal-rate variants of the DMS Lyapunov functional in equilibrium underdamped Langevin models are developed by Cao–Lu–Wang [4] and Roussel–Stoltz [38]; the choice of resolvent shift in those works is essentially fixed in the position-Poincaré units, which is what Section 2 reanalyzes. The small-forcing periodic invariant-density expansion of Iacobucci–Olla–Stoltz [22] both motivates and supplies the natural comparison point for our perturbative centering. The gap-scale shift method extends previous works on specific examples, such as [16, 28]. *Acceleration by lifting.* On acceleration by lifting, the foundational references are Chen–Lovász–Pak [7] in discrete time and Hwang–Hwang–Ma–Sheu [21] and Lelièvre–Nier–Pavliotis [26] in continuous time. Recent operator-theoretic work on second-order lifts includes Eberle–Lörler [14, 15], Eberle–Guillin–Hahn–Lörler–Michel [13], and Brigati–Lörler–Wang [3]. These works establish, among other things, that the $\sqrt{m}$ improvement is the natural ceiling for second-order lifts in equilibrium settings. By contrast, the present paper isolates the centered case in which Q_O arises from adiabatic elimination and feeds directly into the DMS corrector, identifies the gap-scale shift as the calibration that exposes the $\sqrt{\lambda(Q_O)}$ profile in this DMS framework, and shows that the same calibration also fixes the natural scale of the perturbative nonequilibrium loss. The paired overdamped upper bound (Section 5.2) shows that, within the same reference norm, no comparably general ballistic estimate is available outside the intermediate-friction window.

Nonequilibrium Langevin and weighted- L^2 frameworks. For L^2 analysis of nonequilibrium steady states beyond explicit Gibbs weights we rely on Iacobucci–Olla–Stoltz [22], Dietert’s weighted- L^2 framework [10], and the geometric ergodicity results of Monmarché for non-equilibrium kinetic Langevin dynamics [30, 29]. These works establish exponential ergodicity but do not, in general, track the macroscopic gap explicitly through the constants. The present paper is complementary: we restrict to the perturbative regime so that the slow gap remains the only small parameter, but we track its dependence quantitatively through the constants Θ , K_{geom} , $b_{2,\text{eff}}$, and C_{cross} .

Active matter, telegraph, and quantum lifts. The active-particle example uses the standard active Brownian equations and their athermal/no-translational-diffusion specialization [37, 17, 36, 2, 39], with the gravitational or external-field aligning torque familiar from bottom-heavy/gravitactic swimmers [44, 40, 42]. The telegraph example uses the classical Goldstein–Kac velocity-jump process [18, 23], its asymmetric-switching variants [27], circular versions [8], and the equivalent one-dimensional two-velocity run-and-tumble formulation in active-matter theory [41, 6]. The quantum example uses the boundary spin-flip reservoirs and XXZ Hamiltonian familiar from the boundary-driven quantum-transport literature reviewed in [25]; exact NESS for larger boundary-driven XXZ chains form a central theme of Prosen’s work [34, 35], while effective Zeno dynamics for dissipative spin chains and related Schur-complement viewpoints appear in dissipative Schrieffer–Wolff and boundary-Zeno reductions [24, 32, 5]. The closest two-qubit comparison point we are aware of is the open XYZ/XXZ analysis of Popkov–Presilla–Salerno [33]; their dissipator is a single polarizing jump rather than the symmetric-plus-biased pair used here, and consequently their slow space differs from ours, as discussed in Section 6.4. To our knowledge, none of these four model classes has previously been treated within a single hypocoercive framework with explicit gap-tracking constants.

1.4. Organization of the paper. Section 2 carries out the equilibrium Langevin warm-up that motivates the gap-scale shift $\varpi_* = \lambda(Q_O)$. Sections 3 and 4 set up the abstract slow–fast framework, define the effective operator Q_O and the modified DMS functional, and collect the structural assumptions. Section 5 states the abstract lower bound (Theorem 5.2) and the matching overdamped upper bound (Section 5.2). Section 6 verifies the model-specific hypotheses for the four families. The appendices collect the matrix computations and the model-specific verifications.

2. WARM-UP: EQUILIBRIUM LANGEVIN UNDER A LOWER HESSIAN BOUND

This section isolates the recalibration mechanism in the classical equilibrium Langevin model [16], where the slow gap m is sharp and the algebra is fully transparent. The choice $\varpi_* = m$ turns the macroscopic DMS coercive term into an order-one contribution and reduces the lower bound to a 2×2 dissipation-matrix estimate. The argument also pinpoints how nonconvexity enters: under only a Poincaré gap $m > 0$ for the position marginal together with a lower Hessian bound, the loss is carried by the explicit dual-regularity factor $\sqrt{2 + K/(2m)}$.

For simplicity we normalize the inverse temperature to 1 in this warm-up. Let

$$d\rho_\infty(x, v) = d\mu(x) d\kappa(v), \quad d\mu(x) = Z^{-1} e^{-U(x)} dx, \quad d\kappa(v) = (2\pi)^{-d/2} e^{-|v|^2/2} dv,$$

and write $m > 0$ for the exact spectral gap of the overdamped operator $-\Delta_x + \nabla_x U \cdot \nabla_x$ on $L_0^2(\mu)$, equivalently for the optimal constant in the Poincaré inequality

$$(3) \quad \|g\|_{L^2(\mu)}^2 \leq \frac{1}{m} \|\nabla_x g\|_{L^2(\mu)}^2, \quad g \in H^1(\mu), \quad \int g d\mu = 0.$$

We also assume

$$(4) \quad \nabla^2 U(x) \geq -K \text{Id}, \quad x \in \mathbb{R}^d,$$

for some $K \geq 0$. Consider on $L_0^2(\rho_\infty)$ the equilibrium Langevin generator

$$(5) \quad \mathcal{L} = \mathcal{V}_{\text{skew}} + \gamma \mathcal{R}, \quad \mathcal{V}_{\text{skew}} = v \cdot \nabla_x - \nabla_x U \cdot \nabla_v, \quad \mathcal{R} = -v \cdot \nabla_v + \Delta_v.$$

The slow projection is the velocity average

$$P_S f(x) := \int_{\mathbb{R}^d} f(x, v) d\kappa(v), \quad P_F := \text{id} - P_S,$$

so the slow space consists of functions of x alone. The associated positive overdamped operator is

$$(6) \quad \mathcal{L}_{\text{ovd}} := -Q_O = -\Delta_x + \nabla_x U \cdot \nabla_x,$$

acting on $L_0^2(\mu)$. By the definition of m through (3), $\mathcal{L}_{\text{ovd}}$ has gap m on $L_0^2(\mu)$, hence $\lambda(Q_O) = m$.

The required confining Kramers–Fokker–Planck regularity is classical [20, Introduction and §§1–2], [12, §2]. The corresponding semigroup and core properties are recalled in Appendix D; they are used only to justify the density and closure step.

The corrector with shift m is

$$(7) \quad \mathcal{A}_m := (m + \mathcal{L}_{\text{ovd}})^{-1}(\mathcal{V}_{\text{skew}} P_S)^\dagger,$$

and the modified Lyapunov functional is

$$(8) \quad \mathbf{E}_\varepsilon(f) := \frac{1}{2} \|f\|^2 - \varepsilon \Re \langle \mathcal{A}_m f, f \rangle.$$

The minus sign is the usual backward-generator convention for this warm-up. In the abstract theorem below the estimates are written for the adjoint density semigroup, so the DMS cross term appears with the corresponding adjoint sign in (42).

Lemma 2.1 (Corrector estimates for the Langevin warm-up). *For every $\phi \in L_0^2(\rho_\infty)$, the corrector $\mathcal{A}_m$ satisfies*

$$(9) \quad \|\mathcal{A}_m \phi\| \leq \frac{1}{2\sqrt{m}} \|\phi\|,$$

$$(10) \quad \|\mathcal{V}_{\text{skew}} \mathcal{A}_m \phi\| \leq \|\phi\|,$$

$$(11) \quad \|\mathcal{A}_m \mathcal{V}_{\text{skew}} P_F \phi\| \leq \sqrt{2 + \frac{K}{2m}} \|\phi\|.$$

Consequently, if $\varepsilon \leq \sqrt{m}$, then

$$(12) \quad \frac{1 - \varepsilon/\sqrt{m}}{2} \|\phi\|^2 \leq \mathbf{E}_\varepsilon(\phi) \leq \frac{1 + \varepsilon/\sqrt{m}}{2} \|\phi\|^2.$$

Proof. Let $u = \mathcal{A}_m \phi$ and $w = (\mathcal{V}_{\text{skew}} P_S)^\dagger \phi$, so $u = (m + \mathcal{L}_{\text{ovd}})^{-1} w$ and $w \in L_0^2(\mu)$. Since u depends only on x ,

$$\|\mathcal{V}_{\text{skew}} u\|^2 = \|\nabla_x u\|_{L^2(\mu)}^2 = \langle \mathcal{L}_{\text{ovd}} u, u \rangle_{L^2(\mu)}.$$

Taking the $L^2(\mu)$ inner product of $(m + \mathcal{L}_{\text{ovd}})u = w$ with u and applying Cauchy–Schwarz yields

$$m\|u\|^2 + \|\mathcal{L}_{\text{ovd}}^{1/2} u\|^2 \leq \|\phi\| \|\mathcal{L}_{\text{ovd}}^{1/2} u\|.$$

This gives both (9) and (10). For (11), set $B := \mathcal{A}_m \mathcal{V}_{\text{skew}} P_F$. If $g \in \text{Ran}(P_S)$ and $h = (m + \mathcal{L}_{\text{ovd}})^{-1} g$, then an integration by parts shows

$$B^\dagger g = - \sum_{i,j} (v_i v_j - \delta_{ij}) \partial_{ij} h.$$

The Gaussian fourth-moment identity gives $\|B^\dagger g\|^2 = 2\|\nabla_x^2 h\|_{L^2(\mu)}^2$. The weighted Bochner identity under (4) yields

$$\|\nabla_x^2 h\|_{L^2(\mu)}^2 \leq \|\mathcal{L}_{\text{ovd}} h\|_{L^2(\mu)}^2 + K \|\nabla_x h\|_{L^2(\mu)}^2.$$

Therefore

$$\begin{aligned} \|B^\dagger g\|^2 &\leq 2\|\mathcal{L}_{\text{ovd}} h\|_{L^2(\mu)}^2 + 2K \langle \mathcal{L}_{\text{ovd}} h, h \rangle_{L^2(\mu)} \\ &= 2\|\mathcal{L}_{\text{ovd}}(m + \mathcal{L}_{\text{ovd}})^{-1} g\|_{L^2(\mu)}^2 + 2K \langle \mathcal{L}_{\text{ovd}}(m + \mathcal{L}_{\text{ovd}})^{-2} g, g \rangle_{L^2(\mu)} \\ &\leq \left(2 + 2K \sup_{\lambda \geq 0} \frac{\lambda}{(m + \lambda)^2}\right) \|g\|_{L^2(\mu)}^2 = \left(2 + \frac{K}{2m}\right) \|g\|_{L^2(\mu)}^2. \end{aligned}$$

Hence $\|B^\dagger g\| \leq \sqrt{2 + \frac{K}{2m}} \|g\|$, which is (11). The norm equivalence follows from (9) and Cauchy–Schwarz. $\square$

Write $g_S = P_S g$ and $g_F = P_F g$. The Gaussian Poincaré inequality in v gives $\|g_F\|^2 \leq \|\nabla_v g\|^2$. Therefore, if we set

$$X := \|\nabla_v g\|, \quad Y := \|g_S\|, \quad Z := \frac{\gamma}{2\sqrt{m}} + \sqrt{2 + \frac{K}{2m}},$$

then differentiating E_ε along $\partial_t g = \mathcal{L}g$ and using Lemma 2.1 yields

$$(13) \quad \frac{d}{dt} E_\varepsilon(g) \leq - \begin{pmatrix} X \\ Y \end{pmatrix}^T \begin{pmatrix} \gamma - \varepsilon & -\frac{\varepsilon Z}{2} \\ -\frac{\varepsilon Z}{2} & \frac{\varepsilon}{2} \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix}.$$

Indeed, the microscopic dissipation gives $-\gamma X^2$; the macroscopic term is

$$-\varepsilon \left\langle \frac{\mathcal{L}_{\text{ovd}}}{m + \mathcal{L}_{\text{ovd}}} g_S, g_S \right\rangle \leq -\frac{\varepsilon}{2} Y^2$$

because $\mathcal{L}_{\text{ovd}} \geq m$ on $L_0^2(\mu)$; the fast-fast remainder contributes at most εX^2 by (10); the dual regularity estimate (11) contributes $\varepsilon \sqrt{2 + \frac{K}{2m}} XY$; and the correction involving $\gamma \mathcal{R}$ contributes $\varepsilon \gamma (2\sqrt{m})^{-1} XY$ by (9).

Proposition 2.2. *For the corrector $\mathcal{A}_m$, the DMS lower-rate profile (that is, the smallest eigenvalue of the dissipation matrix before converting the Lyapunov functional back to the L^2 norm) satisfies, for every $\gamma > 0$,*

$$(14) \quad \Lambda_{\text{hypo}}^{\text{Lan}}(\gamma) \geq \frac{\gamma}{4 \left((\gamma/(2\sqrt{m}) + \sqrt{2 + \frac{K}{2m}})^2 + 2 \right)}.$$

The displayed lower bound is maximized at $\gamma_{\text{opt}} = \sqrt{16m + 2K}$, yielding

$$(15) \quad \Lambda_{\text{hypo}}^{\text{Lan}}(\gamma_{\text{opt}}) \geq \frac{\sqrt{m}}{4 \left(\sqrt{2 + \frac{K}{2m}} + \sqrt{4 + \frac{K}{2m}} \right)}.$$

Since $m = \lambda(Q_O)$, this is exactly the kinetic scale $\mathcal{O}(\sqrt{\lambda(Q_O)})$ whenever $K = \mathcal{O}(m)$; when $K = 0$, (15) reduces to $\frac{2-\sqrt{2}}{8} \sqrt{m}$.

Proof. Let $M_{\gamma,\varepsilon}$ denote the dissipation matrix in (13). Its determinant is

$$\det M_{\gamma,\varepsilon} = \frac{\gamma\varepsilon}{2} - \frac{\varepsilon^2(Z^2 + 2)}{4},$$

so the natural choice is

$$\varepsilon_* = \frac{\gamma}{Z^2 + 2}.$$

For a positive 2×2 matrix, $\lambda_{\min}(M) \geq \det(M)/\text{tr}(M)$, and here $\text{tr}(M_{\gamma,\varepsilon_*}) = \gamma - \varepsilon_*/2 \leq \gamma$. Hence

$$\lambda_{\min}(M_{\gamma,\varepsilon_*}) \geq \frac{\gamma}{4(Z^2 + 2)}.$$

This gives the stated lower bound. Writing $x = \gamma/(2\sqrt{m})$, the right-hand side becomes

$$\sqrt{m} \frac{x}{2 \left((x + \sqrt{2 + \frac{K}{2m}})^2 + 2 \right)}.$$

A direct derivative computation shows that it is maximized at $x = \sqrt{4 + \frac{K}{2m}}$, i.e. at $\gamma_{\text{opt}} = \sqrt{16m + 2K}$. Substituting this value yields (15). $\square$

Corollary 2.3 (Warm-up semigroup rate at the optimizing friction). *At this maximizing friction, set*

$$\varepsilon_* := \frac{\gamma_{\text{opt}}}{Z(\gamma_{\text{opt}})^2 + 2}.$$

Then the semigroup generated by $\mathcal{L}$ satisfies

$$(16) \quad \|e^{t\mathcal{L}}g\|_{L^2(\rho_\infty)} \leq \sqrt{3} \exp\left(-\frac{\sqrt{m}}{6\left(\sqrt{2+\frac{K}{2m}}+\sqrt{4+\frac{K}{2m}}\right)}t\right) \|g\|_{L^2(\rho_\infty)}, \quad g \in L_0^2(\rho_\infty), \quad t \geq 0.$$

In particular, when $K = 0$ the decay rate is $\frac{2-\sqrt{2}}{12}\sqrt{m}$.

Proof. At $\gamma = \gamma_{\text{opt}}$, Proposition 2.2 gives the matrix-profile bound

$$(17) \quad \Lambda_{\text{hypo}}^{\text{Lan}}(\gamma_{\text{opt}}) \geq \frac{\sqrt{m}}{4\left(\sqrt{2+\frac{K}{2m}}+\sqrt{4+\frac{K}{2m}}\right)}.$$

For the associated optimizer ε_* one has

$$(18) \quad \varepsilon_* = \frac{\sqrt{m}}{\sqrt{2+\frac{K}{2m}}+\sqrt{4+\frac{K}{2m}}} \leq \frac{\sqrt{m}}{2},$$

so Lemma 2.1 yields

$$(19) \quad \frac{1}{4}\|g\|^2 \leq \mathbb{E}_{\varepsilon_*}(g) \leq \frac{3}{4}\|g\|^2.$$

The dissipation estimate (13) with this optimal choice gives

$$(20) \quad \frac{d}{dt}\mathbb{E}_{\varepsilon_*}(g_t) \leq -\Lambda_{\text{hypo}}^{\text{Lan}}(\gamma_{\text{opt}})\|g_t\|^2 \leq -\frac{4}{3}\Lambda_{\text{hypo}}^{\text{Lan}}(\gamma_{\text{opt}})\mathbb{E}_{\varepsilon_*}(g_t).$$

Gronwall's lemma then gives

$$(21) \quad \|g_t\|^2 \leq 3e^{-\frac{4}{3}\Lambda_{\text{hypo}}^{\text{Lan}}(\gamma_{\text{opt}})t}\|g_0\|^2,$$

which is exactly the stated semigroup estimate after taking square roots. $\square$

Within the present DMS determinant–trace estimate, any shift fixed independently of m leaves the macroscopic coercive block of order m and therefore does not recover the kinetic lower-bound scale as $m \downarrow 0$. The warm-up thus already contains the mechanism of the paper: $\varpi_* = m$ restores the kinetic scale when m is the macroscopic gap, while the lower Hessian bound only modifies the explicit dual-regularity constant. The remainder of the paper retains this calibration and incorporates the perturbation $\eta\mathcal{V}_{\text{pert}}$.

3. PRELIMINARIES

The abstract setting retains the two features used in the warm-up: a macroscopic spectral gap, henceforth denoted $\lambda(Q_O)$, and a shifted resolvent corrector on the slow subspace.

3.1. Hilbert-space setup. Let $\mathcal{H}$ be a separable complex Hilbert space with inner product $\langle \cdot, \cdot \rangle$ (conjugate-linear in the first entry) and norm $\|\cdot\|$. We fix a distinguished unit vector $\mathbf{1} \in \mathcal{H}$ and set

$$\mathcal{H}_0 := \{u \in \mathcal{H} : \langle u, \mathbf{1} \rangle = 0\}, \quad \mathcal{H}_1 := \mathbf{1} + \mathcal{H}_0.$$

The generator $\mathcal{L}$ is viewed on this fixed reference space, and $\mathcal{L}^\dagger$ denotes its Hilbert adjoint. In the classical examples below, $\mathcal{H} = L^2(\mu)$ and $\mathbf{1}$ is the constant-one function. Whenever $\mathcal{L}\mathbf{1} = 0$, the subspace $\mathcal{H}_0$ is invariant under $e^{t\mathcal{L}^\dagger}$. If, for a given parameter pair (γ, η) , there exists a stationary vector $h_{\gamma, \eta} \in \mathcal{H}_1$ with $\mathcal{L}^\dagger h_{\gamma, \eta} = 0$, then every $f \in \mathcal{H}_1$ determines a centered datum $g := f - h_{\gamma, \eta} \in$

$\mathcal{H}_0$. All quantitative estimates below are stated on $\mathcal{H}_0$; Lemma 4.2 records the passage back to physical densities or normalized stationary vectors.

Assumption 3.1 (Generation, core, and closure for unbounded realizations). In finite dimensions this assumption is automatic. In unbounded examples we assume that $\mathcal{L}^\dagger$ generates a C_0 -semigroup on $\mathcal{H}$. We also assume a dense invariant core

$$\mathcal{C} \subset \text{dom}(\mathcal{L}) \cap \text{dom}(\mathcal{L}^\dagger) \cap \text{dom}((- \mathcal{R})^{1/2})$$

on which $\mathcal{R}$, $\mathcal{V}_{\text{skew}}$, $\mathcal{V}_{\text{pert}}$, $\mathcal{V}_{\text{pert}}^\dagger$, P_S , P_F , and $\mathcal{R}^+$ preserve $\mathcal{C}$, the adjoint identity

$$\langle \mathcal{V}_{\text{skew}} u, v \rangle = -\langle u, \mathcal{V}_{\text{skew}} v \rangle$$

holds for $u, v \in \mathcal{C}$, and the compositions used below are well defined on $\mathcal{C}$; see [11, 12, 43]. For every $u_0 \in \text{dom}(\mathcal{L}^\dagger)$, the orbit $u(t) = e^{t\mathcal{L}^\dagger} u_0$ belongs to $C([0, \infty); \text{dom}(\mathcal{L}^\dagger)) \cap C^1([0, \infty); \mathcal{H})$ and satisfies $\partial_t u = \mathcal{L}^\dagger u$. We also assume that the differentiated Lyapunov form

$$\mathfrak{d}_{\alpha, \varpi}(f) := \Re \langle \mathcal{L}^\dagger f, f \rangle - \alpha \Re \langle \mathcal{A}_\varpi \mathcal{L}^\dagger f, f \rangle - \alpha \Re \langle \mathcal{A}_\varpi f, \mathcal{L}^\dagger f \rangle$$

and the estimates used to bound it extend continuously from $\mathcal{C} \cap \mathcal{H}_0$ to $\text{dom}(\mathcal{L}^\dagger) \cap \mathcal{H}_0$ in the graph norm. The core compositions used in Lemma 4.9 are assumed to be closable on the domains where they occur; the bounds proved there provide the bounded extensions used later. The semigroup estimates are then extended to all of $\mathcal{H}_0$ by density and strong continuity. For the Langevin family we take $\mathcal{C} = \mathcal{S}(\mathbb{R}^{2d})$; see Appendix D.

3.2. Spectral gaps. Write

$$\mathcal{L}_{\text{sym}} := \frac{1}{2}(\mathcal{L} + \mathcal{L}^\dagger)$$

for the symmetric part of the generator. Whenever $\mathcal{H}_0$ is invariant under $e^{t\mathcal{L}^\dagger}$, we denote by $\mathcal{L}_0^\dagger$ the generator of the restricted semigroup on $\mathcal{H}_0$.

Definition 3.2 (Gap parameters). On $\mathcal{H}_0$ we define the *coercive gap*

$$\lambda_S := \inf_{f \in \mathcal{H}_0 \setminus \{0\}} \frac{-\langle f, \mathcal{L}_{\text{sym}} f \rangle}{\|f\|^2},$$

and the *real-part spectral gap*

$$\lambda(\mathcal{L}) := \inf \{ \Re \mu : \mu \in \text{Spec}(-\mathcal{L}_0^\dagger) \},$$

whenever this restricted generator is defined. Since $\text{Spec}(\mathcal{L}^\dagger) = \overline{\text{Spec}(\mathcal{L})}$, the real-part gap is the same for forward and backward dynamics.

When $\mathcal{L}$ is self-adjoint and non-positive on $\mathcal{H}_0$ (the detailed-balance case), $\lambda(\mathcal{L}) = \lambda_S$. In the non-reversible regime, λ_S need not control relaxation: it may vanish even when the real-part spectral gap $\lambda(\mathcal{L})$ is positive. The skew part $\frac{1}{2}(\mathcal{L} - \mathcal{L}^\dagger)$ couples degenerate directions to dissipative ones, and exponential relaxation is produced through hypocoercivity [43, 11]. Since $\lambda(\mathcal{L})$ is a spectral abscissa, it need not coincide with the optimal decay rate in norm whenever the sharp semigroup estimate requires a prefactor $C > 1$. In the present reference-space setting, the basic estimate takes the form

$$\|e^{t\mathcal{L}^\dagger} g\| \leq C e^{-\lambda t} \|g\|, \quad g \in \mathcal{H}_0,$$

and we refer to the relaxation as *strictly hypocoercive* when $C > 1$ is necessary.

3.3. Slow–fast decomposition and adiabatic elimination. Consider the generator $\mathcal{L}_\gamma = \gamma\mathcal{R} + \mathcal{V}$ with self-adjoint dissipator $\mathcal{R} \leq 0$ and interaction $\mathcal{V}$. Define the *slow* and *fast* subspaces

$$\mathcal{H}_S := \ker(\mathcal{R}), \quad \mathcal{H}_F := \mathcal{H}_S^\perp,$$

with orthogonal projections $P_S, P_F := \text{id} - P_S$. We assume $\mathcal{R}$ has spectral gap $\lambda_R > 0$ on $\mathcal{H}_F$. Any $f \in \mathcal{H}$ decomposes as $f = f_S + f_F$ with $f_S := P_S f$ (slow/macroscopic) and $f_F := P_F f$ (fast/microscopic). For the classical and bounded operator families considered below, the reference stationary mode satisfies $\mathbf{1} \in \mathcal{H}_S$. Therefore,

$$\mathcal{H}_F \subset \mathcal{H}_0, \quad \mathcal{H}_0 = (\mathcal{H}_S \cap \mathcal{H}_0) \oplus \mathcal{H}_F.$$

Hence, if $f \in \mathcal{H}_0$, then both $f_S := P_S f$ and $f_F := P_F f$ remain in $\mathcal{H}_0$.

Definition 3.3 (Centered coupling). The interaction satisfies $P_S \mathcal{V} P_S = 0$.

The centering condition ensures that the effective slow dynamics is second-order in γ^{-1} , generated by slow–fast–slow excursions; without it, an $\mathcal{O}(1)$ tangential drift already appears at the block-operator level.

Definition 3.4 (Effective generator). Let $\mathcal{R}^+$ denote the Moore–Penrose pseudoinverse of $\mathcal{R}$, bounded on $\mathcal{H}_F$ with $\|\mathcal{R}^+\| \leq \lambda_R^{-1}$. Under the centered coupling condition, we define

$$(22) \quad \mathcal{L}_O := -P_S \mathcal{V} \mathcal{R}^+ \mathcal{V} P_S.$$

The operator $\mathcal{L}_O$ is the effective slow dynamics produced by adiabatic elimination. Whenever singular perturbation/adiabatic elimination is justified, the macroscopic-time evolution is governed by $e^{\tau \mathcal{L}_O}$ with $\tau = t/\gamma$, which is the source of the diffusive γ^{-1} bottleneck behind the upper bound below.

When $\mathcal{V} = \mathcal{V}_{\text{skew}} + \eta \mathcal{V}_{\text{pert}}$, we impose the stronger componentwise centering conditions

$$P_S \mathcal{V}_{\text{skew}} P_S = 0, \quad P_S \mathcal{V}_{\text{pert}} P_S = 0.$$

This yields estimates uniform in η and isolates the effective operator Q_O at $\eta = 0$.

4. ABSTRACT FRAMEWORK

The assumptions stated below isolate the structural inputs required by the DMS argument: a fast spectral gap, centered coupling, control of the induced slow–fast flux, and perturbation estimates.

Assumption 4.1 (Mass conservation and a stationary state). The distinguished vector $\mathbf{1}$ belongs to $\mathcal{H}_S \cap \text{dom}(\mathcal{L})$ and satisfies

$$\mathcal{L} \mathbf{1} = 0.$$

Hence the forward semigroup $e^{t\mathcal{L}^\dagger}$ preserves the reference mass:

$$\langle e^{t\mathcal{L}^\dagger} f, \mathbf{1} \rangle = \langle f, \mathbf{1} \rangle, \quad t \geq 0.$$

In particular, the codimension-one subspace $\mathcal{H}_0$ is invariant under $e^{t\mathcal{L}^\dagger}$. For the fixed parameter pair (γ, η) under consideration, there exists a stationary vector/state $h_{\gamma, \eta} \in \mathcal{H}_1 \cap \text{dom}(\mathcal{L}^\dagger)$ such that

$$\mathcal{L}^\dagger h_{\gamma, \eta} = 0.$$

The stationary state enters only through this centering step. All coercive estimates are carried out on $\mathcal{H}_0$; the remaining assumptions describe the slow operator Q_O and the bounds needed for the DMS cross term.

Lemma 4.2 (Centered reduction). *Assume Assumption 4.1. Let $f \in \mathcal{H}_1$ and $f_t := e^{t\mathcal{L}^\dagger} f$. If*

$$g := f - h_{\gamma,\eta}, \quad g_t := f_t - h_{\gamma,\eta},$$

then $g \in \mathcal{H}_0$ and

$$g_t = e^{t\mathcal{L}^\dagger} g \in \mathcal{H}_0.$$

If, in addition, $f \in \text{dom}(\mathcal{L}^\dagger)$, then $\partial_t g_t = \mathcal{L}^\dagger g_t$. Consequently, any semigroup estimate proved on $\mathcal{H}_0$ transfers verbatim to convergence of f_t toward $h_{\gamma,\eta}$.

Proof. Since $\langle f, \mathbf{1} \rangle = \langle h_{\gamma,\eta}, \mathbf{1} \rangle = 1$, one has $g \in \mathcal{H}_0$. Because $\mathcal{L}^\dagger h_{\gamma,\eta} = 0$ and $e^{t\mathcal{L}^\dagger} h_{\gamma,\eta} = h_{\gamma,\eta}$,

$$g_t = e^{t\mathcal{L}^\dagger} f - h_{\gamma,\eta} = e^{t\mathcal{L}^\dagger} (f - h_{\gamma,\eta}) = e^{t\mathcal{L}^\dagger} g.$$

Mass conservation gives $\langle g_t, \mathbf{1} \rangle = 0$, so $g_t \in \mathcal{H}_0$. If $f \in \text{dom}(\mathcal{L}^\dagger)$, then also $g \in \text{dom}(\mathcal{L}^\dagger)$ and the evolution equation follows from the standard differentiability of the semigroup on the operator domain. $\square$

Lemma 4.3 (Centered part generator). *Assume Assumption 4.1. Then $\text{dom}(\mathcal{L}^\dagger) \cap \mathcal{H}_0$ is dense in $\mathcal{H}_0$. Moreover, the restriction of $e^{t\mathcal{L}^\dagger}$ to $\mathcal{H}_0$ is a C_0 -semigroup. Its generator is the part of $\mathcal{L}^\dagger$ in $\mathcal{H}_0$, namely*

$$\mathcal{L}_0^\dagger g := \mathcal{L}^\dagger g, \quad \text{dom}(\mathcal{L}_0^\dagger) = \text{dom}(\mathcal{L}^\dagger) \cap \mathcal{H}_0.$$

Proof. Let $g \in \mathcal{H}_0$. Since $\text{dom}(\mathcal{L}^\dagger)$ is dense in $\mathcal{H}$, choose $f_n \in \text{dom}(\mathcal{L}^\dagger)$ with $f_n \rightarrow g$ in $\mathcal{H}$. Set

$$g_n := f_n - \overline{\langle f_n, \mathbf{1} \rangle} h_{\gamma,\eta}.$$

Because $h_{\gamma,\eta} \in \text{dom}(\mathcal{L}^\dagger) \cap \mathcal{H}_1$, one has $g_n \in \text{dom}(\mathcal{L}^\dagger)$ and

$$\langle g_n, \mathbf{1} \rangle = \langle f_n, \mathbf{1} \rangle - \langle f_n, \mathbf{1} \rangle \langle h_{\gamma,\eta}, \mathbf{1} \rangle = 0,$$

so $g_n \in \text{dom}(\mathcal{L}^\dagger) \cap \mathcal{H}_0$. Since $f_n \rightarrow g$ and $\langle f_n, \mathbf{1} \rangle \rightarrow \langle g, \mathbf{1} \rangle = 0$, we also have $g_n \rightarrow g$ in $\mathcal{H}$. This proves density.

By Assumption 4.1, $\mathcal{H}_0$ is invariant under $e^{t\mathcal{L}^\dagger}$, so the restriction is a C_0 -semigroup on the closed subspace $\mathcal{H}_0$. By the standard restriction theorem for invariant subspaces, its generator is the part of $\mathcal{L}^\dagger$ in $\mathcal{H}_0$ [31, Thm. 1.2.5]. If $g \in \text{dom}(\mathcal{L}^\dagger) \cap \mathcal{H}_0$, then

$$\langle \mathcal{L}^\dagger g, \mathbf{1} \rangle = \langle g, \mathcal{L}\mathbf{1} \rangle = 0$$

because $\mathbf{1} \in \text{dom}(\mathcal{L})$ and $\mathcal{L}\mathbf{1} = 0$. Hence $\mathcal{L}^\dagger g \in \mathcal{H}_0$, and the part-domain is exactly $\text{dom}(\mathcal{L}^\dagger) \cap \mathcal{H}_0$. $\square$

4.1. Abstract assumptions. Let $\mathcal{H} = \mathcal{H}_S \oplus \mathcal{H}_F$ be an orthogonal slow-fast decomposition. We study generators depending on friction $\gamma > 0$ and perturbation magnitude $\eta \geq 0$:

$$(23) \quad \mathcal{L} = \gamma\mathcal{R} + \mathcal{V}.$$

The transport term is decomposed as $\mathcal{V} = \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}$, and P_S, P_F denote the orthogonal projections onto $\mathcal{H}_S, \mathcal{H}_F$. A signed physical amplitude is represented by replacing $\mathcal{V}_{\text{pert}}$ with the signed perturbation direction; the abstract scalar η is always nonnegative.

The estimates use three inputs beyond mass conservation: the self-adjoint slow operator Q_O , quantitative control of the fast pseudoinverse and the induced microscopic fluxes, and perturbation bounds.

Assumption 4.4 (Self-adjoint effective operator on the slow space). There exists a non-positive self-adjoint operator Q_O on $\mathcal{H}_S$ representing the Schur-complement expression

$$(24) \quad Q_O = -P_S \mathcal{V}_{\text{skew}} \mathcal{R}^+ \mathcal{V}_{\text{skew}} P_S$$

on a dense slow core $\mathcal{C}_Q \subset \text{dom}(Q_O) \cap \mathcal{H}_S$. In unbounded examples $\mathcal{C}_Q$ is chosen so that the first-order vectors $\mathcal{R}^+ \mathcal{V}_{\text{skew}} \psi$ with $\psi \in \mathcal{C}_Q$ are admissible test functions for $\mathcal{L}^\dagger$ and $\mathcal{V}_{\text{pert}}^\dagger$. We assume:

- (i) $\mathcal{C}_Q \cap \mathcal{H}_0$ is a core for the part of Q_O in $\mathcal{H}_S \cap \mathcal{H}_0$;
- (ii) for every $\psi \in \text{dom}(Q_O)$, the microscopic flux

$$w_\psi := P_F \mathcal{V}_{\text{skew}} P_S \psi$$

is well defined in $\mathcal{H}_F$ and satisfies

$$(25) \quad \langle \psi, (-Q_O) \psi \rangle = \langle w_\psi, (-\mathcal{R}^+) w_\psi \rangle;$$

- (iii) for every $\psi \in \text{dom}(Q_O)$, the slow image $P_S \mathcal{V}_{\text{skew}} P_F (-\mathcal{R}^+) w_\psi$ is well defined and obeys

$$(26) \quad P_S \mathcal{V}_{\text{skew}} P_F (-\mathcal{R}^+) w_\psi = Q_O \psi.$$

The assumptions below encode four additional inputs only: a spectral gap on the fast space, centered slow-fast transport, a closure estimate for the induced microscopic flux, and perturbation bounds.

Assumption 4.5 (Structural conditions). The generator has the form $\mathcal{L} = \gamma \mathcal{R} + \mathcal{V}_{\text{skew}} + \eta \mathcal{V}_{\text{pert}}$ on $\mathcal{H} = \mathcal{H}_S \oplus \mathcal{H}_F$. We assume:

- (1) **Microscopic spectral gap.** $\mathcal{R}$ is self-adjoint and non-positive on $\mathcal{H}$, with kernel $\ker(\mathcal{R}) = \mathcal{H}_S$, and there exists $\lambda_R > 0$ such that

$$(27) \quad -\langle \mathcal{R} f, f \rangle \geq \lambda_R \|f\|^2, \quad \forall f \in \mathcal{H}_F.$$

- (2) **Centered coupling.**

$$(28) \quad P_S \mathcal{V}_{\text{skew}} P_S = 0, \quad P_S \mathcal{V}_{\text{pert}} P_S = 0,$$

first on the invariant core from Assumption 3.1 and then, by closure, on the domains used below. Since P_S is orthogonal, the second identity gives $P_S \mathcal{V}_{\text{pert}}^\dagger P_S = 0$ on the adjoint domain used in the proof.

- (3) **Flux regularity.** For $\psi \in \text{dom}(Q_O) \subset \mathcal{H}_S$, let w_ψ be the microscopic flux from Assumption 4.4(ii). There exist constants $M_R > 0$ and $\kappa_{\text{ell}}, \kappa_{\text{low}} \geq 0$ such that

$$(29) \quad \|w_\psi\|^2 \leq M_R \langle w_\psi, -\mathcal{R}^+ w_\psi \rangle, \quad \forall \psi \in \text{dom}(Q_O) \subset \mathcal{H}_S.$$

When $\lambda(Q_O) > 0$, these slow-induced fluxes are not identically zero. Combining (29) with $\langle w, -\mathcal{R}^+ w \rangle \leq \lambda_R^{-1} \|w\|^2$ for $w \in \mathcal{H}_F$ then implies $M_R \geq \lambda_R$.

$$(30) \quad \|P_F \mathcal{V}_{\text{skew}} P_F (-\mathcal{R}^+) w_\psi\| \leq \kappa_{\text{ell}} \|(-Q_O) \psi\| + \kappa_{\text{low}} \|(-Q_O)^{1/2} \psi\|.$$

- (4) **Structural decomposition of the perturbation.**

$$(31) \quad \mathcal{V}_{\text{pert}} = \mathcal{K}_{\text{pert}} + \mathcal{S}_{\text{pert}},$$

where

$$\mathcal{K}_{\text{pert}} := \frac{1}{2}(\mathcal{V}_{\text{pert}} - \mathcal{V}_{\text{pert}}^\dagger), \quad \mathcal{S}_{\text{pert}} := \frac{1}{2}(\mathcal{V}_{\text{pert}} + \mathcal{V}_{\text{pert}}^\dagger),$$

with the adjoint taken in $\mathcal{H}$. Assume:

- (a) **Skew perturbation control.** The operator $\mathcal{K}_{\text{pert}}$ is skew-symmetric on its domain. There exist constants $a_1, b_1 \geq 0$ such that for all $f \in \text{dom}((-\mathcal{R})^{1/2})$,

$$(32) \quad \|\mathcal{K}_{\text{pert}} f\| \leq a_1 \|(-\mathcal{R})^{1/2} f\| + b_1 \|f\|.$$

- (b) **Symmetric perturbation control.** $\mathcal{S}_{\text{pert}}$ is symmetric on $\text{dom}((-\mathcal{R})^{1/2})$, and there exist constants $a_2, b_2 \geq 0$ such that for all $f \in \text{dom}((-\mathcal{R})^{1/2})$,

$$(33) \quad \|\mathcal{S}_{\text{pert}} f\| \leq a_2 \|(-\mathcal{R})^{1/2} f\| + b_2 \|f\|.$$

Assumption 4.4 identifies the effective operator on the macroscopic subspace. Item (iii) of Assumption 4.5 is the abstract analogue of the weighted elliptic/Bochner estimate in the warm-up, and item (iv) records the price of nonequilibrium forcing. In (30), κ_{ell} controls the top-order term while κ_{low} records lower-order defects such as the Hessian contribution in the Langevin example. The final input is the positive macroscopic gap in the next assumption.

Assumption 4.6 (Spectral gap of the effective operator). Because $\mathbf{1} \in \mathcal{H}_S$, we write $\mathcal{H}_{S,0} := \mathcal{H}_S \cap \mathcal{H}_0$. We assume that

$$\mathbf{1} \in \text{dom}(Q_O), \quad Q_O \mathbf{1} = 0,$$

and that the bottom of the spectrum of the restricted operator is positive:

$$(34) \quad \lambda(Q_O) := \inf \sigma(-Q_O|_{\mathcal{H}_{S,0}}) > 0.$$

Equivalently,

$$(35) \quad -\langle f, Q_O f \rangle \geq \lambda(Q_O) \|f\|^2, \quad \forall f \in \text{dom}(Q_O) \cap \mathcal{H}_{S,0}.$$

Since Q_O is self-adjoint and $Q_O \mathbf{1} = 0$, the subspace $\mathcal{H}_{S,0} = \mathbf{1}^\perp \cap \mathcal{H}_S$ is Q_O -invariant. Because $\mathbf{1}$ is a unit vector in $\mathcal{H}_S$, $\mathcal{H}_S = \mathbb{C}\mathbf{1} \oplus \mathcal{H}_{S,0}$. Equivalently, the restricted self-adjoint operator $-Q_O|_{\mathcal{H}_{S,0}}$ has spectrum contained in $[\lambda(Q_O), \infty)$. Consequently, $\ker(Q_O) = \mathbb{C}\mathbf{1} \subset \mathcal{H}_S$, and $(\varpi - Q_O)^{-1}$ is bounded on $\mathcal{H}_S$ for every $\varpi > 0$.

The upper bound does not require the bottom of the spectrum to be an eigenvalue. Instead, for every $\varepsilon > 0$ the spectral theorem first gives an approximate bottom vector in the spectral subspace near $\lambda(Q_O)$, and the core property then permits graph-norm approximation by a normalized $\phi_\varepsilon \in \mathcal{C}_Q \cap \mathcal{H}_{S,0}$ with $\|(-Q_O)\phi_\varepsilon\| \leq \lambda(Q_O) + \varepsilon$ and $\|(-Q_O)^{1/2}\phi_\varepsilon\| \leq \sqrt{\lambda(Q_O) + \varepsilon}$; the proof later sends $\varepsilon \downarrow 0$.

The abstract domain input is exhausted by Assumptions 3.1 and 4.4: in the unbounded case the Lyapunov differentiation is carried out first on the invariant core and then extended by graph-norm continuity. For the Langevin family this verification is given in Appendix D.

4.2. The corrector. The corrector is the abstract counterpart of the resolvent used in the warm-up. It solves a shifted equation on the macroscopic subspace and produces the cross term in the modified energy functional.

Lemma 4.7 (Resolvent of the effective operator). *Let Q_O be as in Assumptions 4.4 and 4.6. Then for every shift parameter $\varpi > 0$,*

- (i) *the operator $(\varpi - Q_O)$ is strictly positive on $\mathcal{H}_S$ with spectrum contained in $[\varpi, \infty)$;*
- (ii) *the resolvent $(\varpi - Q_O)^{-1}$ is a bounded, self-adjoint, positive operator on $\mathcal{H}_S$ with $\|(\varpi - Q_O)^{-1}\| \leq \varpi^{-1}$;*
- (iii) *on the mean-zero subspace $\mathcal{H}_{S,0}$, the sharper bound $\|(\varpi - Q_O)^{-1}|_{\mathcal{H}_{S,0}}\| \leq (\varpi + \lambda(Q_O))^{-1}$ holds;*
- (iv) *$(\varpi - Q_O)^{-1}$ maps $\mathcal{H}_S$ into $\text{dom}(Q_O)$.*

Proof. Since $-Q_O \geq 0$ is self-adjoint and $Q_O \mathbf{1} = 0$, the subspace $\mathcal{H}_{S,0} = \mathbf{1}^\perp \cap \mathcal{H}_S$ is Q_O -invariant. Assumption 4.6 therefore gives

$$\sigma(-Q_O|_{\mathcal{H}_{S,0}}) \subset [\lambda(Q_O), \infty), \quad \sigma(-Q_O|_{\mathbb{C}\mathbf{1}}) = \{0\}.$$

Therefore $\sigma(\varpi - Q_O) \subset [\varpi, \varpi + \|Q_O\|]$ in finite dimensions, or $\sigma(\varpi - Q_O) \subset [\varpi, \infty)$ in general. Items (i)–(iii) are direct consequences; item (iv) is the standard resolvent mapping property of the closed self-adjoint operator Q_O . $\square$

Definition 4.8 (Corrector). Let $\varpi > 0$ be a shift parameter. In Section 5 we will choose $\varpi_* = \lambda(Q_O)$. Initially define the **corrector** on the invariant core by

$$(36) \quad \mathcal{A}_\varpi := (\varpi - Q_O)^{-1} P_S \mathcal{V}_{\text{skew}}(-\mathcal{R}^+) P_F.$$

Lemma 4.9 shows that this core operator extends uniquely to a bounded operator $\mathcal{H} \rightarrow \mathcal{H}_S$; we keep the same notation for the extension.

Because dissipation vanishes on $\mathcal{H}_S$, the plain energy is blind to the macroscopic modes; the role of $\mathcal{A}_\varpi$ is to convert coercivity of $-Q_O$ into a bounded cross term. On a macroscopic spectral mode of $-Q_O$ at eigenvalue $\lambda > 0$, the principal DMS contribution carries the factor $\lambda/(\varpi + \lambda)$, so the choice $\varpi = \lambda(Q_O)$ guarantees that every mode in $\mathcal{H}_{S,0}$ contributes at least $1/2$.

Lemma 4.9 (Structural estimates for the DMS corrector). *Under Assumptions 4.4 and 4.5, the core operator $\mathcal{A}_\varpi$ extends uniquely to a bounded operator $\mathcal{H} \rightarrow \mathcal{H}_S$, still denoted by $\mathcal{A}_\varpi$. This extension satisfies the following structural estimates, with constants depending solely on the structural parameters $(M_R, \lambda_R, \kappa_{\text{ell}}, \kappa_{\text{low}}, a_1, b_1, a_2, b_2)$ and with the only shift dependence displayed explicitly below:*

(1) **Corrector Boundedness:**

$$(37) \quad \|\mathcal{A}_\varpi\| \leq k_{\mathcal{A}} \varpi^{-1/2}, \quad \text{with } k_{\mathcal{A}} := \frac{1}{2\sqrt{\lambda_R}}.$$

(2) **Regularity Estimates (Forward and Dual):** *The corrector acts as a bounded regularizer for the skew-symmetric transport in both directions:*

$$(38a) \quad \|\mathcal{V}_{\text{skew}} \mathcal{A}_\varpi\| \leq k_{\text{reg}}, \quad \text{with } k_{\text{reg}} := \sqrt{M_R/\lambda_R},$$

$$(38b) \quad \|\mathcal{A}_\varpi \mathcal{V}_{\text{skew}} P_F\| \leq k_{\text{dual}}(\varpi), \quad \text{with } k_{\text{dual}}(\varpi) := \kappa_{\text{ell}} + \frac{\kappa_{\text{low}}}{2\sqrt{\varpi}}.$$

The dual estimate (38b) is a direct consequence of the flux regularity condition in Assumption 4.5.3.

(3) **Perturbation Control:** *In the unbounded setting, the following compositions are first defined on the invariant core and are closable under Assumption 3.1. The estimates below give the bounded extensions used in the Lyapunov argument. The relative bounds in Assumption 4.5.4 are one route to these estimates in the examples.*

$$(39a) \quad \|\mathcal{V}_{\text{pert}} \mathcal{A}_\varpi\|, \quad \|\mathcal{V}_{\text{pert}}^\dagger \mathcal{A}_\varpi\| \leq k_P(\varpi),$$

$$(39b) \quad \|\mathcal{A}_\varpi \mathcal{V}_{\text{pert}}\|, \quad \|\mathcal{A}_\varpi \mathcal{V}_{\text{pert}}^\dagger\| \leq \tilde{k}_P(\varpi).$$

with control constants

$$(40) \quad k_P(\varpi) := (b_1 + b_2) k_{\mathcal{A}} \varpi^{-1/2},$$

$$(41) \quad \tilde{k}_P(\varpi) := \left((b_1 + b_2) k_{\mathcal{A}} + \frac{a_1 + a_2}{2} \right) \varpi^{-1/2}.$$

Proof. See Appendix A. □

4.3. The modified Lyapunov functional.

Definition 4.10 (Modified Lyapunov functional). The functional is the DMS modified L^2 energy [11, 12]. Because the quantitative proof is carried out for the forward density semigroup $e^{t\mathcal{L}^\dagger}$, the DMS cross term enters with the sign adapted to the adjoint dynamics. For any $f \in \mathcal{H}$ and any parameter $\alpha > 0$, we define the functional

$$(42) \quad \mathcal{E}_{\alpha, \varpi}(f) := \frac{1}{2} \|f\|^2 - \alpha \Re \langle \mathcal{A}_\varpi f, f \rangle.$$

The sign is most easily verified on a slow vector. For $f_S \in \mathcal{H}_{S,0}$, one has

$$\mathcal{A}_\varpi \mathcal{V}_{\text{skew}} f_S = (\varpi - Q_O)^{-1} Q_O f_S,$$

and consequently the transport contribution in the adjoint evolution contains

$$\alpha \Re \langle \mathcal{A}_\varpi \mathcal{V}_{\text{skew}} f_S, f_S \rangle = -\alpha \langle (\varpi - Q_O)^{-1} (-Q_O) f_S, f_S \rangle \leq 0.$$

The minus sign in (42) is therefore the one that turns the adjoint transport term into macroscopic dissipation.

The functional defines a norm equivalent to the Hilbert norm whenever $\alpha\|\mathcal{A}_\varpi\|$ is sufficiently small.

Proposition 4.11 (Norm equivalence). *If the parameter α satisfies $\alpha\|\mathcal{A}_\varpi\| < 1/2$, then for all $f \in \mathcal{H}$:*

$$(43) \quad \frac{1}{2}(1 - 2\alpha\|\mathcal{A}_\varpi\|)\|f\|^2 \leq \mathcal{E}_{\alpha,\varpi}(f) \leq \frac{1}{2}(1 + 2\alpha\|\mathcal{A}_\varpi\|)\|f\|^2.$$

Proof. Immediate from $|\Re\langle \mathcal{A}_\varpi f, f \rangle| \leq \|\mathcal{A}_\varpi\| \|f\|^2$ and the definition of $\mathcal{E}_{\alpha,\varpi}$. $\square$

5. MAIN ABSTRACT RESULTS

This section presents the two abstract estimates used in the examples: an explicit DMS lower bound and a matching overdamped upper bound in the same reference norm.

5.1. Explicit lower bound. Throughout this subsection the shift is fixed at $\varpi_* = \lambda(Q_O)$; only the Lyapunov weight α is optimized. As in the warm-up, the proof reduces to a 2×2 dissipation matrix. The matrix computation is carried out in Appendix B; the constants required for the theorem are recorded here.

Explicit constants. We fix $\varpi_* := \lambda(Q_O)$. Three quantities govern the theorem: the coupling scale $Z(\gamma)$, the unperturbed lower-rate profile $\Lambda_{\text{hypo}}(\gamma)$, and the perturbative loss $B_{\text{pert}}(\gamma)$. To write them compactly, set

$$(44) \quad \Theta := \frac{\sqrt{M_R}}{2}, \quad K_{\text{geom}} := \kappa_{\text{ell}} + \frac{\kappa_{\text{low}}}{2\sqrt{\lambda(Q_O)}}, \quad b_{2,\text{eff}} := b_2 + \frac{a_2}{2}(1 + \lambda_R),$$

Here Θ measures the size of the slow-fast flux, and $K_{\text{geom}} = k_{\text{dual}}(\lambda(Q_O))$ is the effective transport constant at the chosen shift.

For each friction $\gamma > 0$, define the *coupling scale*

$$(45) \quad Z(\gamma) := \gamma\Theta\lambda(Q_O)^{-1/2} + K_{\text{geom}} = \kappa_{\text{ell}} + (\gamma\Theta + \frac{1}{2}\kappa_{\text{low}})\lambda(Q_O)^{-1/2},$$

which measures the strength of slow-fast interaction at friction γ . We then set

$$(46) \quad \alpha_*(\gamma) := \frac{\gamma\lambda_R}{Z(\gamma)^2 + 2k_{\text{reg}}},$$

$$(47) \quad \Lambda_{\text{hypo}}(\gamma) := \frac{\frac{1}{4}\gamma\lambda_R}{Z(\gamma)^2 + 2k_{\text{reg}}},$$

$$(48) \quad B_{\text{pert}}(\gamma) := b_{2,\text{eff}} + \alpha_*(\gamma) C_{\text{cross}} \lambda(Q_O)^{-1/2},$$

where $C_{\text{cross}} := 2(b_1 + b_2)k_{\mathcal{A}} + (a_1 + a_2)$ is the perturbation cross-term constant (see Appendix B for the derivation), with $k_{\mathcal{A}}$ the resolvent norm bound from Lemma 4.9. Thus $\alpha_*(\gamma)$ is the optimized Lyapunov weight, $\Lambda_{\text{hypo}}(\gamma)$ is the unperturbed lower-rate profile, and $B_{\text{pert}}(\gamma)$ is the coefficient of the $\mathcal{O}(\eta)$ loss. At the optimizing friction, the factor $\lambda(Q_O)^{-1/2}$ in B_{pert} is offset by $\alpha_*(\gamma)$; see Corollary 5.3.

We also isolate the explicit admissibility restriction coming from the symmetric perturbation in the basic energy estimate:

$$(49) \quad \eta_{\text{sym}}(\gamma) := \begin{cases} +\infty, & a_2 = 0, \\ 2\gamma/a_2, & a_2 > 0. \end{cases}$$

Equivalently, this is the condition under which the coefficient of $\|(-\mathcal{R})^{1/2}f\|^2$ remains nonnegative in the basic energy estimate after the perturbative terms are bounded by η ; see Appendix B.

Finally, define the Lyapunov-distortion parameter and norm-equivalence prefactor by

$$(50) \quad \delta_*(\gamma) := 2\alpha_*(\gamma)\|\mathcal{A}_{\varpi_*}\|,$$

$$(51) \quad C_*(\gamma) := \sqrt{\frac{1 + \delta_*(\gamma)}{1 - \delta_*(\gamma)}}.$$

The appendix proves the uniform bound $\delta_*(\gamma) \leq 2^{-1/2}$ for all $\gamma > 0$. Hence Proposition 4.11 applies to the optimized Lyapunov functional and $C_*(\gamma) \leq 1 + \sqrt{2}$.

Definition 5.1 (Perturbative window produced by the estimate). For fixed $\gamma > 0$, define

$$\eta^*(\gamma) := \min \left\{ \eta_{\text{sym}}(\gamma), \frac{\Lambda_{\text{hypo}}(\gamma)}{2B_{\text{pert}}(\gamma)} \right\}.$$

The quotient is interpreted as $+\infty$ if $B_{\text{pert}}(\gamma) = 0$. This threshold controls only the DMS coercivity argument.

Theorem 5.2 (Explicit hypocoercive decay estimate). *Let $\varpi_* := \lambda(Q_O) > 0$ be the slow-gap scale from Assumption 4.6, and let $\eta^*(\gamma)$ be the perturbative window from Definition 5.1. Fix $\gamma > 0$ and $0 \leq \eta \leq \eta^*(\gamma)$. Suppose Assumptions 3.1, 4.1, 4.4, 4.5, and 4.6 hold for this parameter pair, with the structural constants in Assumption 4.5 fixed independently of η on the considered window. Then the semigroup on $\mathcal{H}_0$ satisfies*

$$(52) \quad \|e^{t\mathcal{L}^\dagger} g\| \leq C_*(\gamma) e^{-\lambda_{\eta^*} t} \|g\| \leq (1 + \sqrt{2}) e^{-\lambda_{\eta^*} t} \|g\|, \quad \forall g \in \mathcal{H}_0, \quad \forall t \geq 0.$$

The decay rate satisfies

$$(53) \quad \lambda_{\eta} \geq \frac{\Lambda_{\text{hypo}}(\gamma) - \eta B_{\text{pert}}(\gamma)}{1 + \delta_*(\gamma)} \geq (2 - \sqrt{2}) (\Lambda_{\text{hypo}}(\gamma) - \eta B_{\text{pert}}(\gamma)).$$

By Lemma 4.2, the same bound yields exponential convergence of $f_t = e^{t\mathcal{L}^\dagger} f$ toward $h_{\gamma, \eta}$ for every $f \in \mathcal{H}_1$.

Proof. By Lemma 4.2, it suffices to work on $\mathcal{H}_0$. Let $g_t = e^{t\mathcal{L}^\dagger} g$ with $g \in \mathcal{H}_0$. Differentiating the Lyapunov functional $\mathcal{E}_{\alpha, \varpi_*}(g_t)$ on the core gives a 2×2 dissipation matrix in the fast and slow components. The choice $\varpi_* = \lambda(Q_O)$ makes the slow diagonal entry order one, while Assumption 4.5.4 contributes only the explicit $\mathcal{O}(\eta)$ correction encoded in $B_{\text{pert}}(\gamma)$.

Choosing $\alpha = \alpha_*(\gamma)$ optimizes the determinant of this matrix and yields the profile $\Lambda_{\text{hypo}}(\gamma)$. The condition $\eta \leq \eta^*(\gamma)$ guarantees positivity of the optimized matrix, so

$$\frac{d}{dt} \mathcal{E}_{\alpha_*(\gamma), \varpi_*}(g_t) \leq -(\Lambda_{\text{hypo}}(\gamma) - \eta B_{\text{pert}}(\gamma)) \|g_t\|^2.$$

The uniform bound $\delta_*(\gamma) \leq 2^{-1/2}$ and Proposition 4.11 convert this differential inequality into the stated semigroup estimate on $\mathcal{H}_0$. The full matrix computation and the passage from the core computation to the semigroup estimate are given in Appendix B. $\square$

The theorem is pointwise in γ . Corollary 5.3 optimizes the unperturbed profile over γ under uniformity assumptions on the structural constants and tracks the perturbative window at that friction.

Corollary 5.3 (Optimal friction and kinetic scaling). *Keep the notation of Theorem 5.2. Assume Assumptions 3.1, 4.4, 4.5, and 4.6. If one wants to interpret the resulting formulas as centered-semigroup bounds, assume in addition that Assumption 4.1 holds at the frictions under consideration, in particular at γ_{opt} . The conclusion is not a general small-gap statement: it requires the gap-rescaled lower-order flux defect in item (iii) below to remain bounded. Assume furthermore that, uniformly in the family,*

- (i) Θ , k_{reg} , and λ_R are bounded above and below by positive constants;
- (ii) κ_{ell} is uniformly bounded;
- (iii) the gap-rescaled lower-order quantity $\kappa_{\text{low}}/\sqrt{\lambda(Q_O)}$ remains bounded as $\lambda(Q_O) \downarrow 0$, equivalently $K_{\text{geom}} = \mathcal{O}(1)$; for the Langevin family this is precisely the regime $K \leq K_0 \lambda(Q_O) = K_0 m$;
- (iv) $b_{2,\text{eff}}$ and C_{cross} are uniformly bounded.

For the forcing-window conclusion assume, in addition, that the perturbative loss is uniformly non-degenerate: $b_{2,\text{eff}} + C_{\text{cross}} \geq c_0 > 0$.

Then the unperturbed rate function $\Lambda_{\text{hypo}}(\gamma)$ defined in (47) attains its global maximum at

$$(54) \quad \gamma_{\text{opt}} = \frac{\sqrt{\lambda(Q_O)}}{\Theta} \sqrt{K_{\text{geom}}^2 + 2k_{\text{reg}}} \asymp \sqrt{\lambda(Q_O)}.$$

At this friction scale, the maximal lower bound exhibits kinetic scaling:

$$(55) \quad \max_{\gamma > 0} \Lambda_{\text{hypo}}(\gamma) \asymp \sqrt{\lambda(Q_O)}.$$

Moreover, the mixing weight in (48) satisfies

$$(56) \quad \alpha_*(\gamma_{\text{opt}}) \asymp \sqrt{\lambda(Q_O)}, \quad \alpha_*(\gamma_{\text{opt}}) \lambda(Q_O)^{-1/2} = \mathcal{O}(1),$$

so the factor $\lambda(Q_O)^{-1/2}$ in $B_{\text{pert}}(\gamma)$ is offset at the optimal friction. Consequently, $B_{\text{pert}}(\gamma_{\text{opt}}) = \mathcal{O}(1)$ and the threshold produced by the estimate satisfies

$$(57) \quad \eta^*(\gamma_{\text{opt}}) \asymp \sqrt{\lambda(Q_O)}.$$

This last conclusion is a small-forcing statement: a nonequilibrium strength fixed independently of $\lambda(Q_O)$ is not covered in the small-gap limit by the present DMS coercivity estimate.

Proof. Write

$$(58) \quad \Lambda_{\text{hypo}}(\gamma) = \frac{\lambda_R}{4} \frac{\gamma}{(\gamma \Theta \lambda(Q_O)^{-1/2} + K_{\text{geom}})^2 + 2k_{\text{reg}}}.$$

Set $x := \gamma \Theta \lambda(Q_O)^{-1/2}$ and $C_0 := 2k_{\text{reg}}$. Then $\Lambda_{\text{hypo}}(\gamma) = \frac{\lambda_R}{4\Theta} \sqrt{\lambda(Q_O)} \frac{x}{(x + K_{\text{geom}})^2 + C_0}$. A direct derivative computation shows that the function $x \mapsto \frac{x}{(x + K_{\text{geom}})^2 + C_0}$ is maximized at $x = \sqrt{K_{\text{geom}}^2 + C_0}$, giving the stated γ_{opt} and the scaling $\max_{\gamma} \Lambda_{\text{hypo}}(\gamma) \asymp \sqrt{\lambda(Q_O)}$.

For the perturbation term, by (48), $B_{\text{pert}}(\gamma) = b_{2,\text{eff}} + \alpha_*(\gamma) C_{\text{cross}} \lambda(Q_O)^{-1/2}$. At $\gamma = \gamma_{\text{opt}}$ we have $x = \mathcal{O}(1)$ and $Z(\gamma_{\text{opt}}) = x + K_{\text{geom}} = \mathcal{O}(1)$, hence $\alpha_*(\gamma_{\text{opt}}) = \frac{\gamma_{\text{opt}} \lambda_R}{Z(\gamma_{\text{opt}})^2 + 2k_{\text{reg}}} = \mathcal{O}(\sqrt{\lambda(Q_O)})$, and in fact $\alpha_*(\gamma_{\text{opt}}) \lambda(Q_O)^{-1/2}$ is bounded above and below by positive constants determined by the structural constants. Since $b_{2,\text{eff}} + C_{\text{cross}} \geq c_0 > 0$ by assumption, we obtain

$$(59) \quad B_{\text{pert}}(\gamma_{\text{opt}}) = b_{2,\text{eff}} + \alpha_*(\gamma_{\text{opt}}) C_{\text{cross}} \lambda(Q_O)^{-1/2} \asymp 1,$$

Finally, the matrix-controlled threshold satisfies $\Lambda_{\text{hypo}}(\gamma_{\text{opt}})/(2B_{\text{pert}}(\gamma_{\text{opt}})) \asymp \sqrt{\lambda(Q_O)}$. If $a_2 = 0$, then $\eta_{\text{sym}}(\gamma_{\text{opt}}) = +\infty$ and therefore $\eta^*(\gamma_{\text{opt}}) \asymp \sqrt{\lambda(Q_O)}$. If $a_2 > 0$, then $\eta_{\text{sym}}(\gamma_{\text{opt}}) = 2\gamma_{\text{opt}}/a_2 \asymp \sqrt{\lambda(Q_O)}$. In either case,

$$\eta^*(\gamma_{\text{opt}}) = \min \left\{ \eta_{\text{sym}}(\gamma_{\text{opt}}), \frac{\Lambda_{\text{hypo}}(\gamma_{\text{opt}})}{2B_{\text{pert}}(\gamma_{\text{opt}})} \right\} \asymp \sqrt{\lambda(Q_O)}.$$

□

For the equilibrium Langevin warm-up of Section 2, the general formulas reproduce the same $\sqrt{\lambda(Q_O)}$ scaling. Proposition 2.2–Corollary 2.3 retain the combined Langevin constant $\sqrt{2 + \frac{K}{2m}}$, whereas the abstract argument passes through the split quantity $K_{\text{geom}} = \kappa_{\text{ell}} + \kappa_{\text{low}}/(2\sqrt{\lambda(Q_O)})$.

5.2. Diffusive upper bound in the overdamped regime. The upper bound starts from the elementary fact that an exponential estimate on $\mathcal{H}_0$ controls the injectivity modulus of $\mathcal{L}^\dagger$ in the same norm. Testing this modulus on approximate bottom modes of the self-adjoint operator $-Q_O$ gives the overdamped scale; no eigenfunction at the bottom of the macroscopic spectrum is required. Contractivity improves the prefactor but is not needed for the scale.

Define the *singular-value gap of the centered forward generator*

$$(60) \quad s(\mathcal{L}^\dagger) := s(\mathcal{L}_0^\dagger) := \inf_{g \in \mathcal{H}_0 \cap \text{dom}(\mathcal{L}^\dagger) \setminus \{0\}} \frac{\|\mathcal{L}^\dagger g\|}{\|g\|},$$

where $\mathcal{L}_0^\dagger$ is the part generator from Lemma 4.3. In finite dimensions $s(\mathcal{L}^\dagger)$ is the smallest singular value of the centered generator; in the unbounded setting it is the corresponding injectivity modulus.

Theorem 5.4 (Diffusive upper bound in the overdamped regime). *Fix the parameter pair (γ, η) and assume Assumptions 3.1, 4.1, 4.4, 4.5, and 4.6. Let $C \geq 1$ and $\nu > 0$ be such that*

$$(61) \quad \|e^{t\mathcal{L}^\dagger} g\| \leq C e^{-\nu t} \|g\|, \quad \forall g \in \mathcal{H}_0, \quad \forall t \geq 0.$$

Then

$$(62) \quad \begin{aligned} \nu &\leq C s(\mathcal{L}^\dagger) \\ &\leq C \left[\frac{(1 + \kappa_{\text{ell}}) \lambda(Q_O) + \kappa_{\text{low}} \sqrt{\lambda(Q_O)}}{\gamma} + \eta(b_1 + b_2) + \eta(a_1 + a_2) \frac{\sqrt{\lambda(Q_O)}}{\gamma} \right], \end{aligned}$$

where κ_{ell} and κ_{low} are the flux-regularity constants from Assumption 4.5.3 and a_1, a_2, b_1, b_2 are from Assumption 4.5.4. If, in addition, $e^{t\mathcal{L}^\dagger}$ is contractive on $\mathcal{H}_0$, the prefactor C in the two displayed bounds can be replaced by $1 + \log C$. Consequently, along any family of parameter pairs satisfying the hypotheses uniformly, with $\lambda(Q_O)$ and the structural constants uniformly bounded, and obeying $\eta \leq c/\gamma$, the right-hand side has diffusive overdamped scale $\mathcal{O}(\gamma^{-1})$ as $\gamma \rightarrow \infty$.

By Lemma 4.2, the same upper bound applies to convergence of f_t toward $h_{\gamma, \eta}$ for initial data in $\mathcal{H}_1$.

Proof. Lemma C.1 shows that every exponential estimate on $\mathcal{H}_0$ satisfies

$$\nu \leq C s(\mathcal{L}^\dagger).$$

It therefore remains to bound the singular-value gap $s(\mathcal{L}^\dagger)$. For that step, test $\mathcal{L}^\dagger$ on first-order corrections of normalized slow vectors ϕ_ε satisfying $\|(-Q_O)\phi_\varepsilon\| \leq \lambda(Q_O) + \varepsilon$. The two-term flux estimate from Assumption 4.5.3 controls the only nontrivial microscopic remainder, and Assumption 4.5.4 controls the perturbation through the $(-\mathcal{R})^{1/2}$ bounds. This yields the displayed upper bound on $s(\mathcal{L}^\dagger)$ after letting $\varepsilon \downarrow 0$, and therefore gives the bound on ν . If the semigroup is contractive, Corollary C.2 gives the improved prefactor $1 + \log C$. The detailed singular-value computation is given in Appendix C. $\square$

Theorem 5.4 gives the explicit $1/\gamma$ ceiling in the fixed reference norm under the same structural assumptions as the lower bound. Contractivity affects only the prefactor.

For families with $\kappa_{\text{low}}/\sqrt{\lambda(Q_O)} = \mathcal{O}(1)$, the $1/\gamma$ coefficient is of order $\lambda(Q_O)$. Hence the upper bound has the same family scaling as the window $\gamma \asymp \sqrt{\lambda(Q_O)}$ from Corollary 5.3.

6. EXAMPLES

We test the abstract hypotheses in four settings on increasingly compact state spaces, ending with a finite-dimensional quantum model. The nonequilibrium Langevin family on $\mathbb{R}^{2d}$ is the principal unbounded example. The active-Brownian example uses the standard no-translational-diffusion ABP kinematics together with the gravitactic aligning torque, so its nonequilibrium perturbation belongs to the active-matter model class rather than to a bespoke generator modification. The

circular-telegraph example uses the canonical equal-speed Goldstein–Kac/run-and-tumble dynamics with unequal switching rates; the switching asymmetry produces an explicit stationary current. The quantum example uses the boundary spin-flip reservoirs and XXZ Hamiltonian familiar from boundary-driven open spin systems; in the strong-boundary or Zeno scaling, its Schur complement is a non-self-adjoint Lindblad-type slow generator. The accompanying figures are numerical illustrations of the predicted three-regime profile and small-gap scaling; they are not used as evidence for the operator-norm bounds.

6.1. Nonequilibrium Langevin family. We consider a nonequilibrium underdamped Langevin family on $\mathbb{R}^{2d}$ with confining potential, exact position spectral gap $m > 0$, lower Hessian bound $\nabla^2 U \geq -K \text{Id}$, and a non-gradient forcing satisfying the relative-bounded perturbation hypothesis (71). Proposition 6.2 verifies the structural assumptions with constants independent of m , while the transport constant at $\varpi_* = \lambda(Q_O) = m$ remains uniform whenever $K \leq K_0 m$. Together with the stationary-density input recorded as Assumption 6.1 below, Theorem 5.2 then delivers the centered semigroup estimate. In the structural estimate, $\kappa_{\text{low}} \leq \sqrt{2K}$, so the lower-order defect disappears in the convex case. Throughout this example we fix $d \geq 2$, $\beta > 0$, $m \in (0, 1]$, and $K \geq 0$, and take $U \in C^\infty(\mathbb{R}^d)$ to satisfy

$$(63) \quad \nabla^2 U(q) \geq -K \text{Id} \quad \text{for all } q \in \mathbb{R}^d,$$

with all derivatives of positive order growing at most polynomially at infinity. We also take $F \in C^\infty(\mathbb{R}^d; \mathbb{R}^d)$ to be not of gradient form, with all derivatives of positive order growing at most polynomially at infinity. The polynomial-growth assumptions on U and F are used only to place the model within the standard confining Kramers–Fokker–Planck core framework; the explicit constants below enter through the relative bounds (71). These operator bounds are the actual hypothesis on the size of the force in the chosen reference space. Since the bounds use only momentum dissipation, bounded smooth non-gradient forces are the canonical admissible class; unbounded forces require a direct verification of (71).

Lifted dynamics. The microscopic configuration $(q_t, p_t) \in \mathbb{R}^d \times \mathbb{R}^d$ evolves according to the underdamped Langevin SDE

$$(64) \quad \begin{aligned} dq_t &= p_t dt, \\ dp_t &= -\nabla U(q_t) dt + \eta F(q_t) dt - \gamma p_t dt + \sqrt{2\gamma/\beta} dW_t, \end{aligned}$$

with W a standard d -dimensional Brownian motion. The corresponding backward generator on smooth observables is

$$\mathcal{L} = p \cdot \nabla_q - \nabla_q U \cdot \nabla_p + \eta F(q) \cdot \nabla_p + \gamma(-p \cdot \nabla_p + \beta^{-1} \Delta_p),$$

in agreement with the decomposition (70) below.

Setup. Let

$$(65) \quad d\mu(q, p) = Z^{-1} \exp\left(-\beta\left(\frac{1}{2}|p|^2 + U(q)\right)\right) dq dp = d\mu_q(q) d\mu_p(p),$$

where $\mu_q(dq) \propto e^{-\beta U(q)} dq$, and work in $\mathcal{H} = L^2(\mu)$ with slow projection

$$(66) \quad P_S f = \Pi_p f := \int_{\mathbb{R}^d} f(\cdot, p) d\mu_p(p).$$

We assume that the position marginal μ_q has exact spectral gap $m > 0$, equivalently that the optimal Poincaré constant is m^{-1} :

$$(67) \quad \beta^{-1} \|\nabla_q \phi\|_{L^2(\mu_q)}^2 \geq m \left\| \phi - \int \phi d\mu_q \right\|_{L^2(\mu_q)}^2, \quad \phi \in H^1(\mu_q).$$

Uniform convexity $\nabla^2 U \geq m \text{Id}$ is one sufficient special case by Bakry–Émery [1], but the verification below only uses (67) together with the lower Hessian bound. For the unbounded realization we use the Schwartz core

$$(68) \quad \mathcal{C}_{\text{Lan}} := \mathcal{S}(\mathbb{R}_q^d \times \mathbb{R}_p^d).$$

The appendix records the qualitative realization input used for both $\mathcal{L}$ and $\mathcal{L}^\dagger$: the equilibrium part is covered by standard Kramers–Fokker–Planck core theory, and the perturbation preserves $\mathcal{C}_{\text{Lan}}$ because F has polynomially growing coefficients. Thus the proposition below verifies the structural hypotheses under Assumption 3.1. The additional C^∞ and polynomial-growth assumptions on U and F are used only for this qualitative core input. At the level of Assumption 4.5, the quantitative constants below depend only on $(\beta, A_\nabla, B_\nabla, A_p, B_p, K)$ and are independent of m ; after choosing $\varpi_* = \lambda(Q_O) = m$, the effective transport quantity K_{geom} depends only on K/m . In the regime $K \leq K_0 m$, it is therefore controlled solely by K_0 . We apply the abstract theorem on the explicit reference mass-zero space

$$L_0^2(\mu) := \left\{ g \in L^2(\mu) : \int_{\mathbb{R}^{2d}} g d\mu = 0 \right\}, \quad L_1^2(\mu) := \mathbf{1} + L_0^2(\mu).$$

Assumption 6.1 (Stationary state in the reference space). For the parameter pair (γ, η) under consideration, the dynamics admits a stationary density relative to μ ,

$$\psi_{\gamma, \eta} = h_{\gamma, \eta} \mu, \quad h_{\gamma, \eta} \in L_1^2(\mu) \cap \text{dom}(\mathcal{L}^\dagger), \quad \mathcal{L}^\dagger h_{\gamma, \eta} = 0.$$

This is Assumption 4.1 written for the present family. In the equilibrium case $\eta = 0$ it is the standard Maxwell–Boltzmann density and is automatic; for $\eta > 0$ a periodic small-forcing analogue is established by Iacobucci–Olla–Stoltz [22, §2.3]. In the confining $\mathbb{R}^{2d}$ setting the existence and reference-space integrability of $h_{\gamma, \eta}$ are model-specific issues that we do not address here; the DMS coercivity argument supplies the rate, not the stationary density.

Generator decomposition. The Langevin generator is written as

$$(69) \quad \mathcal{L} = \mathcal{V}_{\text{skew}} + \gamma \mathcal{R} + \eta \mathcal{V}_{\text{pert}},$$

with

$$(70) \quad \mathcal{V}_{\text{skew}} := p \cdot \nabla_q - \nabla_q U \cdot \nabla_p, \quad \mathcal{R} := -p \cdot \nabla_p + \beta^{-1} \Delta_p, \quad \mathcal{V}_{\text{pert}} f := F(q) \cdot \nabla_p f.$$

We require the operator-relative bounds

$$(71) \quad \begin{aligned} \|F(q) \cdot \nabla_p f\|_{L^2(\mu)} &\leq A_\nabla \|(-\mathcal{R})^{1/2} f\|_{L^2(\mu)} + B_\nabla \|f\|_{L^2(\mu)}, \\ \|F(q) \cdot p f\|_{L^2(\mu)} &\leq A_p \|(-\mathcal{R})^{1/2} f\|_{L^2(\mu)} + B_p \|f\|_{L^2(\mu)}, \end{aligned}$$

for every $f \in \text{dom}((-\mathcal{R})^{1/2})$, with constants $A_\nabla, B_\nabla, A_p, B_p \geq 0$ independent of m and K . These are the model-specific bounds corresponding to Assumption 4.5.4. Because the right-hand side contains only momentum dissipation, they require F to act as a bounded multiplier in the position variable in typical product-test situations; bounded forces are the basic sufficient case. More general forces are admissible only insofar as (71) holds uniformly.

Effective generator $\mathcal{L}_O$ (explicit formula). Since $\mathcal{R}(p_i) = -p_i$ and $\int p_i d\mu_p = 0$, Definition 3.4 gives, for $\phi = \phi(q)$,

$$(72) \quad \mathcal{L}_O \phi = \beta^{-1} \Delta_q \phi - (\nabla_q U - \eta F(q)) \cdot \nabla_q \phi,$$

and therefore

$$(73) \quad Q_O \phi = \beta^{-1} \Delta_q \phi - \nabla_q U \cdot \nabla_q \phi.$$

The slow reference measure is $\mu_q(dq) \propto e^{-\beta U(q)} dq$. By the assumed Poincaré inequality (67),

$$(74) \quad -\langle \phi, Q_O \phi \rangle_{L^2(\mu_q)} = \beta^{-1} \|\nabla_q \phi\|_{L^2(\mu_q)}^2 \geq m \|\phi - \int \phi d\mu_q\|_{L^2(\mu_q)}^2.$$

Hence

$$(75) \quad \ker(Q_O) = \mathbb{C}\mathbf{1}, \quad \lambda(Q_O) = m.$$

For non-gradient F , the drift $\nabla_q U - \eta F(q)$ is generically not of gradient form for $\eta \neq 0$, so the effective generator $\mathcal{L}_O$ is a nonequilibrium slow generator except in compatible special cases.

Effective slow dynamics. Here (72) generates the overdamped Smoluchowski SDE

$$(76) \quad dq_t = -(\nabla U(q_t) - \eta F(q_t)) dt + \sqrt{2/\beta} d\widetilde{W}_t,$$

on $\mathbb{R}^d$, with $\widetilde{W}$ a standard d -dimensional Brownian motion. This is the $\gamma \rightarrow \infty$ adiabatic limit of (64) after the diffusive time rescaling $t \mapsto \gamma t$, in agreement with the abstract Schur-complement reduction of Definition 3.4. The non-gradient drift ηF generically produces a nonzero stationary current on the slow space when $\eta \neq 0$, provided the slow invariant density exists.

Non-self-adjointness on the slow space. In $L^2(\mu_q)$, integration by parts gives

$$(77) \quad (F \cdot \nabla_q)^\dagger = -F \cdot \nabla_q - (\nabla \cdot F) + \beta \nabla_q U \cdot F.$$

Hence $\mathcal{L}_O$ is generically non-self-adjoint as soon as $\eta \neq 0$. For the concrete force used in the numerics below,

$$(78) \quad F(q) = (-\tanh q_2, \tanh q_1),$$

one has $\nabla \cdot F = 0$. For the radial quadratic potential used in the numerical illustration below, $U(q) = \frac{m}{2}|q|^2$, however,

$$(79) \quad \nabla_q U(q) \cdot F(q) = m(-q_1 \tanh q_2 + q_2 \tanh q_1),$$

which is not identically zero. Thus the slow effective operator is non-self-adjoint for this numerical force whenever $\eta \neq 0$.

Verification of Assumptions 3.1, 4.4, 4.5, and 4.6. The structural hypotheses hold with quantitative constants depending on $(\beta, A_\nabla, B_\nabla, A_p, B_p, K)$ at the assumption level, and with a transport quantity at $\varpi_* = \lambda(Q_O)$ depending on K/m . In particular, if $K \leq K_0 m$, then K_{geom} is uniform in m .

Proposition 6.2 (Verification for the lower-Hessian family). *For the family (70) with $U \in C^\infty(\mathbb{R}^d)$ satisfying $\nabla^2 U \geq -K \text{Id}$ and with all derivatives of positive order growing at most polynomially, whose position marginal μ_q has spectral gap $m > 0$, and force F satisfying (71), Assumptions 3.1, 4.4, 4.5, and 4.6 hold in $L^2(\mu)$ for the common-core realization described above and in Appendix D. Moreover, one may choose structural constants satisfying*

$$(80) \quad \lambda_R = 1, \quad M_R = 1, \quad \kappa_{\text{ell}} \leq \sqrt{2}, \quad \kappa_{\text{low}} \leq \sqrt{2K},$$

and one may take

$$(81) \quad a_1 = A_\nabla + \frac{\beta}{2} A_p, \quad b_1 = B_\nabla + \frac{\beta}{2} B_p, \quad a_2 = \frac{\beta}{2} A_p, \quad b_2 = \frac{\beta}{2} B_p.$$

In particular, the perturbation constants (a_1, b_1, a_2, b_2) depend only on $(\beta, A_\nabla, B_\nabla, A_p, B_p)$ and are independent of (m, K) . More precisely, the appendix proves the weighted elliptic estimate in the form

$$(82) \quad \|P_F \mathcal{V}_{\text{skew}} P_F(-\mathcal{R}^+) w_\psi\|^2 \leq 2 \|(-Q_O) \psi\|^2 + 2K \|(-Q_O)^{1/2} \psi\|^2,$$

hence

$$(83) \quad \|P_F \mathcal{V}_{\text{skew}} P_F(-\mathcal{R}^+) w_\psi\| \leq \sqrt{2} \|(-Q_O) \psi\| + \sqrt{2K} \|(-Q_O)^{1/2} \psi\|,$$

for slow inputs ψ . Thus Assumption 4.5.3 holds with the displayed values of κ_{ell} and κ_{low} . In particular, if $K \leq K_0 m$, then the transport factor at $\varpi_* = \lambda(Q_O)$ satisfies

$$(84) \quad K_{\text{geom}} = k_{\text{dual}}(\lambda(Q_O)) \leq \sqrt{2} + \sqrt{K_0/2}$$

uniformly in m ; the special case $K = m$ gives $K_{\text{geom}} \leq \sqrt{2} + 1/\sqrt{2}$. This abstract constant is expressed in the split two-term form of the general theorem rather than the recombined warm-up factor $\sqrt{2 + \frac{K}{2m}}$ from Proposition 2.2. The extra polynomial-growth assumptions on the derivatives of U enter only in the qualitative Schwartz-core verification recalled in Appendix D; they do not affect these quantitative constants.

Corollary 6.3 (Structural kinetic scaling for the lower-Hessian family). *Fix $K_0 \geq 0$ and assume $K \leq K_0 m$. For the Langevin family above, under the same common-core realization convention, the structural hypotheses controlling γ_{opt} and $\max_{\gamma > 0} \Lambda_{\text{hypo}}(\gamma)$ in Corollary 5.3 are satisfied uniformly as $m \downarrow 0$. Consequently,*

$$(85) \quad \gamma_{\text{opt}} \asymp \sqrt{m}, \quad \max_{\gamma > 0} \Lambda_{\text{hypo}}(\gamma) \asymp \sqrt{m}.$$

These are structural rate-profile statements. Together with Assumption 6.1, Theorem 5.2 converts this scaling into a convergence estimate on $L_0^2(\mu)$ for parameter pairs in the DMS perturbative window $0 \leq \eta \leq \eta^*(\gamma)$. If the family also satisfies the non-degeneracy condition $b_{2,\text{eff}} + C_{\text{cross}} > 0$, the perturbative forcing window obeys

$$(86) \quad \eta^*(\gamma_{\text{opt}}) \asymp \sqrt{m}.$$

Thus the lower-Hessian regime $\nabla^2 U \geq -m \text{Id}$ falls within the $\sqrt{\lambda(Q_O)}$ structural scaling, and the same scaling holds for the semigroup whenever a stationary density in the reference space is available.

The proof of Proposition 6.2 is given in Appendix D. Together with Assumption 6.1, it yields the hypotheses of Theorem 5.2 for this family on the intersection of the realization and DMS windows.

Truncated numerical illustrations. These computations are not used in the proof and are not claimed to approximate the infinite-dimensional spectral gap uniformly; they illustrate only the three-regime profile predicted by the abstract bounds. We test the quadratic instance $U(q) = \frac{m}{2}|q|^2$ in $d = 2$, with $F(q) = (-\tanh q_2, \tanh q_1)$ and the scaling-consistent choice $\eta = c\sqrt{m}$, $c \in \{0, \frac{1}{2}, 1\}$. The generator is discretized in a tensor-product Hermite–Galerkin basis of dimension 256.

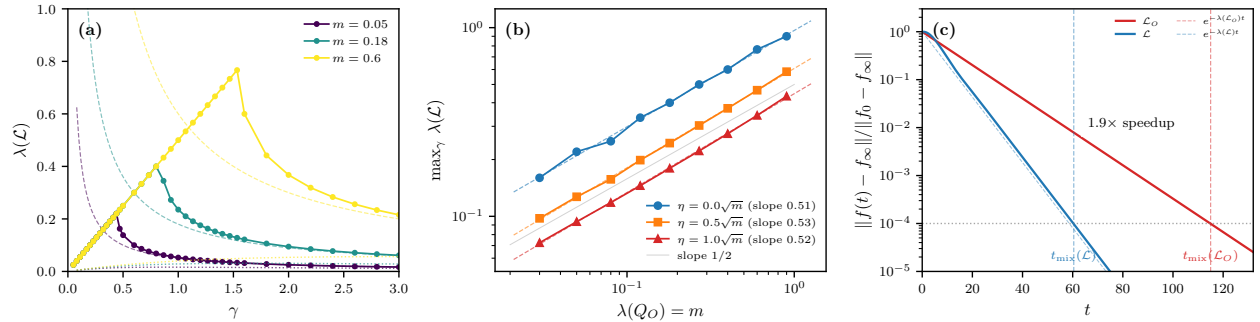


FIGURE 1. Numerical experiment on the truncated Hermite–Galerkin model for Example 6.1 ($\beta = 1$, $m \in [0.03, 0.90]$, 45-point γ -grid). (a) Real-part eigenvalue gap versus γ at $\eta = 0$; dashed: overdamped envelope m/γ ; dotted: DMS lower bound with $\varpi_* = m$. (b) Sampled peak real-part eigenvalue gap versus m on log–log scale for $c \in \{0, \frac{1}{2}, 1\}$; fitted slopes ≈ 0.51 – 0.53 . (c) Decay of one plotted trajectory at $m = 0.08$, $c = \frac{1}{2}$, $\gamma^* = 0.5$; vertical markers: ε -decay times ($\varepsilon = 10^{-4}$).

Figure 1 shows the three-regime structure and the predicted $\sqrt{\lambda(Q_O)}$ scaling for the real-part eigenvalue gaps of the truncations: on the sampled γ -grid, the fitted log-log slopes of the peak gaps are 0.51, 0.53, and 0.52 for $c = 0, \frac{1}{2}, 1$. At the representative grid point $m = 0.08, c = \frac{1}{2}$, the truncated lifted real-part eigenvalue gap is about 0.157, compared with $\lambda(\mathcal{L}_O) \approx 0.080$; the displayed trajectory gives decay times approximately 60 versus 115. The computation is finite-dimensional and non-normal, so the figure should be read only as a numerical illustration.

6.2. Gravitactic active Brownian particle without translational diffusion. This example is anchored to the standard two-dimensional active Brownian particle (ABP) with rotational Brownian noise and no translational Brownian forcing. This no-translational-diffusion, or athermal/high-Péclet, specialization is common in the ABP literature [37, 17, 36, 2, 39]. In our notation the unforced kinematics are

$$\dot{x}(t) = v_0 \cos \theta(t), \quad \dot{y}(t) = v_0 \sin \theta(t), \quad \dot{\theta}(t) = \sqrt{2D_R} \xi(t),$$

where v_0 is the swim speed, D_R is the rotational diffusion constant, and ξ is standard white noise. A bottom-heavy or gravitactic active particle in an external field has the usual overdamped aligning torque generated by an orienting potential proportional to $-\cos \theta$; see, for example, the bottom-heavy swimmer and external-field models in [44, 40, 42]. We put this standard gravitactic ABP on a large torus and write the weak aligning-torque strength as η , obtaining

$$(87) \quad \begin{aligned} dx_t &= v_0 \tau(\theta_t) dt, \\ d\theta_t &= -\eta \sin \theta_t dt + \sqrt{2\gamma} dW_t, \end{aligned}$$

with $\tau(\theta) = (\cos \theta, \sin \theta)$. In the notation of the cited ABP papers, the rotational diffusion coefficient is γ when $\eta = 0$; in this paper the symbol η is reserved for the perturbation amplitude. The torus $\mathbb{T}_L^2$ is only a finite-volume boundary condition for the deterministic spatial kinematics. Adding a term $\sqrt{2D_t} dB_t$ in x_t would create a direct slow diffusion $D_t \Delta_x$ and would no longer be the centered second-order lift treated by the abstract theorem.

Setup. Let $L \geq 1$ and set

$$\mathbb{T}_L^2 := \mathbb{R}^2 / (2\pi L\mathbb{Z})^2, \quad d\mu_L(x, \theta) = \frac{dx}{(2\pi L)^2} \frac{d\theta}{2\pi}$$

on $\mathbb{T}_L^2 \times \mathbb{S}^1$. We work in $\mathcal{H} = L^2(\mu_L)$ with slow projection

$$P_S f = \Pi_\theta f := \int_{\mathbb{S}^1} f(\cdot, \theta) \frac{d\theta}{2\pi}.$$

Write

$$L_0^2(\mu_L) := \left\{ g \in L^2(\mu_L) : \int g d\mu_L = 0 \right\}, \quad L_1^2(\mu_L) := \mathbf{1} + L_0^2(\mu_L).$$

With $\tau(\theta) = (\cos \theta, \sin \theta)$ and $\tau^\perp(\theta) = (-\sin \theta, \cos \theta)$, the backward generator is

$$(88) \quad \mathcal{L}f = \gamma \partial_{\theta\theta} f + v_0 \tau(\theta) \cdot \nabla_x f - \eta \sin \theta \partial_\theta f.$$

Equivalently,

$$(89) \quad \mathcal{L} = \mathcal{V}_{\text{skew}} + \gamma \mathcal{R} + \eta \mathcal{V}_{\text{pert}},$$

where

$$(90) \quad \begin{aligned} \mathcal{R} &:= \partial_{\theta\theta}, \\ \mathcal{V}_{\text{skew}} f &:= v_0 \tau(\theta) \cdot \nabla_x f, \\ \mathcal{V}_{\text{pert}} f &:= -\sin \theta \partial_\theta f. \end{aligned}$$

Centered coupling is immediate: $\int_{\mathbb{S}^1} \tau d\theta = 0$, and the torque operator $\mathcal{V}_{\text{pert}}$ annihilates slow functions. The adjoint stationary density is the von Mises angular law,

$$(91) \quad h_{\gamma, \eta}(x, \theta) = \frac{\exp((\eta/\gamma) \cos \theta)}{I_0(\eta/\gamma)},$$

where $I_0(a) = \int_{\mathbb{S}^1} e^{a \cos \theta} d\theta / (2\pi)$. Hence $h_{\gamma, \eta} \in L_1^2(\mu_L)$ for every $\gamma > 0$ and $\eta \geq 0$, and Assumption 4.1 holds.

Effective generator $\mathcal{L}_O$ (explicit formula). For a slow test function $\psi = \psi(x)$,

$$w_\psi := P_F \mathcal{V}_{\text{skew}} P_S \psi = v_0 \tau(\theta) \cdot \nabla_x \psi, \quad \mathcal{R}^+ w_\psi = -v_0 \tau(\theta) \cdot \nabla_x \psi.$$

Definition 3.4 therefore gives

$$(92) \quad Q_O \psi = \frac{v_0^2}{2} \Delta_x \psi.$$

The full effective operator contains the angular-bias cross term

$$(93) \quad \begin{aligned} \mathcal{L}_O \psi &= -P_S (\mathcal{V}_{\text{skew}} + \eta \mathcal{V}_{\text{pert}}) \mathcal{R}^+ (\mathcal{V}_{\text{skew}} + \eta \mathcal{V}_{\text{pert}}) P_S \psi \\ &= Q_O \psi + \eta \frac{v_0}{2} \partial_{x_1} \psi. \end{aligned}$$

Indeed, $\mathcal{V}_{\text{pert}} P_S = 0$, while

$$P_S \mathcal{V}_{\text{pert}} \mathcal{R}^+ \mathcal{V}_{\text{skew}} P_S \psi = -\frac{v_0}{2} \partial_{x_1} \psi.$$

The outer minus sign in Definition 3.4 therefore gives the positive sign of the drift in (93). The perturbation $(\eta v_0/2) \partial_{x_1}$ is skew-adjoint in $L^2(dx/(2\pi L)^2)$. It is a nonzero constant drift around the torus and is not generated by a periodic potential, so $\mathcal{L}_O$ is a genuinely non-self-adjoint slow generator whenever $\eta \neq 0$.

Effective slow dynamics. The operator $\mathcal{L}_O$ in (93), multiplied by the kinetic time scale γ^{-1} , generates the effective Smoluchowski SDE on $\mathbb{T}_L^2$,

$$(94) \quad dx_t = \frac{\eta v_0}{2\gamma} e_1 dt + \frac{v_0}{\sqrt{\gamma}} d\widetilde{W}_t,$$

where $\widetilde{W}$ is a standard 2-dimensional Brownian motion. At $\eta = 0$, the generator diffusion coefficient is $v_0^2/(2\gamma) = v_0^2/(2D_R)$ if D_R denotes the rotational diffusion coefficient; equivalently, the covariance diffusivity tensor is $(v_0^2/\gamma) \text{Id}$. This is the standard long-time ABP effective diffusivity. The present finite-volume example supplies the spectral comparison point: for the same no-translational-diffusion ABP mechanism, the Schur complement gives the slow gap $\lambda(Q_O) = v_0^2/(2L^2)$, the overdamped finite-volume rate has diffusive scale $\lambda(Q_O)/\gamma$, and Corollary 6.5 gives an optimized hypocoercive lower-rate profile of order $\sqrt{\lambda(Q_O)} \asymp L^{-1}$. The weak gravitactic torque adds a controlled nonequilibrium current, rather than a direct slow diffusion.

The exact angular stationary law of the biased lift is the von Mises density (91). Therefore the exact average swim velocity in the e_1 direction is

$$v_0 \mathbb{E}_{h_{\gamma, \eta}}[\cos \theta] = v_0 \frac{I_1(\eta/\gamma)}{I_0(\eta/\gamma)} = \frac{\eta v_0}{2\gamma} + O((\eta/\gamma)^3).$$

The Schur-complement drift $(\eta v_0/2\gamma) e_1$ in (94) is precisely this weak-bias linear response.

The spectrum of $-Q_O$ on mean-zero slow functions is

$$\left\{ \frac{v_0^2 |n|^2}{2L^2} : n \in \mathbb{Z}^2 \setminus \{0\} \right\},$$

Hence

$$\ker(Q_O) = \mathbb{C} \mathbf{1}, \quad \lambda(Q_O) = \frac{v_0^2}{2L^2}.$$

Verification of Assumptions 3.1, 4.4, 4.5, and 4.6. The model is infinite-dimensional but compact, and the verification is a Fourier calculation.

Proposition 6.4 (Gravitactic ABP verification). *For the family (88) on $L^2(\mu_L)$ with $L \geq 1$ and fixed $v_0 > 0$, Assumptions 3.1, 4.1, 4.4, 4.5, and 4.6 hold. One may choose*

$$(95) \quad \lambda_R = 1, \quad M_R = 1, \quad \kappa_{\text{ell}} \leq 1/\sqrt{2}, \quad \kappa_{\text{low}} = 0,$$

and

$$(96) \quad a_1 = 1, \quad b_1 = \frac{1}{2}, \quad a_2 = 0, \quad b_2 = \frac{1}{2}.$$

Moreover,

$$(97) \quad \ker(Q_O) = \mathbb{C}\mathbf{1}, \quad \lambda(Q_O) = \frac{v_0^2}{2L^2}.$$

For every $\gamma > 0$ and $\eta \geq 0$, the adjoint dynamics has the smooth positive stationary density (91). The effective slow generator is (93); its perturbative part is nonzero for $\eta \neq 0$.

Corollary 6.5 (Gravitactic ABP scaling). *For the family above, the hypotheses controlling γ_{opt} and $\max_{\gamma>0} \Lambda_{\text{hypo}}(\gamma)$ in Corollary 5.3 hold uniformly as $L \rightarrow \infty$ with v_0 fixed. Indeed, $\Theta = 1/2$, $k_{\text{reg}} = 1$, $K_{\text{geom}} \leq 1/\sqrt{2}$, $b_{2,\text{eff}} = 1/2$, and $C_{\text{cross}} = 2$, while $\lambda(Q_O) = v_0^2/(2L^2)$. Consequently,*

$$(98) \quad \gamma_{\text{opt}} \asymp L^{-1}, \quad \max_{\gamma>0} \Lambda_{\text{hypo}}(\gamma) \asymp L^{-1} \asymp \sqrt{\lambda(Q_O)}.$$

The perturbative aligning-torque window at the optimizing friction obeys

$$(99) \quad \eta^*(\gamma_{\text{opt}}) \asymp L^{-1} \asymp \sqrt{\lambda(Q_O)}.$$

The proof of Proposition 6.4 is given in Appendix E. The example is centered at the generator level, uses the fixed-speed ABP equations without translational diffusion and the standard gravitactic aligning torque, adds only a finite-volume boundary condition, and has a nonzero nonequilibrium effective perturbation. References whose ABP equations include independent translational Brownian noise are not used as model sources for the slow-fast part of this example, because that additional slow noise would violate the centered second-order structure being verified here.

Truncated numerical illustrations. Here we discretize (88) in a Fourier basis in both x and θ , with $v_0 = 1$. On the grid $L \in [1.8, 7.2]$ we take the scaling-consistent choice $\eta = c/L$ with $c \in \{0, \frac{1}{2}, 1\}$, equivalently $\eta = \sqrt{2}c\sqrt{\lambda(Q_O)}$. The angular-bias contribution to (93) is a constant skew drift, so $\lambda(\mathcal{L}_O) = \lambda(Q_O) = 1/(2L^2)$ for this family; this equality is special to the constant-drift example. The lifted operator is truncated to spatial modes $|n_i| \leq 2$ and angular modes $|k| \leq 5$, and panel (b) is plotted against $\lambda(\mathcal{L}_O)$. The calculation is finite-dimensional and non-normal, so it is only an illustration of the scaling statement.

Figure 2 again shows the three-regime structure for the real-part eigenvalue gaps of the truncations. Since $\lambda(\mathcal{L}_O) = \lambda(Q_O) = 1/(2L^2)$ here, panel (b) is organized by the same slow scale used in the theorem. On this grid the fitted slopes of the sampled peak gaps are approximately 0.53, 0.50, and 0.50 for $c = 0, \frac{1}{2}$, and 1, respectively. At the representative grid point $L = 4.2$, $c = \frac{1}{2}$, the sampled lifted real-part eigenvalue gap is about 0.0877, whereas $\lambda(\mathcal{L}_O) \approx 0.0283$; the displayed trajectory gives decay times approximately 109 versus 325.

6.3. Biased circular telegraph/run-and-tumble process. The Goldstein–Kac telegraph process is the classical velocity-jump model with velocities $\pm c$ and Poisson reversals [18, 23]. The same two-state dynamics is the one-dimensional run-and-tumble particle used in active-matter theory [41, 6]. Unequal reversal rates are the standard asymmetric telegraph extension [27], and De Gregorio–Iafate treat the corresponding random evolutions on a circle [8]. We use the canonical equal-speed, unequal-rate version directly.

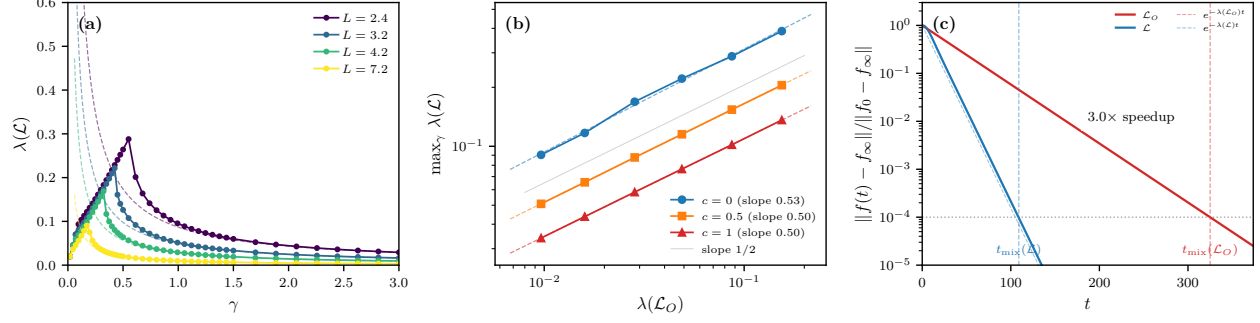


FIGURE 2. Truncated Fourier computations for the gravitactic active-Brownian family from Example 6.2 ($|n_i| \leq 2$, $|k| \leq 5$, $L \in [1.8, 7.2]$). (a) Real-part eigenvalue gap versus γ at $\eta = 0$ for several values of L ; dashed curves show the overdamped envelope $\lambda(\mathcal{L}_O)/\gamma$. (b) Log-log plot of the sampled peak real-part eigenvalue gap versus $\lambda(\mathcal{L}_O) = 1/(2L^2)$ for $\eta = c/L$ with $c \in \{0, \frac{1}{2}, 1\}$. (c) Decay of one plotted trajectory at $L = 4.2$, $c = \frac{1}{2}$, and sampled optimal $\gamma^* \approx 0.323$; vertical markers indicate the ε -decay times ($\varepsilon = 10^{-4}$).

Let $c > 0$ and $0 \leq \eta < \gamma$. The angular coordinate $\theta_t \in \mathbb{T} := \mathbb{R}/(2\pi\mathbb{Z})$ and velocity state $\sigma_t \in \{+1, -1\}$ evolve as

$$(100) \quad \begin{aligned} \dot{\theta}_t &= c\sigma_t, \\ +1 &\longrightarrow -1 \quad \text{at rate } \frac{\gamma + \eta}{2}, \\ -1 &\longrightarrow +1 \quad \text{at rate } \frac{\gamma - \eta}{2}. \end{aligned}$$

Thus η is the standard switching-rate bias. The condition $\eta < \gamma$ is exactly the positivity condition for the transition rates; the abstract theorem additionally imposes its own perturbative DMS window.

Setup. Let

$$d\mu_{\text{tel}}(\theta, \sigma) = \frac{d\theta}{2\pi} \frac{\delta_{+1}(d\sigma) + \delta_{-1}(d\sigma)}{2},$$

and work in $\mathcal{H} = L^2(\mu_{\text{tel}})$. The slow projection is the velocity-state average

$$P_S f(\theta) := \frac{1}{2}f(\theta, +1) + \frac{1}{2}f(\theta, -1),$$

so $\mathcal{H}_S$ consists of functions of θ alone and $P_S \sigma = 0$. The backward generator of (100) is

$$(101) \quad \mathcal{L}f(\theta, \sigma) = c\sigma\partial_\theta f(\theta, \sigma) + q_{+1}^{\gamma, \eta}(f(\theta, \bar{\sigma}) - f(\theta, \sigma)),$$

where $\bar{\sigma} = -\sigma$, $q_{+1}^{\gamma, \eta} = (\gamma + \eta)/2$, and $q_{-1}^{\gamma, \eta} = (\gamma - \eta)/2$. Equivalently,

$$(102) \quad \mathcal{L} = \mathcal{V}_{\text{skew}} + \gamma\mathcal{R} + \eta\mathcal{V}_{\text{pert}},$$

where

$$(103) \quad \begin{aligned} \mathcal{V}_{\text{skew}} f &:= c\sigma\partial_\theta f, \\ \mathcal{R}f(\theta, +1) &:= \frac{1}{2}(f(\theta, -1) - f(\theta, +1)), & \mathcal{R}f(\theta, -1) &:= \frac{1}{2}(f(\theta, +1) - f(\theta, -1)), \\ \mathcal{V}_{\text{pert}} f(\theta, +1) &:= \frac{1}{2}(f(\theta, -1) - f(\theta, +1)), \\ \mathcal{V}_{\text{pert}} f(\theta, -1) &:= -\frac{1}{2}(f(\theta, +1) - f(\theta, -1)). \end{aligned}$$

The componentwise centering is exact: $P_S \mathcal{V}_{\text{skew}} P_S = 0$ because $P_S \sigma = 0$, and $P_S \mathcal{V}_{\text{pert}} P_S = 0$ because the switching difference annihilates slow functions.

The stationary density relative to μ_{tel} is

$$(104) \quad h_{\gamma, \eta}(\theta, +1) = 1 - \frac{\eta}{\gamma}, \quad h_{\gamma, \eta}(\theta, -1) = 1 + \frac{\eta}{\gamma}.$$

Indeed, this is the stationary law of the two-state velocity chain, tensored with the uniform angular law, written relative to the uniform reference measure above; it is strictly positive for $0 \leq \eta < \gamma$. The corresponding stationary current is $-\eta c/\gamma$, so the lifted process is nonequilibrium whenever $\eta > 0$. Hence Assumption 4.1 holds in the fixed reference space.

Effective generator $\mathcal{L}_O$ (explicit formula). For a slow test function $\psi = \psi(\theta)$,

$$w_\psi = P_F \mathcal{V}_{\text{skew}} P_S \psi = c\sigma \partial_\theta \psi, \quad \mathcal{R}^+ w_\psi = -c\sigma \partial_\theta \psi.$$

Therefore

$$(105) \quad Q_O \psi = c^2 \partial_{\theta\theta} \psi.$$

The full Schur-complement operator is

$$(106) \quad \begin{aligned} \mathcal{L}_O \psi &= -P_S (\mathcal{V}_{\text{skew}} + \eta \mathcal{V}_{\text{pert}}) \mathcal{R}^+ (\mathcal{V}_{\text{skew}} + \eta \mathcal{V}_{\text{pert}}) P_S \psi \\ &= c^2 \partial_{\theta\theta} \psi - \eta c \partial_\theta \psi. \end{aligned}$$

Thus, on the kinetic time scale, the effective angular diffusion is

$$(107) \quad d\theta_t = -\frac{\eta c}{\gamma} dt + \sqrt{\frac{2c^2}{\gamma}} dB_t \quad \text{on } \mathbb{T}.$$

The perturbative term $-\eta c \partial_\theta$ is a nonzero constant drift around the circle. It is skew-adjoint in $L^2(d\theta/(2\pi))$ and is not the gradient of a periodic potential, so the effective slow dynamics has a nonzero stationary current whenever $\eta \neq 0$.

Equivalently, if $p_\pm$ are the forward densities and $\rho = p_+ + p_-$, $j = c(p_+ - p_-)$, then the exact telegraph equations are

$$\partial_t \rho + \partial_\theta j = 0, \quad \partial_t j = -c^2 \partial_{\theta\theta} \rho - \eta c \rho - \gamma j.$$

The quasi-steady closure $j = -(c^2/\gamma) \partial_\theta \rho - (\eta c/\gamma) \rho$ gives precisely the Fokker–Planck equation adjoint to (107). Thus the Schur complement recovers the standard drifted diffusion limit of the biased telegraph process.

The spectrum of $-Q_O$ on mean-zero slow functions is

$$\{c^2 n^2 : n \in \mathbb{Z} \setminus \{0\}\}.$$

Consequently,

$$(108) \quad \ker(Q_O) = \mathbb{C}\mathbf{1}, \quad \lambda(Q_O) = c^2.$$

Verification of Assumptions 3.1, 4.4, 4.5, and 4.6. The fast space is one-dimensional over each angular point, so all constants are obtained by a two-state calculation.

Proposition 6.6 (Circular telegraph verification). *For the biased circular telegraph family (101) on $L^2(\mu_{\text{tel}})$ with $c > 0$, $\gamma > 0$, and $0 \leq \eta < \gamma$, Assumptions 3.1, 4.1, 4.4, 4.5, and 4.6 hold. One may choose*

$$(109) \quad \lambda_R = 1, \quad M_R = 1, \quad \kappa_{\text{ell}} = 0, \quad \kappa_{\text{low}} = 0,$$

and

$$(110) \quad a_1 = 0, \quad b_1 = \frac{1}{2}, \quad a_2 = 0, \quad b_2 = \frac{1}{2}.$$

Moreover,

$$(111) \quad \ker(Q_O) = \mathbb{C}\mathbf{1}, \quad \lambda(Q_O) = c^2.$$

For every $\gamma > 0$ and $0 \leq \eta < \gamma$, the adjoint dynamics has the positive stationary density (104). The effective slow generator is (106); its perturbative drift is nonzero for $\eta \neq 0$.

Corollary 6.7 (Biased circular telegraph scaling). *Fix $\bar{c} > 0$ and set $c = s\bar{c}$ with $s \downarrow 0$. For the family above, the hypotheses controlling γ_{opt} and $\max_{\gamma>0} \Lambda_{\text{hypo}}(\gamma)$ in Corollary 5.3 hold uniformly along this fixed-shape scaling. Consequently,*

$$(112) \quad \gamma_{\text{opt}} \asymp s, \quad \max_{\gamma>0} \Lambda_{\text{hypo}}(\gamma) \asymp s \asymp \sqrt{\lambda(Q_O)}.$$

The DMS perturbative window at the optimizing friction obeys

$$(113) \quad \eta^*(\gamma_{\text{opt}}) \asymp s \asymp \sqrt{\lambda(Q_O)}.$$

After intersecting with the positivity condition for the switching rates, $0 \leq \eta < \gamma_{\text{opt}}$, the admissible switching-bias range is still of order s .

The proof of Proposition 6.6 is given in Appendix F. This example is fully compact and has no unbounded-domain input: the core is the trigonometric polynomials in θ with two velocity signs. It also shows a finite-state fast mechanism where the bounded switching-rate bias is centered but produces a genuinely nonequilibrium effective current.

Truncated numerical illustrations. Here the Fourier modes in θ are exact two-state blocks. We set $c = s$ for $s \in [0.08, 0.78]$ and use the scaling-consistent choice $\eta = \rho c = \rho \sqrt{\lambda(Q_O)}$ with $\rho \in \{0, \frac{1}{2}, 1\}$. The truncation keeps modes $|n| \leq 60$. Since the perturbative part of (106) is a constant skew drift, $\lambda(\mathcal{L}_O) = \lambda(Q_O) = c^2$ for these runs; this equality is special to the constant-drift example. Panel (b) is therefore plotted against $\lambda(\mathcal{L}_O)$, as in the preceding examples. The computation is finite-dimensional and non-normal, so it is only an illustration of the scaling statement.

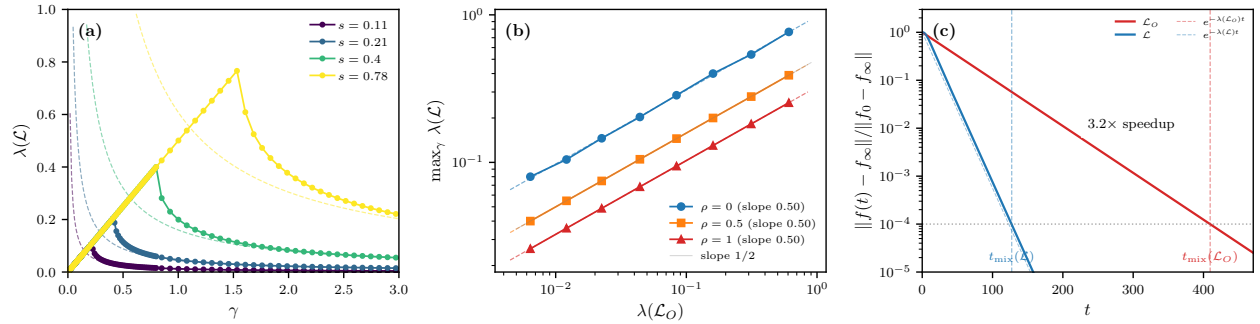


FIGURE 3. Truncated Fourier computations for the biased circular telegraph/run-and-tumble family from Example 6.3 ($|n| \leq 60$, $c = s$). (a) Real-part eigenvalue gap versus γ at $\eta = 0$ for several values of s ; dashed curves show the overdamped envelope c^2/γ . (b) Log-log plot of the sampled peak real-part eigenvalue gap versus $\lambda(\mathcal{L}_O) = c^2$ for $\eta = \rho c$ with $\rho \in \{0, \frac{1}{2}, 1\}$. (c) Decay of one plotted trajectory at $s = 0.15$, $\rho = \frac{1}{2}$, and sampled optimal $\gamma^* \approx 0.308$; vertical markers indicate the ε -decay times ($\varepsilon = 10^{-4}$).

Figure 3 shows the same three-regime gap profile as the two diffusion examples. Along the fixed-shape scaling $c = s$, the fitted slopes of the sampled peak gaps are approximately 0.501, 0.500, and 0.500 for $\rho = 0$, $\frac{1}{2}$, and 1, respectively. At the representative grid point $s = 0.15$, $\rho = \frac{1}{2}$, one has $c = 0.15$, $\eta = 0.075$, and the sampled lifted real-part eigenvalue gap is about 0.0749, compared with $\lambda(\mathcal{L}_O) = 0.0225$; the displayed trajectory gives decay times approximately 127 versus 409.

6.4. Boundary-driven two-qubit XXZ Lindblad model. This example is a finite-dimensional open quantum system. It uses the standard boundary-driven spin-chain ingredients reviewed in [25]: an XXZ Hamiltonian and incoherent spin-flip reservoirs at a boundary. Exact nonequilibrium steady states for larger boundary-driven XXZ chains are well studied, notably in Prosen's matrix-product solutions [34, 35]. The closest two-qubit comparison we found is Popkov–Presilla–Salerno [33], who study an open two-qubit XYZ/XXZ exchange model with one qubit coupled to a polarizing Lindblad bath and analyze its NESS relaxation, Liouvillian exceptional-point manifolds, and effective Zeno regime. Their Hamiltonian and one-boundary geometry are close to ours, but their dissipator is a single polarizing jump, whereas the model below uses the symmetric pair of spin-flip jumps plus a small centered rate bias. This distinction is essential for the present framework: the symmetric reservoir leaves the protected slow space as $I \otimes \mathcal{B}(\mathbb{C}^2)$, and the bias enters as a centered perturbation of that reservoir. We take the smallest nontrivial two-qubit member because it can be verified completely in the present Hilbert-space framework without importing any unproved many-body estimates. The strong boundary reservoir plays the role of the fast variable, so the construction is also a concrete instance of the Zeno-limit Schur complement used in dissipative perturbation theory [24, 5].

Setup. Let $\mathfrak{h} = (\mathbb{C}^2)^{\otimes 2}$ and let $\sigma_1, \sigma_2, \sigma_3$ be the Pauli matrices. We write

$$\sigma_{\pm} := \frac{1}{2}(\sigma_1 \pm i\sigma_2).$$

The Hilbert space for the abstract theorem is the observable space

$$\mathcal{H} := \mathcal{B}(\mathfrak{h}), \quad \langle A, B \rangle := 4^{-1} \text{tr}(A^* B), \quad \mathbf{1} := I \otimes I.$$

For a jump operator L define the Heisenberg Lindblad dissipator

$$\mathcal{D}_L(A) := 2L^*AL - L^*LA - AL^*L.$$

Fix an anisotropy $\Delta > 0$ and coupling strength $J > 0$. The Hamiltonian is the two-site XXZ Hamiltonian

$$(114) \quad H_J := J(\sigma_1 \otimes \sigma_1 + \sigma_2 \otimes \sigma_2 + \Delta \sigma_3 \otimes \sigma_3).$$

The boundary reservoir acts only on the first spin. Its symmetric infinite-temperature part and centered bias direction are

$$(115) \quad \begin{aligned} \mathcal{R}A &:= \frac{1}{2}\mathcal{D}_{\sigma_+ \otimes I}(A) + \frac{1}{2}\mathcal{D}_{\sigma_- \otimes I}(A), \\ \mathcal{V}_{\text{pert}}A &:= \frac{1}{2}\mathcal{D}_{\sigma_+ \otimes I}(A) - \frac{1}{2}\mathcal{D}_{\sigma_- \otimes I}(A), \\ \mathcal{V}_{\text{skew}}A &:= i[H_J, A]. \end{aligned}$$

Thus the lifted Heisenberg generator is

$$(116) \quad \mathcal{L}A = \gamma \mathcal{R}A + i[H_J, A] + \eta \mathcal{V}_{\text{pert}}A.$$

Equivalently, for $0 \leq \eta < \gamma$ the dissipative boundary rates are

$$\frac{\gamma + \eta}{2}\mathcal{D}_{\sigma_+ \otimes I} + \frac{\gamma - \eta}{2}\mathcal{D}_{\sigma_- \otimes I},$$

which is the usual left-boundary spin-injection and spin-removal reservoir, restricted here to two spins. The condition $\eta < \gamma$ is the physical positivity condition for the two rates; the abstract theorem additionally imposes its own DMS perturbative window.

Let P_S be the partial trace over the first spin against the normalized trace,

$$P_SA := \frac{1}{2}I \otimes \text{tr}_1 A.$$

Then $\mathcal{H}_S = \ker(\mathcal{R}) = I \otimes \mathcal{B}(\mathbb{C}^2)$ and

$$\mathcal{H}_0 = \{A \in \mathcal{B}(\mathfrak{h}) : \text{tr } A = 0\}.$$

The core is the whole finite-dimensional space, so Assumption 3.1 is automatic.

Remark (Why the example is two-site). The direct nearest-neighbour N -site extension with the same single left-boundary reservoir is not an immediate example of the present centered theorem. If

$$H_N = J \sum_{r=1}^{N-1} \left(\sigma_1^{(r)} \sigma_1^{(r+1)} + \sigma_2^{(r)} \sigma_2^{(r+1)} + \Delta \sigma_3^{(r)} \sigma_3^{(r+1)} \right)$$

and the dissipator acts only on site 1, then $\ker(\mathcal{R}) = I_1 \otimes \mathcal{B}((\mathbb{C}^2)^{\otimes(N-1)})$. For a slow observable $A = I_1 \otimes B$ and $N \geq 3$,

$$P_S i[H_N, A] = I_1 \otimes i[H_{2:N}, B], \quad H_{2:N} := J \sum_{r=2}^{N-1} \left(\sigma_1^{(r)} \sigma_1^{(r+1)} + \sigma_2^{(r)} \sigma_2^{(r+1)} + \Delta \sigma_3^{(r)} \sigma_3^{(r+1)} \right),$$

which is nonzero for generic B . Thus $P_S \mathcal{V}_{\text{skew}} P_S = 0$ fails as soon as there is a bulk bond entirely inside the protected algebra. Longer boundary-Zeno spin chains are natural and have their own effective projected Hamiltonian dynamics [32, 5], but accommodating that direct slow skew term would require an extension of the abstract framework rather than a verification of the theorem as stated here.

Effective generator $\mathcal{L}_O$ (explicit formula). Use the slow Pauli basis

$$E_0 := I \otimes I, \quad E_j := I \otimes \sigma_j, \quad j = 1, 2, 3.$$

The Schur complement gives

$$(117) \quad Q_O E_0 = 0, \\ Q_O E_1 = -J^2(4 + 2\Delta^2)E_1, \quad Q_O E_2 = -J^2(4 + 2\Delta^2)E_2, \quad Q_O E_3 = -8J^2 E_3.$$

The full effective generator contains the boundary-bias cross term

$$(118) \quad \begin{aligned} \mathcal{L}_O E_0 &= 0, \\ \mathcal{L}_O E_1 &= -J^2(4 + 2\Delta^2)E_1 - 2\eta J \Delta E_2, \\ \mathcal{L}_O E_2 &= -J^2(4 + 2\Delta^2)E_2 + 2\eta J \Delta E_1, \\ \mathcal{L}_O E_3 &= -8J^2 E_3. \end{aligned}$$

Thus the perturbative part of the slow operator is the nonzero rotation

$$2\eta J \Delta (E_1 \partial_{E_2} - E_2 \partial_{E_1})$$

on the slow coherences when $\eta \Delta J \neq 0$. In matrix form on $\text{span}\{E_1, E_2, E_3\}$,

$$\mathcal{L}_O = \begin{pmatrix} -J^2(4 + 2\Delta^2) & 2\eta J \Delta & 0 \\ -2\eta J \Delta & -J^2(4 + 2\Delta^2) & 0 \\ 0 & 0 & -8J^2 \end{pmatrix}$$

with the convention that columns give input basis vectors. This matrix is non-self-adjoint whenever $\eta \Delta J \neq 0$. Since the stationary state of the effective slow dynamics is the normalized identity but the slow generator has a rotational current on the E_1 – E_2 coherences, the example is genuinely nonequilibrium at the effective level. The reference gap is

$$(119) \quad \lambda(Q_O) = J^2 \min\{4 + 2\Delta^2, 8\}.$$

Effective slow quantum dynamics. On the kinetic time scale, the slow observable evolution is generated by $\gamma^{-1} \mathcal{L}_O$. Its reference dissipative part damps the three traceless Pauli observables

of the second spin at rates $J^2(4 + 2\Delta^2)/\gamma$, $J^2(4 + 2\Delta^2)/\gamma$, and $8J^2/\gamma$. The weak boundary bias produces the coherent precession

$$\frac{2\eta J \Delta}{\gamma} (E_1 \partial_{E_2} - E_2 \partial_{E_1}),$$

which is the leading Zeno-limit current induced on the protected second spin by the biased boundary reservoir. If $\Delta = 0$ the bias cross term vanishes; this is why the genuinely nonequilibrium example uses the XXZ, not the XX, Hamiltonian.

The adjoint Schrödinger generator is a finite-dimensional Lindblad generator for $0 \leq \eta < \gamma$, hence compactness of the density-matrix simplex gives at least one stationary density matrix. If $\rho_{\gamma,\eta}$ is such a state, then $h_{\gamma,\eta} := 4\rho_{\gamma,\eta}$ belongs to $\mathcal{H}_1$ and satisfies $\mathcal{L}^\dagger h_{\gamma,\eta} = 0$. This verifies Assumption 4.1.

Verification of Assumptions 3.1, 4.4, 4.5, and 4.6. All computations are finite-dimensional Pauli-matrix identities.

Proposition 6.8 (Boundary-driven quantum XXZ verification). *For the two-qubit Lindblad family (116) with $J > 0$, $\Delta > 0$, $\gamma > 0$, and $0 \leq \eta < \gamma$, Assumptions 3.1, 4.1, 4.4, 4.5, and 4.6 hold in $\mathcal{B}((\mathbb{C}^2)^{\otimes 2})$ equipped with the normalized Hilbert–Schmidt inner product, for parameter pairs in the DMS perturbative window of Theorem 5.2. One may choose*

(120)

$$\lambda_R = 1, \quad M_R = \max \left\{ 1, \frac{4 + 4\Delta^2}{4 + 2\Delta^2} \right\} \leq 2, \quad \kappa_{\text{ell}} \leq \max \left\{ 1, \frac{6\Delta}{4 + 2\Delta^2} \right\} \leq \frac{3\sqrt{2}}{4}, \quad \kappa_{\text{low}} = 0,$$

and

$$(121) \quad a_1 = 0, \quad b_1 = 1, \quad a_2 = 0, \quad b_2 = 1.$$

Moreover,

$$(122) \quad \ker(Q_O) = \mathbb{C}\mathbf{1}, \quad \lambda(Q_O) = J^2 \min\{4 + 2\Delta^2, 8\}.$$

The effective slow generator is (118); its perturbative part is nonzero for $\eta\Delta J \neq 0$.

Corollary 6.9 (Boundary-driven quantum XXZ scaling). *Fix an anisotropy $\Delta > 0$. Along the weak-coupling scaling $J \downarrow 0$, the hypotheses controlling γ_{opt} and $\max_{\gamma>0} \Lambda_{\text{hypo}}(\gamma)$ in Corollary 5.3 hold uniformly. Consequently,*

$$(123) \quad \gamma_{\text{opt}} \asymp J, \quad \max_{\gamma>0} \Lambda_{\text{hypo}}(\gamma) \asymp J \asymp \sqrt{\lambda(Q_O)}.$$

The DMS perturbative window at the optimizing friction obeys

$$(124) \quad \eta^*(\gamma_{\text{opt}}) \asymp J \asymp \sqrt{\lambda(Q_O)}.$$

After intersecting with the positivity condition for the boundary spin-flip rates, $0 \leq \eta < \gamma_{\text{opt}}$, the admissible boundary-bias range is still of order J .

The proof of Proposition 6.8 is given in Appendix G. The example does not alter the quantum dynamics to fit the framework: the reservoir is precisely the standard pair of boundary spin-flip jump operators, the Hamiltonian is the two-site XXZ Hamiltonian, and only the rates are written in symmetric-plus-biased coordinates. The perturbation is centered in the abstract sense but its Schur-complement contribution is nonzero.

Truncated numerical illustrations. There is no truncation error in this example: the observable space has dimension 16. We fix $\Delta = 0.7$, use the weak-coupling grid $J \in [0.04, 0.30]$, and take the scaling-consistent boundary bias $\eta = \rho\sqrt{\lambda(Q_O)}$ with $\rho \in \{0, \frac{1}{2}, 1\}$. The figure scans the real-part eigenvalue gap of the exact 16×16 Heisenberg generator. The slow effective real-part gap is $\lambda(\mathcal{L}_O) = \lambda(Q_O)$ because the perturbative slow term in (118) is skew-adjoint.

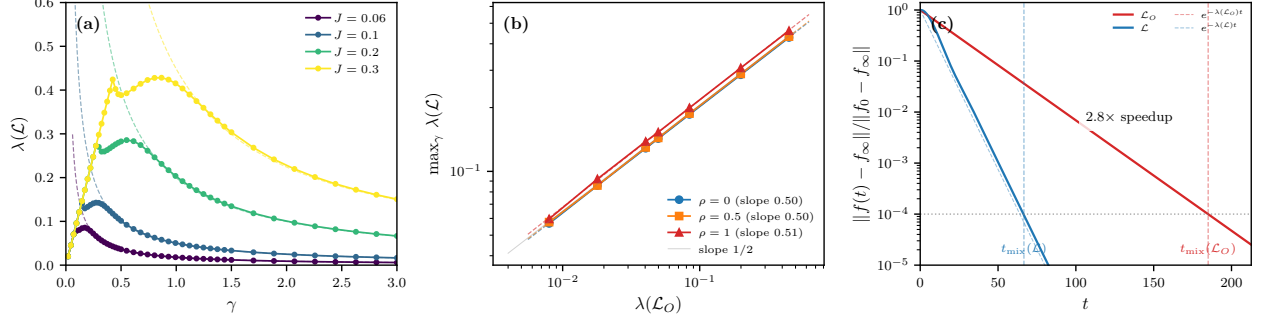


FIGURE 4. Exact finite-dimensional computations for the boundary-driven two-qubit XXZ family from Example 6.4 ($\Delta = 0.7$, dimension 16). (a) Real-part eigenvalue gap versus γ at $\eta = 0$ for several values of J ; dashed curves show the overdamped envelope $\lambda(\mathcal{L}_O)/\gamma$. (b) Log-log plot of the sampled peak real-part eigenvalue gap versus $\lambda(\mathcal{L}_O) = \lambda(Q_O)$ for $\eta = \rho\sqrt{\lambda(Q_O)}$ with $\rho \in \{0, \frac{1}{2}, 1\}$. (c) Decay of one plotted centered trajectory at $J = 0.10$, $\rho = \frac{1}{2}$, and sampled optimal γ^* ; vertical markers indicate the ε -decay times ($\varepsilon = 10^{-4}$).

Figure 4 shows the same three-regime profile as the classical examples. Along the weak-coupling scaling, the sampled peak gaps have log-log slopes reported in the figure output and close to 1/2 when plotted against $\lambda(Q_O)$. Since the matrix is non-normal for $\eta > 0$, the plotted trajectory is an illustration of typical decay rather than a sharp norm estimate.

7. CONCLUSION

We have established a quantitative hypocoercive framework for centered second-order lifts whose effective macroscopic dynamics need not be reversible. The starting point is the classical observation that, in such lifts, the slow variables relax only through a slow-fast-slow excursion encoded by the self-adjoint Schur-complement operator Q_O on $\mathcal{H}_S$. Our central technical contribution is the recognition that, within the modified- L^2 method of Dolbeault–Mouhot–Schmeiser, the resolvent shift in the corrector must be calibrated to the macroscopic gap, $\varpi_* = \lambda(Q_O)$, in order for the kinetic acceleration scale to survive in the small-gap limit. A shift fixed independently of $\lambda(Q_O)$ suppresses the weakest macroscopic mode by an avoidable factor of $\lambda(Q_O)$ inside the DMS dissipation matrix and degrades the optimized rate from $\sqrt{\lambda(Q_O)}$ to a strictly slower scale; the warm-up of Section 2 makes this loss fully explicit on the equilibrium underdamped Langevin model.

With the gap-scale calibration in hand, Theorem 5.2 converts the DMS Lyapunov computation into a 2×2 dissipation-matrix problem whose solution is uniform in the macroscopic gap and linear in the nonequilibrium amplitude η . Corollary 5.3 then shows that

$$\gamma_{\text{opt}} \asymp \sqrt{\lambda(Q_O)}, \quad \sup_{\gamma > 0} \Lambda_{\text{hypo}}(\gamma) \asymp \sqrt{\lambda(Q_O)}, \quad \eta^*(\gamma_{\text{opt}}) \asymp \sqrt{\lambda(Q_O)},$$

matching the classical kinetic acceleration law and pinning the natural perturbative window to the same scale. The matching diffusive upper bound (Theorem 5.4), proved via a singular-value-gap argument that does not require contractivity, complements this lower bound: it shows that within the same reference norm the square-root profile cannot be obtained at large friction, and singles out the intermediate-friction window $\gamma \asymp \sqrt{\lambda(Q_O)}$ as the unique regime in which the present method delivers the ballistic profile.

The four examples developed in Section 6—a confining nonequilibrium underdamped Langevin family on $\mathbb{R}^{2d}$, a gravitactic active Brownian particle without translational diffusion on the torus, a biased circular telegraph (run-and-tumble) process on $\mathbb{S}^1$, and a boundary-driven two-qubit XXZ Lindblad model—demonstrate that the calibration mechanism is intrinsic to centered second-order

lifts and is not tied to any particular stochastic representation. Each example required a model-specific verification of the structural inputs (Assumptions 3.1, 4.1, 4.4, 4.5, and 4.6) and produced an explicit nonequilibrium effective slow generator: a non-gradient drift on $\mathbb{R}^d$, a constant skew current on the torus, an asymmetric advection on the circle, and a Zeno-limit coherent precession on a protected qubit. The same gap-rescaled $\sqrt{\lambda(Q_O)}$ rate is recovered along the natural small-gap scaling in each case, and the perturbative forcing window is also matched to the same scale. The accompanying numerics confirm the predicted three-regime profile (overdamped, intermediate, and underdamped) with sampled peak-gap exponents close to $1/2$.

Discussion and outlook. Several directions deserve further investigation.

(i) *Beyond the perturbative window.* The present analysis controls the nonequilibrium perturbation linearly, through the explicit constant $B_{\text{pert}}(\gamma)$, and the admissible window scales as $\eta \leq \eta^*(\gamma_{\text{opt}}) \asymp \sqrt{\lambda(Q_O)}$. A genuinely nonperturbative treatment, valid for forcing amplitudes of order one or larger, would require either a weighted- L^2 refinement in the spirit of Dietert [10] and Monmarché [30, 29] or a Schur-complement reformulation that does not rely on closeness to the equilibrium reference Q_O . We do not know whether the $\sqrt{\lambda(Q_O)}$ profile persists in this regime; the upper-bound mechanism (Theorem 5.4) suggests that it should, since the diffusive ceiling is unaffected by η in leading order, but a corresponding lower bound is open.

(ii) *Constructive design of lifts.* The framework is model-agnostic but does not yet supply a constructive recipe for choosing the auxiliary fast space and factorization. A promising question is whether, given a target overdamped operator with gap m , one can systematically design a centered second-order lift whose structural constants $(\Theta, k_{\text{reg}}, \kappa_{\text{ell}}, \kappa_{\text{low}}, a_i, b_i)$ remain informative as $m \downarrow 0$. This is particularly relevant in high-dimensional or many-body settings, where naive lifts (such as Hamiltonian Monte Carlo with a fixed momentum mass) typically degrade in dimension. The constants K_{geom} and C_{cross} identified here provide a quantitative target for such design.

(iii) *Sharper prefactors and contractivity.* Theorem 5.2 carries the prefactor $C_*(\gamma) \leq 1 + \sqrt{2}$ and the rate constant $c = 2 - \sqrt{2}$, both inherited from the norm-equivalence step at the optimal Lyapunov weight. When the centered semigroup is contractive on $\mathcal{H}_0$, the upper bound improves from C to $1 + \log C$, but the lower bound retains the $1 + \sqrt{2}$ prefactor. Identifying classes of nonequilibrium models for which contractivity holds—and removing the resulting prefactor gap—would tighten the comparison between the lower and upper bounds and clarify when the two coincide up to constants.

(iv) *Many-body and mean-field extensions.* The two-qubit XXZ model is the smallest nontrivial member of a family of boundary-driven spin chains for which exact NESS are known [34, 35]. Whether the gap-rescaled $\sqrt{\lambda(Q_O)}$ rate persists for longer chains or in the thermodynamic limit, and whether the structural constants of Assumption 4.5 can be controlled uniformly along such families, are natural questions. For the natural single-boundary nearest-neighbour chain, the projected bulk Hamiltonian already produces a direct slow skew term, so this is not merely a matter of estimating the present constants uniformly; it requires a version of the framework that allows nonzero slow Hamiltonian dynamics. Analogous extensions to many-body active-matter and McKean–Vlasov NESS would test the same calibration mechanism in the mean-field and propagation-of-chaos regimes.

(v) *Discrete-time and numerical analogues.* Finally, the calibration $\varpi_* = \lambda(Q_O)$ is, in the underlying Markov-chain language of [7, 9], a precise prescription for choosing the reference time scale on which to apply the lift. We expect that the same recalibration is relevant for the analysis of non-reversible MCMC schemes, splitting integrators for underdamped Langevin dynamics in the small-gap regime, and lifted samplers for active-matter free-energy computations. Quantifying this connection is left to future work.

APPENDIX A. PROOF OF LEMMA 4.9

This appendix establishes the structural bounds of Lemma 4.9. The proof follows the order in which the Lyapunov argument uses them: first the solver and regularity bounds for $\mathcal{A}_\varpi$, then the interaction estimates with $\mathcal{R}$, $\mathcal{V}_{\text{skew}}$, and $\mathcal{V}_{\text{pert}}$.

Proof. In the unbounded setting, all identities below are first established on the dense invariant core from Section 3.1; the resulting operator bounds then extend to the stated energy spaces by density and continuity. On the slow core from Assumption 4.4, the effective operator is represented by

$$Q_O = -P_S \mathcal{V}_{\text{skew}} \mathcal{R}^+ \mathcal{V}_{\text{skew}} P_S.$$

By Assumption 4.5.1, the pseudoinverse is bounded on $\mathcal{H}_F$ with $\|\mathcal{R}^+\| \leq \lambda_R^{-1}$, hence

$$(125) \quad \langle g, -\mathcal{R}^+ g \rangle \leq \frac{1}{\lambda_R} \|g\|^2, \quad \forall g \in \mathcal{H}_F.$$

The complementary (lower) control needed below is supplied by the flux conditioning assumption (29), which applies to fluxes of the form $w = w_\psi = P_F \mathcal{V}_{\text{skew}} P_S \psi$.

A.1. Proof of Properties 1 and 2 (Boundedness and Regularity). Solver boundedness and forward regularity. Let f belong to the invariant core and define $u := \mathcal{A}_\varpi f \in \mathcal{H}_S$. The defining equation is $(\varpi - Q_O)u = P_S \mathcal{V}_{\text{skew}} (-\mathcal{R}^+) P_F f$. Taking the inner product with u and using the skew-adjointness $\mathcal{V}_{\text{skew}}^\dagger = -\mathcal{V}_{\text{skew}}$:

$$(126) \quad \varpi \|u\|^2 + \langle u, -Q_O u \rangle = \langle u, P_S \mathcal{V}_{\text{skew}} (-\mathcal{R}^+) P_F f \rangle = -\langle \mathcal{V}_{\text{skew}} u, (-\mathcal{R}^+) P_F f \rangle.$$

Let $v := \mathcal{V}_{\text{skew}} u$. Since $P_S \mathcal{V}_{\text{skew}} P_S = 0$ (Centered Coupling), we strictly have $v \in \mathcal{H}_F$. Using Assumption 4.4(ii) with $w_u = P_F \mathcal{V}_{\text{skew}} P_S u = v$ and the pseudoinverse bound (125):

$$(127) \quad \langle u, -Q_O u \rangle = \langle v, -\mathcal{R}^+ v \rangle = \|(-\mathcal{R}^+)^{1/2} v\|^2.$$

Substituting this into the energy identity and applying Cauchy–Schwarz yields:

$$(128) \quad \varpi \|u\|^2 + \|(-\mathcal{R}^+)^{1/2} v\|^2 \leq \|(-\mathcal{R}^+)^{1/2} v\| \|(-\mathcal{R}^+)^{1/2} P_F f\|.$$

Writing $a := \|(-\mathcal{R}^+)^{1/2} v\|$ and $b := \|(-\mathcal{R}^+)^{1/2} P_F f\|$, we have $\varpi \|u\|^2 + a^2 \leq ab$. Completing the square,

$$(129) \quad a^2 - ab = (a - \frac{1}{2}b)^2 - \frac{1}{4}b^2 \geq -\frac{1}{4}b^2,$$

so $\varpi \|u\|^2 \leq \frac{1}{4}b^2$, hence

$$(130) \quad \|u\| \leq \frac{1}{2} \varpi^{-1/2} \|(-\mathcal{R}^+)^{1/2} P_F f\| \leq \frac{1}{2\sqrt{\lambda_R}} \varpi^{-1/2} \|f\|.$$

The first inequality is the spectral bound $\varpi^{-1/2} \cdot \frac{1}{2}$, which can equivalently be derived from the spectral calculus: writing $Q_O = -\int_0^\infty \lambda dE_\lambda$ on $\mathcal{H}_{S,0}$, one has $\|(\varpi - Q_O)^{-1} (-Q_O)^{1/2}\| = \sup_{\lambda \geq 0} \frac{\sqrt{\lambda}}{\varpi + \lambda}$, and the calculus identity $\sup_{\lambda \geq 0} \sqrt{\lambda}/(\varpi + \lambda) = 1/(2\sqrt{\varpi})$ (attained at $\lambda = \varpi$) gives the stated bound. The second inequality uses $\|(-\mathcal{R}^+)^{1/2} P_F f\| \leq \lambda_R^{-1/2} \|f\|$. Moreover, $a \leq b$. By construction of the resolvent, $u \in \text{dom}(Q_O)$. Then the vector $v = \mathcal{V}_{\text{skew}} u$ is a slow-induced flux of the form w_ψ with $\psi \in \text{dom}(Q_O)$; we may therefore apply the flux conditioning bound (29) to v to obtain

$$(131) \quad \|v\|^2 \leq M_R \langle v, -\mathcal{R}^+ v \rangle = M_R a^2 \leq M_R b^2 \leq \frac{M_R}{\lambda_R} \|f\|^2,$$

so $\|v\| \leq \sqrt{M_R/\lambda_R} \|f\|$. These bounds depend only on $\|f\|$, so the core operator extends uniquely by continuity to a bounded operator $\mathcal{H} \rightarrow \mathcal{H}_S$, still denoted $\mathcal{A}_\varpi$. This proves $\|\mathcal{A}_\varpi\| \leq k_{\mathcal{A}} \varpi^{-1/2}$ and $\|\mathcal{V}_{\text{skew}} \mathcal{A}_\varpi\| \leq k_{\text{reg}}$, with

$$(132) \quad k_{\mathcal{A}} := \frac{1}{2\sqrt{\lambda_R}}, \quad k_{\text{reg}} := \sqrt{M_R/\lambda_R}.$$

Dual regularity. On the invariant core, define $\mathcal{T} := \mathcal{A}_\varpi \mathcal{V}_{\text{skew}} P_F$. To prove boundedness, it suffices to identify and estimate the operator appearing in the dual pairing against slow test functions. We first compute the adjoint of the bounded solver:

$$(133) \quad \begin{aligned} \mathcal{A}_\varpi^\dagger &= ((\varpi - Q_O)^{-1} P_S \mathcal{V}_{\text{skew}} (-\mathcal{R}^+) P_F)^\dagger \\ &= P_F (-\mathcal{R}^+) \mathcal{V}_{\text{skew}}^\dagger P_S (\varpi - Q_O)^{-1} \\ &= -P_F (-\mathcal{R}^+) \mathcal{V}_{\text{skew}} P_S (\varpi - Q_O)^{-1}. \end{aligned}$$

Here we used that $(\varpi - Q_O)^{-1}$, P_S , P_F , and $-\mathcal{R}^+$ are self-adjoint, and the only minus sign comes from $\mathcal{V}_{\text{skew}}^\dagger = -\mathcal{V}_{\text{skew}}$. Consequently, for every core vector f and every $g \in \mathcal{H}_S$,

$$(134) \quad \langle \mathcal{T} f, g \rangle = \langle f, P_F \mathcal{V}_{\text{skew}} P_F (-\mathcal{R}^+) \mathcal{V}_{\text{skew}} P_S (\varpi - Q_O)^{-1} g \rangle.$$

We therefore estimate the operator

$$\mathcal{B} := P_F \mathcal{V}_{\text{skew}} P_F (-\mathcal{R}^+) \mathcal{V}_{\text{skew}} P_S (\varpi - Q_O)^{-1} : \mathcal{H}_S \rightarrow \mathcal{H}_F.$$

Let $g \in \mathcal{H}_S$ and $\psi := (\varpi - Q_O)^{-1} g$. Define the macroscopic flux

$$(135) \quad w_\psi := P_F \mathcal{V}_{\text{skew}} P_S \psi \in \mathcal{H}_F.$$

Then $\mathcal{B} g = P_F \mathcal{V}_{\text{skew}} P_F (-\mathcal{R}^+) w_\psi$. Invoking the **Flux Regularity** condition from Assumption 4.5.3:

$$(136) \quad \|P_F \mathcal{V}_{\text{skew}} P_F (-\mathcal{R}^+) w_\psi\| \leq \kappa_{\text{ell}} \|(-Q_O) \psi\| + \kappa_{\text{low}} \|(-Q_O)^{1/2} \psi\|.$$

Substituting ψ back:

$$(137) \quad \begin{aligned} \|\mathcal{B} g\| &\leq \kappa_{\text{ell}} \|(-Q_O)(\varpi - Q_O)^{-1} g\| \\ &\quad + \kappa_{\text{low}} \|(-Q_O)^{1/2} (\varpi - Q_O)^{-1} g\|. \end{aligned}$$

Since Q_O is self-adjoint and non-positive, functional calculus gives

$$(138) \quad \|(-Q_O)(\varpi - Q_O)^{-1}\| = \sup_{\lambda \geq 0} \frac{\lambda}{\varpi + \lambda} \leq 1,$$

$$(139) \quad \|(-Q_O)^{1/2} (\varpi - Q_O)^{-1}\| = \sup_{\lambda \geq 0} \frac{\sqrt{\lambda}}{\varpi + \lambda} = \frac{1}{2\sqrt{\varpi}}.$$

Thus:

$$(140) \quad \|\mathcal{B} g\| \leq \left(\kappa_{\text{ell}} + \frac{\kappa_{\text{low}}}{2\sqrt{\varpi}} \right) \|g\|.$$

Hence $\mathcal{B}$ extends boundedly from $\mathcal{H}_S$ to $\mathcal{H}_F$ with norm at most $k_{\text{dual}}(\varpi)$. Returning to the dual pairing, we obtain

$$|\langle \mathcal{T} f, g \rangle| \leq \|f\| \|\mathcal{B} g\| \leq k_{\text{dual}}(\varpi) \|f\| \|g\|, \quad g \in \mathcal{H}_S,$$

so $\mathcal{T}$ extends uniquely to a bounded operator $\mathcal{H} \rightarrow \mathcal{H}_S$ and $\|\mathcal{A}_\varpi \mathcal{V}_{\text{skew}} P_F\| \leq k_{\text{dual}}(\varpi)$.

A.2. Proof of Property 3 (Perturbation Control). We use the structural decomposition

$$\mathcal{V}_{\text{pert}} = \mathcal{K}_{\text{pert}} + \mathcal{S}_{\text{pert}}$$

from Assumption 4.5, item 4. Recall the bound for the skew part (Assumption 4.5, item 4(a)):

$$(141) \quad \|\mathcal{K}_{\text{pert}} f\| \leq a_1 \|(-\mathcal{R})^{1/2} f\| + b_1 \|f\|, \quad f \in \text{dom}((-\mathcal{R})^{1/2}).$$

For the symmetric part, we use the energy bound (Assumption 4.5.4(b)):

$$(142) \quad \|\mathcal{S}_{\text{pert}} f\| \leq a_2 \|(-\mathcal{R})^{1/2} f\| + b_2 \|f\|, \quad f \in \text{dom}((-\mathcal{R})^{1/2}).$$

Reverse estimates ($\mathcal{V}_{\text{pert}} \mathcal{A}_{\varpi}$ and $\mathcal{V}_{\text{pert}}^{\dagger} \mathcal{A}_{\varpi}$). Let f first belong to the invariant core and set $u := \mathcal{A}_{\varpi} f$. The adjoint is $\mathcal{V}_{\text{pert}}^{\dagger} = -\mathcal{K}_{\text{pert}} + \mathcal{S}_{\text{pert}}$. Applying the triangle inequality and the bounds above:

$$(143) \quad \|\mathcal{V}_{\text{pert}}^{\dagger} u\| \leq \|\mathcal{K}_{\text{pert}} u\| + \|\mathcal{S}_{\text{pert}} u\|$$

$$(144) \quad \leq b_1 \|u\| + b_2 \|u\| + (a_1 + a_2) \|(-\mathcal{R})^{1/2} u\|.$$

Since $u \in \mathcal{H}_S = \ker(\mathcal{R})$ (by definition of $\mathcal{A}_{\varpi}$), one has $(-\mathcal{R})^{1/2} u = 0$, hence

$$(145) \quad \|\mathcal{V}_{\text{pert}}^{\dagger} u\| \leq (b_1 + b_2) \|u\|.$$

On this core this also gives the same estimate for $\mathcal{V}_{\text{pert}} u = (\mathcal{K}_{\text{pert}} + \mathcal{S}_{\text{pert}})u$; the standing closure convention extends the resulting bound to the bounded operator $\mathcal{V}_{\text{pert}} \mathcal{A}_{\varpi}$ on $\mathcal{H}$. Substituting the Solver Bound ($\|u\| \leq k_{\mathcal{A}} \varpi^{-1/2} \|f\|$):

$$(146) \quad \|\mathcal{V}_{\text{pert}} \mathcal{A}_{\varpi} f\|, \|\mathcal{V}_{\text{pert}}^{\dagger} \mathcal{A}_{\varpi} f\| \leq (b_1 + b_2) k_{\mathcal{A}} \varpi^{-1/2} \|f\|.$$

This establishes (39a) with $k_P(\varpi) := (b_1 + b_2) k_{\mathcal{A}} \varpi^{-1/2}$.

Forward estimates ($\mathcal{A}_{\varpi} \mathcal{V}_{\text{pert}}$ and $\mathcal{A}_{\varpi} \mathcal{V}_{\text{pert}}^{\dagger}$). Let $g \in \mathcal{H}_S$ first belong to the slow core, with $\|g\| = 1$, and set $w := \mathcal{A}_{\varpi}^{\dagger} g$. Using the bounds on $\mathcal{K}_{\text{pert}}$ and the energy bound on $\mathcal{S}_{\text{pert}}$, the same argument applies to $\mathcal{V}_{\text{pert}}^{\dagger} = -\mathcal{K}_{\text{pert}} + \mathcal{S}_{\text{pert}}$ and to $\mathcal{V}_{\text{pert}} = \mathcal{K}_{\text{pert}} + \mathcal{S}_{\text{pert}}$:

$$(147) \quad \|\mathcal{V}_{\text{pert}} w\|, \|\mathcal{V}_{\text{pert}}^{\dagger} w\| \leq (b_1 + b_2) \|w\| + (a_1 + a_2) \|(-\mathcal{R})^{1/2} w\|.$$

We compute the norms on the right-hand side.

(1) **Solver term ($\|w\|$):**

$$(148) \quad \|w\| = \|\mathcal{A}_{\varpi}^{\dagger} g\| \leq \|\mathcal{A}_{\varpi}\| \|g\| \leq k_{\mathcal{A}} \varpi^{-1/2}.$$

(2) **Energy term ($\|(-\mathcal{R})^{1/2} w\|$):** Using $w = \mathcal{A}_{\varpi}^{\dagger} g$ and the functional calculus for the non-positive self-adjoint operator Q_O ,

$$(149) \quad \|(-\mathcal{R})^{1/2} w\| = \|(-\mathcal{R})^{1/2} \mathcal{A}_{\varpi}^{\dagger} g\| = \|(-\mathcal{R}^+)^{1/2} P_F \mathcal{V}_{\text{skew}} P_S (\varpi - Q_O)^{-1} g\|.$$

Setting $\psi := (\varpi - Q_O)^{-1} g \in \mathcal{H}_S$, we note the identity (for such slow-induced fluxes)

$$(150) \quad w_{\psi} := P_F \mathcal{V}_{\text{skew}} P_S \psi,$$

$$(151) \quad \|(-\mathcal{R}^+)^{1/2} w_{\psi}\|^2 = \langle w_{\psi}, -\mathcal{R}^+ w_{\psi} \rangle = -\langle \psi, Q_O \psi \rangle = \|(-Q_O)^{1/2} \psi\|^2,$$

hence

$$(152) \quad \|(-\mathcal{R})^{1/2} w\| = \|(-Q_O)^{1/2} (\varpi - Q_O)^{-1} g\| \leq \frac{1}{2} \varpi^{-1/2} \|g\|.$$

(The last step follows from the scalar bound $\sup_{x \geq 0} \frac{\sqrt{x}}{\varpi + x} = \frac{1}{2\sqrt{\varpi}}$.)

Substituting these estimates into (147) yields

$$(153) \quad \|\mathcal{V}_{\text{pert}} \mathcal{A}_{\varpi}^{\dagger} g\|, \|\mathcal{V}_{\text{pert}}^{\dagger} \mathcal{A}_{\varpi}^{\dagger} g\| \leq \left((b_1 + b_2) k_{\mathcal{A}} + \frac{a_1 + a_2}{2} \right) \varpi^{-1/2} \|g\|.$$

By density of the slow core and boundedness of the right-hand side, the estimate extends to all $g \in \mathcal{H}_S$. This establishes (39b) with $\tilde{k}_P(\varpi) := ((b_1 + b_2) k_{\mathcal{A}} + \frac{a_1 + a_2}{2}) \varpi^{-1/2}$. $\square$

APPENDIX B. PROOF OF THEOREM 5.2

This appendix derives the dissipation matrix governing the hypocoercive convergence. The computation rests on the modified Lyapunov functional with resolvent shift $\varpi > 0$ and mixing weight $\alpha > 0$:

$$(154) \quad \mathcal{E}_{\alpha, \varpi}(f) := \frac{1}{2} \|f\|^2 - \alpha \Re \langle \mathcal{A}_{\varpi} f, f \rangle,$$

where $\mathcal{A}_{\varpi}$ denotes the bounded extension of the core DMS corrector $(\varpi - Q_O)^{-1} P_S \mathcal{V}_{\text{skew}} (-\mathcal{R}^+) P_F$ associated with the reference effective dynamics Q_O . The proof proceeds in three stages: we differentiate this functional on the invariant core, condense the resulting estimates into a 2×2 dissipation matrix, and optimize the mixing weight α at the choice $\varpi_* = \lambda(Q_O)$.

B.1. Preliminaries. This appendix uses the structural estimates of Lemma 4.9 (solver bound, regularity, and perturbation control) and the notation of equations (44)–(48). We set $\varpi_* = \lambda(Q_O)$ throughout and use the effective perturbation coefficient

$$(155) \quad b_{2, \text{eff}} := b_2 + \frac{a_2}{2} (1 + \lambda_R),$$

as defined in (44). We also record the energy-admissibility threshold

$$(156) \quad \eta_{\text{sym}}(\gamma) := \begin{cases} +\infty, & a_2 = 0, \\ 2\gamma/a_2, & a_2 > 0, \end{cases}$$

which is exactly the range for η for which the coefficient of $\|(-\mathcal{R})^{1/2} f\|^2$ in the basic energy estimate below remains nonnegative.

B.2. Derivation of Differential Estimates. Standing regularity for the Lyapunov computation. In the bounded-operator setting, for any $f_0 \in \mathcal{H}$ the solution $f_t = e^{t\mathcal{L}^{\dagger}} f_0$ is continuously differentiable and satisfies $\partial_t f_t = \mathcal{L}^{\dagger} f_t$. In unbounded settings, by Assumption 3.1, for every $f_0 \in \text{dom}(\mathcal{L}^{\dagger}) \cap \mathcal{H}_0$ the orbit $f_t = e^{t\mathcal{L}^{\dagger}} f_0$ is classical, and the algebraic identities below are first derived on the invariant core $\mathcal{C}$. The only model-dependent input is the existence of that core. For fixed (α, ϖ) , define on $\text{dom}(\mathcal{L}^{\dagger}) \cap \mathcal{H}_0$

$$(157) \quad \mathfrak{d}_{\alpha, \varpi}(f) := \Re \langle \mathcal{L}^{\dagger} f, f \rangle - \alpha \Re \langle \mathcal{A}_{\varpi} \mathcal{L}^{\dagger} f, f \rangle - \alpha \Re \langle \mathcal{A}_{\varpi} f, \mathcal{L}^{\dagger} f \rangle.$$

Because $\mathcal{A}_{\varpi}$ is bounded, $\mathfrak{d}_{\alpha, \varpi}$ is continuous for the graph norm $\|f\| + \|\mathcal{L}^{\dagger} f\|$. Hence any quadratic-form inequality proved for $\mathfrak{d}_{\alpha, \varpi}$ on $\mathcal{C}$ extends automatically to $\text{dom}(\mathcal{L}^{\dagger}) \cap \mathcal{H}_0$.

Lemma B.1 (Energy identity for the fast dissipator). *Assume $\mathcal{R}$ is self-adjoint and non-positive on $\mathcal{H}$. Then for all $f \in \text{dom}(\mathcal{R})$,*

$$(158) \quad \Re \langle \mathcal{R} f, f \rangle = \langle \mathcal{R} f, f \rangle = -\|(-\mathcal{R})^{1/2} f\|^2.$$

In particular, if $f = f_S + f_F$ with $f_S \in \ker(\mathcal{R})$ and $f_F \in \mathcal{H}_F$, then

$$(159) \quad \Re \langle \mathcal{R} f, f \rangle = \Re \langle \mathcal{R} f_F, f_F \rangle.$$

Proof. Since $\mathcal{R}$ is self-adjoint and non-positive, $-\mathcal{R} \geq 0$ admits a functional calculus square root $(-\mathcal{R})^{1/2}$ and $\langle \mathcal{R} f, f \rangle = -\langle (-\mathcal{R}) f, f \rangle = -\|(-\mathcal{R})^{1/2} f\|^2$. Moreover $\mathcal{R} f_S = 0$ and $\mathcal{R} f = \mathcal{R} f_F$. $\square$

Let f_t denote such a classical solution. By differentiating (42) along the flow and using that $\mathcal{A}_\varpi$ is time-independent, we obtain

$$(160) \quad \frac{d}{dt} \mathcal{E}_{\alpha, \varpi}(f_t) = \Re \langle \partial_t f_t, f_t \rangle - \alpha \Re \langle \mathcal{A}_\varpi \partial_t f_t, f_t \rangle - \alpha \Re \langle \mathcal{A}_\varpi f_t, \partial_t f_t \rangle$$

$$(161) \quad = \Re \langle \mathcal{L}^\dagger f_t, f_t \rangle - \alpha \Re \langle \mathcal{A}_\varpi \mathcal{L}^\dagger f_t, f_t \rangle - \alpha \Re \langle \mathcal{A}_\varpi f_t, \mathcal{L}^\dagger f_t \rangle.$$

Split the right-hand side into the “plain” dissipation coming from the generator itself and the hypo-coercive correction generated by the cross term:

$$(162) \quad \frac{d}{dt} \mathcal{E}_{\alpha, \varpi}(f_t) = \mathcal{D}_I(f_t) + \mathcal{D}_{\text{mod}}(f_t),$$

where we define

$$(163) \quad \mathcal{D}_I(f) := \Re \langle \mathcal{L}^\dagger f, f \rangle,$$

$$(164) \quad \mathcal{D}_{\text{mod}}(f) := -\alpha \Re \langle \mathcal{A}_\varpi \mathcal{L}^\dagger f, f \rangle - \alpha \Re \langle \mathcal{A}_\varpi f, \mathcal{L}^\dagger f \rangle.$$

Expanding $\mathcal{L}^\dagger = \gamma \mathcal{R} - \mathcal{V}_{\text{skew}} + \eta \mathcal{V}_{\text{pert}}^\dagger$ in $\mathcal{D}_I$ then yields the decomposition displayed below.

Natural dissipation ($\mathcal{D}_I$).

$$(165) \quad \mathcal{D}_I = \gamma \Re \langle \mathcal{R} f, f \rangle - \Re \langle \mathcal{V}_{\text{skew}} f, f \rangle + \eta \Re \langle \mathcal{V}_{\text{pert}}^\dagger f, f \rangle.$$

Using Assumption 4.5.1 ($\mathcal{R}$ coercivity) and Assumption 4.5.4 (Perturbation decomposition $\mathcal{V}_{\text{pert}} = \mathcal{K}_{\text{pert}} + \mathcal{S}_{\text{pert}}$): the fast dissipator contributes $\gamma \Re \langle \mathcal{R} f, f \rangle \leq -\gamma \lambda_R \|f_F\|^2$, while $\Re \langle \mathcal{V}_{\text{skew}} f, f \rangle = 0$ and $\Re \langle \mathcal{K}_{\text{pert}} f, f \rangle = 0$ by skew-adjointness. Since $\mathcal{V}_{\text{pert}}^\dagger = -\mathcal{K}_{\text{pert}} + \mathcal{S}_{\text{pert}}$, the forward perturbation contributes the same symmetric part as in the backward decomposition. For the symmetric part, using (33) and Cauchy–Schwarz,

$$(166) \quad \eta \Re \langle \mathcal{S}_{\text{pert}} f, f \rangle \leq \eta \|\mathcal{S}_{\text{pert}} f\| \|f\| \leq \eta \left(b_2 \|f\|^2 + a_2 \|(-\mathcal{R})^{1/2} f\| \|f\| \right)$$

$$(167) \quad \leq \eta \left(\left(b_2 + \frac{a_2^2}{2} \right) \|f\|^2 + \frac{a_2}{2} \|(-\mathcal{R})^{1/2} f\|^2 \right).$$

Combining with the dissipator contribution, note that by Lemma B.1, $\Re \langle \mathcal{R} f, f \rangle = \Re \langle \mathcal{R} f_F, f_F \rangle = -\|(-\mathcal{R})^{1/2} f\|^2$. If $\eta \leq \eta_{\text{sym}}(\gamma)$, i.e. if $\gamma - \eta a_2/2 \geq 0$, then we may use the lower bound $\|(-\mathcal{R})^{1/2} f\|^2 \geq \lambda_R \|f_F\|^2$ without changing the inequality direction. Hence

$$(168) \quad \gamma \Re \langle \mathcal{R} f, f \rangle + \eta \Re \langle \mathcal{S}_{\text{pert}} f, f \rangle \leq -\left(\gamma - \frac{\eta a_2}{2} \right) \|(-\mathcal{R})^{1/2} f\|^2 + \eta \left(b_2 + \frac{a_2^2}{2} \right) \|f\|^2$$

$$(169) \quad \leq -(\gamma \lambda_R - \eta b_{2, \text{eff}}) \|f_F\|^2 + \eta \left(b_2 + \frac{a_2^2}{2} \right) \|f_S\|^2,$$

where we used $\|(-\mathcal{R})^{1/2} f\|^2 \geq \lambda_R \|f_F\|^2$, the decomposition $\|f\|^2 = \|f_S\|^2 + \|f_F\|^2$, and the fact that $\frac{a_2 \lambda_R}{2} + \left(b_2 + \frac{a_2^2}{2} \right) = b_{2, \text{eff}}$ by definition. The fast mode carries the full effective coefficient $b_{2, \text{eff}}$, while the slow mode retains only the smaller $b_2 + \frac{a_2^2}{2}$ (since $f_S \in \ker(\mathcal{R})$ does not benefit from the λ_R -amplification).

Therefore

$$(170) \quad \mathcal{D}_I \leq -(\gamma \lambda_R - \eta b_{2, \text{eff}}) \|f_F\|^2 + \eta \left(b_2 + \frac{a_2^2}{2} \right) \|f_S\|^2.$$

Hypo-coercive correction ($\mathcal{D}_{\text{mod}}$). Recall

$$(171) \quad \mathcal{D}_{\text{mod}}(f) = -\alpha \Re \langle \mathcal{A}_\varpi \mathcal{L}^\dagger f, f \rangle - \alpha \Re \langle \mathcal{A}_\varpi f, \mathcal{L}^\dagger f \rangle.$$

Using $\mathcal{L}^\dagger = \gamma \mathcal{R} - \mathcal{V}_{\text{skew}} + \eta \mathcal{V}_{\text{pert}}^\dagger$ and linearity, we regroup $\mathcal{D}_{\text{mod}}$ according to which part of the generator appears:

$$(172) \quad \mathcal{D}_{\text{mod}}(f) = T_{\mathcal{R}}(f) + T_{\mathcal{V}}(f) + T_{\mathcal{P}}(f),$$

where

$$(173) \quad T_{\mathcal{R}}(f) := -\alpha\gamma \Re\langle \mathcal{A}_{\varpi} \mathcal{R}f, f \rangle - \alpha\gamma \Re\langle \mathcal{A}_{\varpi} f, \mathcal{R}f \rangle,$$

$$(174) \quad T_{\mathcal{V}}(f) := \alpha \Re\langle \mathcal{A}_{\varpi} \mathcal{V}_{\text{skew}} f, f \rangle + \alpha \Re\langle \mathcal{A}_{\varpi} f, \mathcal{V}_{\text{skew}} f \rangle,$$

$$(175) \quad T_{\mathcal{P}}(f) := -\alpha\eta \Re\langle \mathcal{A}_{\varpi} \mathcal{V}_{\text{pert}}^{\dagger} f, f \rangle - \alpha\eta \Re\langle \mathcal{A}_{\varpi} f, \mathcal{V}_{\text{pert}}^{\dagger} f \rangle.$$

Interaction with the fast dissipator. We start from the regrouped definition

$$(176) \quad T_{\mathcal{R}}(f) = -\alpha\gamma \Re\langle \mathcal{A}_{\varpi} \mathcal{R}f, f \rangle - \alpha\gamma \Re\langle \mathcal{A}_{\varpi} f, \mathcal{R}f \rangle.$$

Since $\mathcal{R}f = \mathcal{R}f_F$ and $\mathcal{A}_{\varpi} f \in \mathcal{H}_S = \ker(\mathcal{R})$, the second term vanishes by self-adjointness of $\mathcal{R}$:

$$(177) \quad \Re\langle \mathcal{A}_{\varpi} f, \mathcal{R}f \rangle = \Re\langle \mathcal{R} \mathcal{A}_{\varpi} f, f \rangle = 0.$$

Moreover $\mathcal{A}_{\varpi} \mathcal{R}f_F \in \mathcal{H}_S$ is orthogonal to $f_F \in \mathcal{H}_F$, hence

$$(178) \quad T_{\mathcal{R}}(f) = -\alpha\gamma \Re\langle \mathcal{A}_{\varpi} \mathcal{R}f_F, f_S \rangle.$$

Using $\mathcal{A}_{\varpi} = (\varpi - Q_O)^{-1} P_S \mathcal{V}_{\text{skew}} (-\mathcal{R}^+) P_F$ and the pseudoinverse identity $\mathcal{R}^+ \mathcal{R} = P_F$ on $\text{dom}(\mathcal{R})$, we obtain

$$(179) \quad \mathcal{A}_{\varpi} \mathcal{R}f_F = (\varpi - Q_O)^{-1} P_S \mathcal{V}_{\text{skew}} (-\mathcal{R}^+) \mathcal{R}f_F = -(\varpi - Q_O)^{-1} P_S \mathcal{V}_{\text{skew}} f_F.$$

Therefore, by skew-adjointness of $\mathcal{V}_{\text{skew}}$,

$$(180) \quad |T_{\mathcal{R}}| = \alpha\gamma |\Re\langle (\varpi - Q_O)^{-1} P_S \mathcal{V}_{\text{skew}} f_F, f_S \rangle| = \alpha\gamma |\Re\langle f_F, \mathcal{V}_{\text{skew}} (\varpi - Q_O)^{-1} f_S \rangle|$$

$$(181) \quad \leq \alpha\gamma \|f_F\| \|\mathcal{V}_{\text{skew}} (\varpi - Q_O)^{-1} f_S\|.$$

To bound $\|\mathcal{V}_{\text{skew}} (\varpi - Q_O)^{-1} f_S\|$, set $\psi := (\varpi - Q_O)^{-1} f_S \in \text{dom}(Q_O) \subset \mathcal{H}_S$ and denote its induced microscopic flux

$$(182) \quad w_{\psi} := P_F \mathcal{V}_{\text{skew}} P_S \psi \in \mathcal{H}_F.$$

Since $\psi \in \mathcal{H}_S$, the centered-coupling assumption gives $P_S \mathcal{V}_{\text{skew}} \psi = 0$, hence $w_{\psi} = \mathcal{V}_{\text{skew}} \psi = \mathcal{V}_{\text{skew}} (\varpi - Q_O)^{-1} f_S$. Applying the flux conditioning bound (29) yields

$$(183) \quad \|\mathcal{V}_{\text{skew}} (\varpi - Q_O)^{-1} f_S\|^2 = \|w_{\psi}\|^2 \leq M_R \langle w_{\psi}, -\mathcal{R}^+ w_{\psi} \rangle.$$

The “extra Q_O ” appears when the right-hand side is rewritten using Assumption 4.4(ii). Indeed, for $\psi \in \text{dom}(Q_O)$,

$$(184) \quad \langle w_{\psi}, -\mathcal{R}^+ w_{\psi} \rangle = \langle w_{\psi}, (-\mathcal{R}^+) w_{\psi} \rangle = \langle \psi, (-Q_O) \psi \rangle = \|(-Q_O)^{1/2} \psi\|^2.$$

Therefore

$$(185) \quad \|\mathcal{V}_{\text{skew}} (\varpi - Q_O)^{-1} f_S\| \leq \sqrt{M_R} \|(-Q_O)^{1/2} (\varpi - Q_O)^{-1} f_S\|.$$

Finally, by functional calculus for the self-adjoint operator $-Q_O \geq 0$,

$$(186) \quad \|(-Q_O)^{1/2} (\varpi - Q_O)^{-1}\| = \sup_{x \geq 0} \frac{\sqrt{x}}{\varpi + x} = \frac{1}{2\sqrt{\varpi}},$$

which implies

$$(187) \quad \|\mathcal{V}_{\text{skew}} (\varpi - Q_O)^{-1} f_S\| \leq \frac{\sqrt{M_R}}{2} \varpi^{-1/2} \|f_S\|.$$

Hence

$$(188) \quad |T_{\mathcal{R}}| \leq \alpha\gamma \frac{\sqrt{M_R}}{2} \varpi^{-1/2} \|f_F\| \|f_S\|.$$

Interaction with transport. Starting from the regrouped definition

$$(189) \quad T_{\mathcal{V}}(f) = \alpha \Re\langle \mathcal{A}_{\varpi} \mathcal{V}_{\text{skew}} f, f \rangle + \alpha \Re\langle \mathcal{A}_{\varpi} f, \mathcal{V}_{\text{skew}} f \rangle,$$

we decompose $f = f_S + f_F$ and use that $\mathcal{A}_\varpi$ maps into the slow space and annihilates the slow component: $\mathcal{A}_\varpi = \mathcal{A}_\varpi P_F$ (since P_F is on the right in (36)). Hence $\mathcal{A}_\varpi f = \mathcal{A}_\varpi f_F$. Expanding the *first* bracket gives

$$(190) \quad \Re \langle \mathcal{A}_\varpi \mathcal{V}_{\text{skew}} f, f \rangle = \Re \langle \mathcal{A}_\varpi \mathcal{V}_{\text{skew}} f_S, f_S \rangle + \Re \langle \mathcal{A}_\varpi \mathcal{V}_{\text{skew}} f_F, f_S \rangle$$

$$(191) \quad + \Re \langle \mathcal{A}_\varpi \mathcal{V}_{\text{skew}} f_S, f_F \rangle + \Re \langle \mathcal{A}_\varpi \mathcal{V}_{\text{skew}} f_F, f_F \rangle.$$

The last term vanishes because $\mathcal{A}_\varpi \mathcal{V}_{\text{skew}} f_F \in \mathcal{H}_S$ is orthogonal to $f_F \in \mathcal{H}_F$. Moreover, since $\mathcal{A}_\varpi \mathcal{V}_{\text{skew}} f_S \in \mathcal{H}_S$, we have $\Re \langle \mathcal{A}_\varpi \mathcal{V}_{\text{skew}} f_S, f_F \rangle = 0$. Therefore

$$(192) \quad \Re \langle \mathcal{A}_\varpi \mathcal{V}_{\text{skew}} f, f \rangle = \Re \langle \mathcal{A}_\varpi \mathcal{V}_{\text{skew}} f_S, f_S \rangle + \Re \langle \mathcal{A}_\varpi \mathcal{V}_{\text{skew}} f_F, f_S \rangle.$$

In the *second* bracket we use $\mathcal{A}_\varpi f = \mathcal{A}_\varpi f_F$ and expand $\mathcal{V}_{\text{skew}} f$:

$$(193) \quad \Re \langle \mathcal{A}_\varpi f, \mathcal{V}_{\text{skew}} f \rangle = \Re \langle \mathcal{A}_\varpi f_F, \mathcal{V}_{\text{skew}} f_S \rangle + \Re \langle \mathcal{A}_\varpi f_F, \mathcal{V}_{\text{skew}} f_F \rangle.$$

Putting the pieces together, we obtain the explicit decomposition

$$(194) \quad T_V = \alpha \Re \langle \mathcal{A}_\varpi \mathcal{V}_{\text{skew}} f_S, f_S \rangle + \alpha \Re \langle \mathcal{A}_\varpi \mathcal{V}_{\text{skew}} f_F, f_S \rangle$$

$$(195) \quad + \alpha \Re \langle \mathcal{A}_\varpi f_F, \mathcal{V}_{\text{skew}} f_S \rangle + \alpha \Re \langle \mathcal{A}_\varpi f_F, \mathcal{V}_{\text{skew}} f_F \rangle.$$

Main Coercivity. Since $f_S \in \mathcal{H}_S$ and $P_S \mathcal{V}_{\text{skew}} P_S = 0$, we have $P_F \mathcal{V}_{\text{skew}} f_S = \mathcal{V}_{\text{skew}} f_S$. On the invariant core, the represented identity for Q_O gives

$$(196) \quad \mathcal{A}_\varpi \mathcal{V}_{\text{skew}} f_S = (\varpi - Q_O)^{-1} P_S \mathcal{V}_{\text{skew}} (-\mathcal{R}^+) \mathcal{V}_{\text{skew}} f_S = (\varpi - Q_O)^{-1} Q_O f_S.$$

Hence

$$(197) \quad \alpha \Re \langle \mathcal{A}_\varpi \mathcal{V}_{\text{skew}} f_S, f_S \rangle = -\alpha \langle (\varpi - Q_O)^{-1} (-Q_O) f_S, f_S \rangle.$$

Since $f \in \mathcal{H}_0$ implies $f_S = P_S f \in \mathcal{H}_S \cap \mathcal{H}_0 = \mathcal{H}_{S,0}$, the spectral theorem for the non-negative self-adjoint operator $-Q_O|_{\mathcal{H}_{S,0}}$ yields

$$(198) \quad \alpha \Re \langle \mathcal{A}_\varpi \mathcal{V}_{\text{skew}} f_S, f_S \rangle \leq -\alpha \frac{\lambda(Q_O)}{\varpi + \lambda(Q_O)} \|f_S\|^2.$$

Here Assumption 4.6 identifies $\lambda(Q_O)$ as the bottom of the spectrum of $-Q_O$ on $\mathcal{H}_{S,0}$.

Cross Terms. Since $\mathcal{A}_\varpi f_F \in \mathcal{H}_S$ (by the range of the corrector) and $\mathcal{V}_{\text{skew}} f_S \in \mathcal{H}_F$ (by centering, $P_S \mathcal{V}_{\text{skew}} P_S = 0$), the third inner product vanishes: $\Re \langle \mathcal{A}_\varpi f_F, \mathcal{V}_{\text{skew}} f_S \rangle = 0$. The only surviving cross term is bounded by the dual regularity estimate:

$$(199) \quad \alpha \Re \langle \mathcal{A}_\varpi \mathcal{V}_{\text{skew}} f_F, f_S \rangle \leq \alpha \|\mathcal{A}_\varpi \mathcal{V}_{\text{skew}} P_F\| \|f_F\| \|f_S\| \leq \alpha \left(\kappa_{\text{ell}} + \frac{\kappa_{\text{low}}}{2\sqrt{\varpi}} \right) \|f_F\| \|f_S\|.$$

Fast-fast remainder. Using $\mathcal{V}_{\text{skew}}^\dagger = -\mathcal{V}_{\text{skew}}$ and Cauchy-Schwarz,

$$(200) \quad |\alpha \Re \langle \mathcal{A}_\varpi f_F, \mathcal{V}_{\text{skew}} f_F \rangle| = \alpha |\Re \langle -\mathcal{V}_{\text{skew}} \mathcal{A}_\varpi f_F, f_F \rangle| \leq \alpha \|\mathcal{V}_{\text{skew}} \mathcal{A}_\varpi\| \|f_F\|^2 \leq \alpha k_{\text{reg}} \|f_F\|^2.$$

Interaction with perturbation.

$$(201) \quad T_P = -\alpha \eta \Re \langle \mathcal{A}_\varpi \mathcal{V}_{\text{pert}}^\dagger f, f \rangle - \alpha \eta \Re \langle \mathcal{A}_\varpi f, \mathcal{V}_{\text{pert}}^\dagger f \rangle.$$

We estimate each term using Lemma 4.9 and the centered coupling in Assumption 4.5.2.

Forward term. Since $\mathcal{A}_\varpi \mathcal{V}_{\text{pert}}^\dagger f \in \mathcal{H}_S$, we have $\langle \mathcal{A}_\varpi \mathcal{V}_{\text{pert}}^\dagger f, f \rangle = \langle \mathcal{A}_\varpi \mathcal{V}_{\text{pert}}^\dagger f, f_S \rangle$. Therefore,

$$(202) \quad |\alpha \eta \Re \langle \mathcal{A}_\varpi \mathcal{V}_{\text{pert}}^\dagger f, f \rangle| \leq \alpha \eta \|\mathcal{A}_\varpi \mathcal{V}_{\text{pert}}^\dagger f\| \|f_S\| \leq \alpha \eta \tilde{k}_P(\varpi) \|f\| \|f_S\|$$

$$(203) \quad \leq \alpha \eta \tilde{k}_P(\varpi) (\|f_S\|^2 + \|f_F\| \|f_S\|).$$

Reverse term. Since $\mathcal{A}_\varpi f \in \mathcal{H}_S$ and $P_S \mathcal{V}_{\text{pert}} P_S = 0$, we have $\langle \mathcal{A}_\varpi f, \mathcal{V}_{\text{pert}}^\dagger f \rangle = \langle \mathcal{V}_{\text{pert}} \mathcal{A}_\varpi f, f \rangle = \langle \mathcal{V}_{\text{pert}} \mathcal{A}_\varpi f, f_F \rangle$. Therefore,

$$(204) \quad |\alpha \eta \Re \langle \mathcal{A}_\varpi f, \mathcal{V}_{\text{pert}}^\dagger f \rangle| \leq \alpha \eta \|\mathcal{V}_{\text{pert}} \mathcal{A}_\varpi f\| \|f_F\| \leq \alpha \eta k_P(\varpi) \|f\| \|f_F\|$$

$$(205) \quad \leq \alpha \eta k_P(\varpi) (\|f_F\|^2 + \|f_F\| \|f_S\|).$$

B.3. Dissipation Matrix Construction. Summing all estimates and applying Cauchy–Schwarz to terms of the form

$$|\langle X f_F, f_S \rangle| \leq \|X\| \|f_F\| \|f_S\|$$

yields a closed differential inequality of the form

$$(206) \quad \frac{d}{dt} \mathcal{E}_{\alpha, \varpi}(f_t) \leq -\mathbf{v}(t)^T \mathbf{M} \mathbf{v}(t), \quad \mathbf{v}(t) := \begin{pmatrix} \|P_F f_t\| \\ \|P_S f_t\| \end{pmatrix}.$$

Here $\mathbf{M}$ is a real symmetric 2×2 matrix whose entries depend only on the structural constants and the chosen parameters (α, ϖ) . The entries of $\mathbf{M}$ are explicitly decomposed by powers of ϖ below. The fast block comes from the coercive part $\mathcal{R}$, the slow block from the DMS corrector, and the off-diagonal block from transport and perturbative couplings.

Fast-Fast Block (M_{11}): From Natural Dissipation, the fast–fast transport remainder, and Reverse Perturbation.

$$(207) \quad \begin{aligned} M_{11} &= (\gamma \lambda_R - \eta b_{2, \text{eff}}) - \alpha k_{\text{reg}} - \alpha \eta k_P(\varpi) \\ &= \gamma \lambda_R - \eta b_{2, \text{eff}} - \alpha k_{\text{reg}} - \alpha \eta (b_1 + b_2) k_{\mathcal{A}} \varpi^{-1/2}. \end{aligned}$$

Slow-Slow Block (M_{22}): From Main Coercivity, Natural Dissipation (gain), and both Perturbations.

$$(208) \quad \begin{aligned} M_{22} &= \alpha \frac{\lambda(Q_O)}{\varpi + \lambda(Q_O)} - \eta(b_2 + \frac{a_2}{2}) - \alpha \eta \tilde{k}_P(\varpi) \\ &= \alpha \frac{\lambda(Q_O)}{\varpi + \lambda(Q_O)} - \eta(b_2 + \frac{a_2}{2}) - \alpha \eta \left[\left((b_1 + b_2) k_{\mathcal{A}} + \frac{a_1 + a_2}{2} \right) \varpi^{-1/2} \right]. \end{aligned}$$

Cross Block (M_{12}): From Mixing, Transport Cross terms, and Forward Perturbation.

$$(209) \quad M_{12} = -\frac{1}{2} \alpha \left[\kappa_{\text{ell}} + \left(\frac{\kappa_{\text{low}}}{2} + \gamma \frac{\sqrt{M_R}}{2} + \eta \left(2(b_1 + b_2) k_{\mathcal{A}} + \frac{a_1 + a_2}{2} \right) \right) \varpi^{-1/2} \right].$$

This explicit form allows for direct asymptotic analysis by choosing $\varpi \sim \lambda(Q_O)$, confirming the stability for small η .

B.4. Optimization of the Decay Rate. We work in the unperturbed regime $\eta = 0$ to identify the dominant hypocoercive rate. Let $\mathbf{M}_0(\alpha)$ denote the dissipation matrix at $\eta = 0$. Since $\mathbf{M}_0$ is 2×2 symmetric, its smallest eigenvalue satisfies the determinant–trace bound

$$(210) \quad \lambda_{\min}(\mathbf{M}_0) \geq \frac{\det(\mathbf{M}_0)}{\text{Tr}(\mathbf{M}_0)}.$$

Choice of resolvent shift. The slow-coercivity contribution in M_{22}^0 is $\alpha \lambda(Q_O)/(\varpi + \lambda(Q_O))$, while the gap-dependent part of the cross term $|M_{12}^0|$ is

$$(211) \quad (\gamma \Theta + \frac{1}{2} \kappa_{\text{low}}) \varpi^{-1/2}.$$

A leading-order balance (maximising $\varpi(\varpi + \lambda(Q_O))^{-2}$ in the large- γ rate) motivates the choice

$$(212) \quad \varpi_* := \lambda(Q_O).$$

No separate global optimisation over all admissible shifts is claimed here; ϖ_* is the canonical gap-scale choice that balances the slow coercive block against the cross term inside the determinant-trace estimate used in this paper. At this value, $\lambda(Q_O)/(\varpi_* + \lambda(Q_O)) = \frac{1}{2}$. Using the constants $\Theta, K_{\text{geom}}, Z(\gamma)$ from (44)–(45), the unperturbed matrix entries at $\varpi = \varpi_*$ are:

$$(213) \quad M_{11}^0 = \gamma \lambda_R - \alpha k_{\text{reg}},$$

$$(214) \quad M_{22}^0 = \frac{1}{2} \alpha,$$

$$(215) \quad M_{12}^0 = -\frac{1}{2} \alpha Z(\gamma).$$

Optimal mixing weight. The determinant $D(\alpha) := M_{11}^0 M_{22}^0 - (M_{12}^0)^2$ is a concave quadratic in α :

$$(216) \quad D(\alpha) = \frac{1}{2} \gamma \lambda_R \alpha - \left(\frac{1}{2} k_{\text{reg}} + \frac{1}{4} Z(\gamma)^2 \right) \alpha^2.$$

It is maximised at

$$(217) \quad \alpha_*(\gamma) = \frac{\gamma \lambda_R}{Z(\gamma)^2 + 2k_{\text{reg}}},$$

in agreement with (46), giving

$$(218) \quad D(\alpha_*) = \frac{\frac{1}{4} (\gamma \lambda_R)^2}{Z(\gamma)^2 + 2k_{\text{reg}}}.$$

Trace upper bound and rate estimate. The exact trace is $\text{Tr}(\mathbf{M}_0)|_{\alpha_*} = (\gamma \lambda_R - \alpha_* k_{\text{reg}}) + \frac{1}{2} \alpha_* = \gamma \lambda_R - \alpha_* (k_{\text{reg}} - \frac{1}{2})$. Since $k_{\text{reg}} = \sqrt{M_R/\lambda_R} \geq 1$ (the flux conditioning bound gives $M_R \geq \lambda_R$), we have $k_{\text{reg}} - \frac{1}{2} \geq \frac{1}{2} > 0$, hence

$$(219) \quad \text{Tr}(\mathbf{M}_0)|_{\alpha_*} \leq \gamma \lambda_R.$$

Applying (210) yields

$$(220) \quad \lambda_{\min}(\mathbf{M}_0) \geq \frac{D(\alpha_*)}{\text{Tr}(\mathbf{M}_0)|_{\alpha_*}} \geq \frac{\frac{1}{4} \gamma \lambda_R}{Z(\gamma)^2 + 2k_{\text{reg}}} =: \Lambda_{\text{hypo}}(\gamma),$$

consistent with definition (47). This provides a rigorous lower bound on the unperturbed decay rate for all $\gamma > 0$.

Asymptotic scaling analysis. Equation (220) reveals the precise scaling behavior across regimes without relying on ad-hoc approximations.

Overdamped Limit ($\gamma \rightarrow \infty$). The denominator is dominated by the $\mathcal{O}(\gamma^2)$ term, giving

$$(221) \quad \lim_{\gamma \rightarrow \infty} \gamma \Lambda_{\text{hypo}}(\gamma) = \frac{\frac{1}{4} \lambda_R \lambda(Q_O)}{\Theta^2} = \lambda(Q_O) \frac{\lambda_R}{M_R}.$$

Thus, $\Lambda_{\text{hypo}}(\gamma) = \lambda(Q_O) \frac{\lambda_R}{M_R} \gamma^{-1} + \mathcal{O}(\gamma^{-2})$. In particular, $\Lambda_{\text{hypo}}(\gamma) \asymp \lambda(Q_O) \gamma^{-1}$.

Underdamped/Intermediate Regime. For fixed $\lambda(Q_O)$ and sufficiently small γ , equivalently $\gamma \ll \sqrt{\lambda(Q_O)}$ in small-gap families, the denominator is controlled by the γ -independent coefficients K_{geom} and $2k_{\text{reg}}$. These arise from the slow-fast coupling through $\mathcal{V}_{\text{skew}}$ and prevent the estimate from degenerating even though $\mathcal{R}$ does not dissipate on $\mathcal{H}_S$ directly. The transport mixes slow directions into fast dissipative directions, yielding $\Lambda_{\text{hypo}}(\gamma) \asymp \gamma$ for small γ .

The stability condition follows from an explicit bound on the perturbation matrix $\mathbf{E}(\gamma)$ as a function of γ , with the dependence on the macroscopic spectral gap $\varpi_* = \lambda(Q_O)$ kept visible.

Explicit norm of the perturbation matrix. Recalling that $\mathbf{M} \geq \mathbf{M}_0 - \eta \mathbf{E}$ in quadratic-form order, the perturbation matrix $\mathbf{E}$ has entries read off from the full $\mathbf{M}$ computation: The off-diagonal entry is one half of the coefficient multiplying $\|f_F\| \|f_S\|$ in the corresponding quadratic form.

$$(222) \quad E_{11} = b_{2,\text{eff}} + \alpha_* k_P(\varpi_*),$$

$$(223) \quad E_{22} = \left(b_2 + \frac{a_2}{2}\right) + \alpha_* \tilde{k}_P(\varpi_*),$$

$$(224) \quad E_{12} = \frac{1}{2} \alpha_* (k_P(\varpi_*) + \tilde{k}_P(\varpi_*)).$$

Here E_{11} contains the full effective coefficient $b_{2,\text{eff}} = b_2 + \frac{a_2}{2}(1 + \lambda_R)$, while E_{22} contains only the slow-mode coefficient $b_2 + \frac{a_2}{2}$; the extra $\frac{a_2 \lambda_R}{2}$ absorbed into $b_{2,\text{eff}}$ arises in the fast block from the bound $\|(-\mathcal{R})^{1/2} f\|^2 \geq \lambda_R \|f_F\|^2$ applied to the symmetric perturbation in $\mathcal{D}_I$ (see (170)). Since $\mathbf{E}$ is symmetric, we bound its spectral norm by the maximal absolute row sum (Gershgorin):

$$(225) \quad \|\mathbf{E}(\gamma)\| \leq \max\{E_{11} + |E_{12}|, E_{22} + |E_{12}|\}.$$

Using $b_{2,\text{eff}} \geq b_2 + \frac{a_2}{2}$ and $k_P(\varpi_*) \leq \tilde{k}_P(\varpi_*)$, both row sums admit the same uniform upper bound:

$$(226) \quad \begin{aligned} \|\mathbf{E}(\gamma)\| &\leq b_{2,\text{eff}} + \alpha_*(\gamma) \left[\tilde{k}_P(\varpi_*) + \frac{1}{2} (k_P(\varpi_*) + \tilde{k}_P(\varpi_*)) \right] \\ &\leq b_{2,\text{eff}} + 2\alpha_*(\gamma) \tilde{k}_P(\varpi_*). \end{aligned}$$

Recall from Lemma 4.9 that the bounds k_P and $\tilde{k}_P$ depend on $\varpi_*^{-1/2}$. Define the structural coefficient independent of ϖ and γ by

$$(227) \quad C_{\text{cross}} := 2(b_1 + b_2)k_A + (a_1 + a_2).$$

Indeed,

$$2\tilde{k}_P(\varpi_*) = 2\left((b_1 + b_2)k_A + \frac{a_1 + a_2}{2}\right)\varpi_*^{-1/2} = C_{\text{cross}} \varpi_*^{-1/2},$$

so the bound (226) becomes

$$(228) \quad \|\mathbf{E}(\gamma)\| \leq b_{2,\text{eff}} + \alpha_*(\gamma) C_{\text{cross}} \varpi_*^{-1/2}.$$

Substituting $\alpha_*(\gamma)$ from (46), the bound becomes

$$(229) \quad \|\mathbf{E}(\gamma)\| \leq b_{2,\text{eff}} + \alpha_*(\gamma) C_{\text{cross}} \lambda(Q_O)^{-1/2} = B_{\text{pert}}(\gamma),$$

consistent with definition (48).

Friction-dependent threshold. The stability condition $\lambda_{\min}(\mathbf{M}) > 0$ is ensured by $\eta \|\mathbf{E}(\gamma)\| \leq \frac{1}{2} \Lambda_{\text{hypo}}(\gamma)$, which yields the exact matrix threshold

$$(230) \quad \eta^\sharp(\gamma) := \frac{\Lambda_{\text{hypo}}(\gamma)}{2\|\mathbf{E}(\gamma)\|}.$$

Since $\|\mathbf{E}(\gamma)\| \leq B_{\text{pert}}(\gamma)$, the explicit threshold used in Theorem 5.2 is

$$(231) \quad \eta_{\text{mat}}(\gamma) := \frac{\Lambda_{\text{hypo}}(\gamma)}{2B_{\text{pert}}(\gamma)} \leq \eta^\sharp(\gamma).$$

Together with the energy-admissibility condition $\eta \leq \eta_{\text{sym}}(\gamma)$ coming from (170), this gives the theorem threshold

$$(232) \quad \eta^*(\gamma) := \min\{\eta_{\text{sym}}(\gamma), \eta_{\text{mat}}(\gamma)\}.$$

Substituting the explicit forms of Λ_{hypo} (47) and B_{pert} (48) yields

$$(233) \quad \eta_{\text{mat}}(\gamma) = \frac{\frac{1}{4} \gamma \lambda_R}{(Z(\gamma)^2 + 2k_{\text{reg}})(2b_{2,\text{eff}} + 2\alpha_*(\gamma) C_{\text{cross}} \lambda(Q_O)^{-1/2})}.$$

Asymptotic analysis of the threshold. We analyze first the scaling of $\eta_{\text{mat}}(\gamma)$ in the limiting regimes. Note that $Z(\gamma) \asymp \gamma$ as $\gamma \rightarrow \infty$.

Overdamped Limit ($\gamma \rightarrow \infty$). The numerator scales as $\mathcal{O}(\gamma)$. In the denominator, the term $2b_{2,\text{eff}}Z(\gamma)^2$ scales as $\mathcal{O}(\gamma^2)$, while the skew-term scales as $\mathcal{O}(\gamma)$. If $b_{2,\text{eff}} > 0$ (symmetric perturbation present), the $\mathcal{O}(\gamma^2)$ term dominates and $\eta_{\text{mat}}(\gamma) \asymp \gamma^{-1}$. If $b_{2,\text{eff}} = 0$ but $C_{\text{cross}} > 0$, the denominator is dominated by the skew-interaction term $\mathcal{O}(\gamma)$ and $\eta_{\text{mat}}(\gamma) \asymp 1$. If both $b_{2,\text{eff}}$ and C_{cross} vanish, this perturbation produces no loss in the present estimate and the matrix threshold is infinite.

Optimal friction scale ($\gamma \sim \sqrt{\lambda(Q_O)}$). At the optimal friction γ_{opt} , determined in the optimization step below, we have $Z(\gamma_{\text{opt}}) = \mathcal{O}(1)$. Consider the gap-dependent term in the denominator:

$$(234) \quad \gamma_{\text{opt}} \cdot C_{\text{cross}} \lambda(Q_O)^{-1/2} \asymp \sqrt{\lambda(Q_O)} \cdot \lambda(Q_O)^{-1/2} \asymp \mathcal{O}(1).$$

Thus the gap-dependent factor is offset at the optimal friction, and the matrix-controlled threshold has the expected scale:

$$(235) \quad \eta_{\text{mat}}(\gamma_{\text{opt}}) \asymp \sqrt{\lambda(Q_O)}.$$

Since $\eta_{\text{sym}}(\gamma_{\text{opt}}) = +\infty$ when $a_2 = 0$ and $\eta_{\text{sym}}(\gamma_{\text{opt}}) = 2\gamma_{\text{opt}}/a_2 \asymp \sqrt{\lambda(Q_O)}$ when $a_2 > 0$, the overall threshold $\eta^*(\gamma_{\text{opt}}) = \min\{\eta_{\text{sym}}(\gamma_{\text{opt}}), \eta_{\text{mat}}(\gamma_{\text{opt}})\}$ has the same $\sqrt{\lambda(Q_O)}$ scaling.

From coercivity to global decay. Having established the positivity of the dissipation matrix, we convert the algebraic estimate into a global semigroup decay bound. This step is an explicit algebraic substitution that keeps the gap dependence of the perturbation term visible.

Uniform norm equivalence. Recall the distortion parameter

$$(236) \quad \delta_*(\gamma) := 2\alpha_*(\gamma) \|\mathcal{A}_{\varpi_*}\|.$$

Using $\|\mathcal{A}_{\varpi_*}\| \leq k_{\mathcal{A}} \varpi_*^{-1/2}$ with $k_{\mathcal{A}} = (2\sqrt{\lambda_R})^{-1}$ and the explicit weight $\alpha_*(\gamma) = \frac{\gamma\lambda_R}{Z(\gamma)^2 + 2k_{\text{reg}}}$, we obtain

$$(237) \quad \delta_*(\gamma) = \frac{\gamma\lambda_R}{Z(\gamma)^2 + 2k_{\text{reg}}} \frac{1}{\sqrt{\lambda_R} \varpi_*} = \frac{\gamma\sqrt{\lambda_R} \varpi_*^{-1/2}}{(\gamma\Theta\varpi_*^{-1/2} + K_{\text{geom}})^2 + 2k_{\text{reg}}}.$$

Setting $x := \gamma\Theta\varpi_*^{-1/2}$ gives

$$(238) \quad \delta_*(\gamma) = \frac{\sqrt{\lambda_R}}{\Theta} \cdot \frac{x}{(x + K_{\text{geom}})^2 + 2k_{\text{reg}}}.$$

The elementary optimisation

$$(239) \quad \sup_{x>0} \frac{x}{(x + K_{\text{geom}})^2 + 2k_{\text{reg}}} = \frac{1}{2(K_{\text{geom}} + \sqrt{K_{\text{geom}}^2 + 2k_{\text{reg}}})}$$

together with $M_R \geq \lambda_R$ and $\sqrt{K_{\text{geom}}^2 + 2k_{\text{reg}}} \geq \sqrt{2k_{\text{reg}}} \geq \sqrt{2}$ (the latter since $k_{\text{reg}} = \sqrt{M_R/\lambda_R} \geq 1$) therefore yields

$$(240) \quad \delta_*(\gamma) \leq \frac{\sqrt{\lambda_R}}{\sqrt{M_R}(K_{\text{geom}} + \sqrt{K_{\text{geom}}^2 + 2k_{\text{reg}}})} \leq 2^{-1/2},$$

and therefore

$$(241) \quad \alpha_*(\gamma) \|\mathcal{A}_{\varpi_*}\| = \frac{\delta_*(\gamma)}{2} \leq 2^{-3/2} < \frac{1}{2}.$$

Thus Proposition 4.11 applies with $(\alpha, \varpi) = (\alpha_*, \varpi_*)$. Hence the norm-equivalence prefactor satisfies

$$(242) \quad C_*(\gamma) := \sqrt{\frac{1 + \delta_*(\gamma)}{1 - \delta_*(\gamma)}} \leq 1 + \sqrt{2}.$$

Thus, for all centered data $g \in \mathcal{H}_0$,

$$(243) \quad \frac{1}{2}(1 - \delta_*(\gamma)) \|g\|^2 \leq \mathcal{E}_{\alpha_*, \varpi_*}(g) \leq \frac{1}{2}(1 + \delta_*(\gamma)) \|g\|^2.$$

Explicit calculation of the decay rate. Set

$$(244) \quad c(\gamma, \eta) := \Lambda_{\text{hypo}}(\gamma) - \eta \|\mathbf{E}(\gamma)\|.$$

The algebraic dissipation estimate proved above on the core is

$$(245) \quad \mathfrak{d}_{\alpha_*, \varpi_*}(g) \leq -c(\gamma, \eta) \|g\|^2, \quad g \in \mathcal{C} \cap \mathcal{H}_0.$$

By graph-norm continuity of $\mathfrak{d}_{\alpha_*, \varpi_*}$, the same inequality holds for every $g \in \text{dom}(\mathcal{L}^\dagger) \cap \mathcal{H}_0$. Therefore, for any $g_0 \in \text{dom}(\mathcal{L}^\dagger) \cap \mathcal{H}_0$ and $g_t = e^{t\mathcal{L}^\dagger} g_0$,

$$(246) \quad \frac{d}{dt} \mathcal{E}_{\alpha_*, \varpi_*}(g_t) \leq -c(\gamma, \eta) \|g_t\|^2.$$

Next we use the norm equivalence from Proposition 4.11 in the convenient form

$$(247) \quad \frac{1}{2}(1 - \delta_*(\gamma)) \|g\|^2 \leq \mathcal{E}_{\alpha_*, \varpi_*}(g) \leq \frac{1}{2}(1 + \delta_*(\gamma)) \|g\|^2,$$

where $\delta_*(\gamma) := 2\alpha_*(\gamma) \|\mathcal{A}_{\varpi_*}\| \in (0, 1)$. The upper bound implies

$$(248) \quad \|g\|^2 \geq \frac{2}{1 + \delta_*(\gamma)} \mathcal{E}_{\alpha_*, \varpi_*}(g).$$

Substituting this into the dissipation inequality gives a closed differential inequality for the Lyapunov functional:

$$(249) \quad \frac{d}{dt} \mathcal{E}_{\alpha_*, \varpi_*}(g_t) \leq -\frac{2c(\gamma, \eta)}{1 + \delta_*(\gamma)} \mathcal{E}_{\alpha_*, \varpi_*}(g_t).$$

By Grönwall's inequality,

$$(250) \quad \mathcal{E}_{\alpha_*, \varpi_*}(g_t) \leq \exp\left(-\frac{2c(\gamma, \eta)}{1 + \delta_*(\gamma)} t\right) \mathcal{E}_{\alpha_*, \varpi_*}(g_0).$$

Finally, converting back to the L^2 norm using the two-sided equivalence yields

$$(251) \quad \|g_t\|^2 \leq \frac{2}{1 - \delta_*(\gamma)} \mathcal{E}_{\alpha_*, \varpi_*}(g_t) \leq \frac{1 + \delta_*(\gamma)}{1 - \delta_*(\gamma)} e^{-\frac{2c(\gamma, \eta)}{1 + \delta_*(\gamma)} t} \|g_0\|^2.$$

Therefore

$$(252) \quad \|g_t\| \leq \sqrt{\frac{1 + \delta_*(\gamma)}{1 - \delta_*(\gamma)}} e^{-\lambda_\eta t} \|g_0\|, \quad \lambda_\eta := \frac{c(\gamma, \eta)}{1 + \delta_*(\gamma)}.$$

Substituting the explicit expressions gives the unperturbed rate

$$(253) \quad \Lambda_{\text{hypo}}(\gamma) = \frac{\frac{1}{4}\lambda_R \gamma}{Z(\gamma)^2 + 2k_{\text{reg}}}, \quad \text{with } Z(\gamma) = \gamma \Theta \lambda(Q_O)^{-1/2} + K_{\text{geom}}.$$

For the perturbation norm, we use the sharp bound (228) at $\varpi_* = \lambda(Q_O)$ to track the gap-dependent factor:

$$(254) \quad \|\mathbf{E}(\gamma)\| \leq b_{2,\text{eff}} + \underbrace{\frac{\gamma\lambda_R}{Z(\gamma)^2 + 2k_{\text{reg}}}}_{\alpha_*(\gamma)} C_{\text{cross}} \lambda(Q_O)^{-1/2}.$$

Substituting these into the expression for λ_η and using the explicit distortion bound on $\delta_*(\gamma)$:

$$(255) \quad \lambda_\eta \geq \frac{\frac{1}{4}\lambda_R \gamma}{(1 + \delta_*(\gamma))(Z(\gamma)^2 + 2k_{\text{reg}})} - \frac{\eta}{1 + \delta_*(\gamma)} \left(b_{2,\text{eff}} + \frac{\gamma\lambda_R}{Z(\gamma)^2 + 2k_{\text{reg}}} C_{\text{cross}} \lambda(Q_O)^{-1/2} \right).$$

Equivalently,

$$(256) \quad \lambda_\eta \geq \frac{\Lambda_{\text{hypo}}(\gamma) - \eta B_{\text{pert}}(\gamma)}{1 + \delta_*(\gamma)} \geq (2 - \sqrt{2})(\Lambda_{\text{hypo}}(\gamma) - \eta B_{\text{pert}}(\gamma)),$$

using $1 + \delta_*(\gamma) \leq 1 + 2^{-1/2}$, i.e. $(1 + \delta_*(\gamma))^{-1} \geq (1 + 2^{-1/2})^{-1} = 2 - \sqrt{2}$. The preceding inequality is the lower bound used in Theorem 5.2. The factor $C_{\text{cross}}\lambda(Q_O)^{-1/2}$ is multiplied by the mixing weight $\alpha_*(\gamma) = \gamma\lambda_R/(Z(\gamma)^2 + 2k_{\text{reg}})$, which scales as $\sqrt{\lambda(Q_O)}$ at the optimal friction scale, so the perturbation contribution remains bounded.

Finally, converting back to the L^2 norm via Grönwall's lemma yields the global estimate on $\mathcal{H}_0$:

$$(257) \quad \|e^{t\mathcal{L}^\dagger} g\|_{\mathcal{H}} \leq C_*(\gamma) e^{-\lambda_\eta t} \|g\|_{\mathcal{H}} \leq (1 + \sqrt{2}) e^{-\lambda_\eta t} \|g\|_{\mathcal{H}}.$$

This has been proved for $g \in \text{dom}(\mathcal{L}^\dagger) \cap \mathcal{H}_0$. By Lemma 4.3, this domain is dense in $\mathcal{H}_0$. Hence for arbitrary $g \in \mathcal{H}_0$, choose $g^{(n)} \in \text{dom}(\mathcal{L}^\dagger) \cap \mathcal{H}_0$ with $g^{(n)} \rightarrow g$ in $\mathcal{H}$, apply the estimate to each $g^{(n)}$, and pass to the limit using strong continuity of $e^{t\mathcal{L}^\dagger}$ and the norm equivalence above. The passage from centered data in $\mathcal{H}_0$ to physical densities/states in $\mathcal{H}_1$ is then exactly Lemma 4.2. This completes the proof.

B.5. Optimization over Friction γ : Kinetic Acceleration. We determine γ_{opt} maximising $\Lambda_{\text{hypo}}(\gamma)$. From (220), setting $x := \Theta\lambda(Q_O)^{-1/2}\gamma$, the rate function reads

$$(258) \quad \Lambda_{\text{hypo}}(\gamma) = \frac{\frac{1}{4}\lambda_R}{\Theta} \sqrt{\lambda(Q_O)} \cdot \frac{x}{(x + K_{\text{geom}})^2 + 2k_{\text{reg}}}.$$

Optimal friction. The function $x \mapsto x/[(x + K_{\text{geom}})^2 + C_0]$ with $C_0 := 2k_{\text{reg}}$ is maximised when $(x + K_{\text{geom}})^2 + C_0 = 2x(x + K_{\text{geom}})$, equivalently $x^2 = K_{\text{geom}}^2 + C_0$. Translating back:

$$(259) \quad \gamma_{\text{opt}} = \frac{\sqrt{K_{\text{geom}}^2 + 2k_{\text{reg}}}}{\Theta} \sqrt{\lambda(Q_O)} =: C_{\text{lift}} \sqrt{\lambda(Q_O)},$$

where $C_{\text{lift}} := \Theta^{-1} \sqrt{K_{\text{geom}}^2 + 2k_{\text{reg}}}$.

Maximal rate. At $\gamma = \gamma_{\text{opt}}$ one has $x_* := \Theta\lambda(Q_O)^{-1/2}\gamma_{\text{opt}} = \sqrt{K_{\text{geom}}^2 + C_0}$. Substituting into the denominator: $(x_* + K_{\text{geom}})^2 + C_0 = 2(K_{\text{geom}}^2 + C_0 + K_{\text{geom}}x_*)$. Hence

$$(260) \quad \Lambda_{\text{max}} := \Lambda_{\text{hypo}}(\gamma_{\text{opt}}) = \frac{\frac{1}{4}\lambda_R}{2\Theta(x_* + K_{\text{geom}})} \sqrt{\lambda(Q_O)} \asymp \sqrt{\lambda(Q_O)}.$$

This confirms the kinetic acceleration: $\Lambda_{\text{max}} \asymp \sqrt{\lambda(Q_O)}$, compared to the overdamped upper bound $\Lambda_{\text{hypo}}(\gamma) \asymp \lambda(Q_O)/\gamma$ as $\gamma \rightarrow \infty$.

Threshold at optimal friction. The admissible perturbation range in Theorem 5.2 is determined as follows. From (230), the exact matrix threshold depends on $\|\mathbf{E}(\gamma)\|$, while the theorem uses the

explicit upper bound $B_{\text{pert}}(\gamma)$. The dominant term involves the mixing weight $\alpha_*(\gamma)$. Evaluating at γ_{opt} :

$$(261) \quad \alpha_*(\gamma_{\text{opt}}) = \frac{\gamma_{\text{opt}} \lambda_R}{Z(\gamma_{\text{opt}})^2 + 2k_{\text{reg}}}.$$

Using $\gamma_{\text{opt}} = C_{\text{lift}} \sqrt{\lambda(Q_O)}$ and $Z(\gamma_{\text{opt}}) = \sqrt{K_{\text{geom}}^2 + 2k_{\text{reg}}} + K_{\text{geom}} = \mathcal{O}(1)$, the numerator of $\alpha_*(\gamma_{\text{opt}})$ is $\gamma_{\text{opt}} \lambda_R \asymp \sqrt{\lambda(Q_O)}$ and the denominator $Z(\gamma_{\text{opt}})^2 + 2k_{\text{reg}} = \mathcal{O}(1)$ (independent of $\lambda(Q_O)$). Hence $\alpha_*(\gamma_{\text{opt}}) \asymp \sqrt{\lambda(Q_O)}$, so $\alpha_*(\gamma_{\text{opt}}) \cdot \lambda(Q_O)^{-1/2} = \mathcal{O}(1)$. Applying (228):

$$(262) \quad \|\mathbf{E}(\gamma_{\text{opt}})\| \leq b_{2,\text{eff}} + \alpha_*(\gamma_{\text{opt}}) C_{\text{cross}} \lambda(Q_O)^{-1/2} = \mathcal{O}(1).$$

Since the uniform non-degeneracy condition gives $b_{2,\text{eff}} + C_{\text{cross}} \geq c_0 > 0$ and

$$B_{\text{pert}}(\gamma_{\text{opt}}) = b_{2,\text{eff}} + \alpha_*(\gamma_{\text{opt}}) C_{\text{cross}} \lambda(Q_O)^{-1/2} = \mathcal{O}(1),$$

and $\alpha_*(\gamma_{\text{opt}}) \lambda(Q_O)^{-1/2} \asymp 1$, we have $B_{\text{pert}}(\gamma_{\text{opt}}) \asymp 1$. Since $\Lambda_{\text{max}} \asymp \sqrt{\lambda(Q_O)}$, the matrix-controlled threshold satisfies

$$(263) \quad \eta_{\text{mat}}(\gamma_{\text{opt}}) = \frac{\Lambda_{\text{max}}}{2B_{\text{pert}}(\gamma_{\text{opt}})} \asymp \sqrt{\lambda(Q_O)}.$$

Together with $\eta_{\text{sym}}(\gamma_{\text{opt}}) = +\infty$ when $a_2 = 0$ and $\eta_{\text{sym}}(\gamma_{\text{opt}}) = 2\gamma_{\text{opt}}/a_2 \asymp \sqrt{\lambda(Q_O)}$ when $a_2 > 0$, this yields

$$(264) \quad \eta^*(\gamma_{\text{opt}}) = \min\{\eta_{\text{sym}}(\gamma_{\text{opt}}), \eta_{\text{mat}}(\gamma_{\text{opt}})\} \asymp \sqrt{\lambda(Q_O)}.$$

This confirms that the accelerated regime retains an $\mathcal{O}(\sqrt{\lambda(Q_O)})$ admissible perturbation range.

APPENDIX C. UPPER BOUND VIA SINGULAR VALUE GAP

This appendix proves Theorem 5.4.

C.1. Exponential decay implies a singular value-gap bound. By Lemma 4.3, the restriction of $e^{t\mathcal{L}^\dagger}$ to $\mathcal{H}_0$ is a C_0 -semigroup with generator

$$\mathcal{L}_0^\dagger := \mathcal{L}^\dagger|_{\text{dom}(\mathcal{L}^\dagger) \cap \mathcal{H}_0}.$$

For a (possibly unbounded) closed operator T on $\mathcal{H}_0$, define its injectivity modulus

$$(265) \quad s(T) := \inf_{g \in \text{dom}(T) \setminus \{0\}} \frac{\|Tg\|}{\|g\|}.$$

Lemma C.1 (From exponential decay to a singular value-gap bound). *If there exist constants $C \geq 1$ and $\nu > 0$ such that*

$$(266) \quad \|e^{t\mathcal{L}^\dagger} g\| \leq C e^{-\nu t} \|g\|, \quad \forall t \geq 0, \forall g \in \mathcal{H}_0,$$

then 0 belongs to the resolvent set of $\mathcal{L}_0^\dagger$ and

$$(267) \quad \nu \leq C s(\mathcal{L}^\dagger).$$

Proof. For $\lambda > 0$, define the resolvent regularization

$$(268) \quad u_\lambda := - \int_0^\infty e^{-\lambda t} e^{t\mathcal{L}^\dagger} g \, dt.$$

This Bochner integral is absolutely convergent because

$$(269) \quad \|e^{-\lambda t} e^{t\mathcal{L}^\dagger} g\| \leq C e^{-(\lambda+\nu)t} \|g\|,$$

and the right-hand side is integrable on $[0, \infty)$. Therefore,

$$(270) \quad \|u_\lambda\| \leq C \int_0^\infty e^{-\nu t} dt \|g\| = \frac{C}{\nu} \|g\|.$$

By the Hille–Yosida resolvent formula for the centered semigroup, $u_\lambda \in \text{dom}(\mathcal{L}_0^\dagger) = \text{dom}(\mathcal{L}^\dagger) \cap \mathcal{H}_0$ and

$$(271) \quad (\lambda - \mathcal{L}_0^\dagger)(-u_\lambda) = g, \quad \text{equivalently} \quad \mathcal{L}_0^\dagger u_\lambda = g + \lambda u_\lambda.$$

The family $(u_\lambda)_{\lambda>0}$ is uniformly bounded in $\mathcal{H}_0$ by the estimate above and converges, as $\lambda \downarrow 0$, to

$$(272) \quad u := - \int_0^\infty e^{t\mathcal{L}^\dagger} g dt$$

by dominated convergence, using the integrable majorant $Ce^{-\nu t}\|g\|$. Since $\lambda u_\lambda \rightarrow 0$ in $\mathcal{H}_0$ and $\mathcal{L}_0^\dagger$ is closed, the relation $\mathcal{L}_0^\dagger u_\lambda = g + \lambda u_\lambda$ passes to the limit and gives $u \in \text{dom}(\mathcal{L}_0^\dagger)$ together with $\mathcal{L}_0^\dagger u = g$. Injectivity follows from the same exponential estimate: if $\mathcal{L}_0^\dagger v = 0$, then $e^{t\mathcal{L}^\dagger} v = v$ for all $t \geq 0$, whereas the bound $\|v\| \leq Ce^{-\nu t}\|v\|$ forces $v = 0$ as $t \rightarrow \infty$. Thus $(\mathcal{L}_0^\dagger)^{-1}$ exists and satisfies $\|(\mathcal{L}_0^\dagger)^{-1}\| \leq C/\nu$, hence $s(\mathcal{L}^\dagger) = s(\mathcal{L}_0^\dagger) = \|(\mathcal{L}_0^\dagger)^{-1}\|^{-1} \geq \nu/C$, i.e. $\nu \leq C s(\mathcal{L}^\dagger)$. $\square$

Corollary C.2 (Contractive semigroup bound). *Under the hypotheses of Lemma C.1, assume in addition that $e^{t\mathcal{L}^\dagger}$ is a contraction semigroup on $\mathcal{H}_0$. Then*

$$(273) \quad \nu \leq (1 + \log C) s(\mathcal{L}^\dagger).$$

Proof. Let $T := (\log C)/\nu$ and define u_λ as in Lemma C.1. Since $e^{-\lambda t} \leq 1$, contractivity on $[0, T]$ and the exponential bound on $[T, \infty)$ give

$$(274) \quad \|u_\lambda\| \leq \int_0^T \|e^{-\lambda t} e^{t\mathcal{L}^\dagger} g\| dt + \int_T^\infty \|e^{-\lambda t} e^{t\mathcal{L}^\dagger} g\| dt$$

$$(275) \quad \leq T \|g\| + C \int_T^\infty e^{-\nu t} dt \|g\|$$

$$(276) \quad = \frac{\log C}{\nu} \|g\| + \frac{Ce^{-\nu T}}{\nu} \|g\|$$

$$(277) \quad = \frac{1 + \log C}{\nu} \|g\|.$$

Passing to $\lambda \downarrow 0$ exactly as in Lemma C.1 yields

$$(278) \quad \|(\mathcal{L}_0^\dagger)^{-1}\| \leq \frac{1 + \log C}{\nu},$$

and therefore $\nu \leq (1 + \log C) s(\mathcal{L}^\dagger)$. $\square$

Lemma C.3 (Relaxation time is bounded below by the singular value gap). *Assume $\ker(\mathcal{L}^\dagger) \cap \mathcal{H}_0 = \{0\}$. Assume moreover that $e^{t\mathcal{L}^\dagger}$ is a contraction semigroup on $\mathcal{H}_0$, i.e. $\|e^{t\mathcal{L}^\dagger} f\| \leq \|f\|$ for all $t \geq 0$ and $f \in \mathcal{H}_0$. Assume there exist constants $C \geq 1$ and $\nu > 0$ such that*

$$(279) \quad \|e^{t\mathcal{L}^\dagger} f\| \leq C e^{-\nu t} \|f\|, \quad \forall t \geq 0, \forall f \in \mathcal{H}_0.$$

Define the L^2 relaxation time

$$(280) \quad t_{\text{rel}}(\mathcal{L}^\dagger) := \inf \left\{ t \geq 0; \|e^{t\mathcal{L}^\dagger} f\| \leq e^{-1} \|f\| \text{ for all } f \in \mathcal{H}_0 \right\}.$$

Then

$$(281) \quad t_{\text{rel}}(\mathcal{L}^\dagger) \geq \frac{1}{2 s(\mathcal{L}^\dagger)}.$$

This L^2 bound is a continuous-time analogue of discrete-time relaxation estimates based on singular values for non-reversible Markov chains.

Proof. Let $\tau > t_{\text{rel}}(\mathcal{L}^\dagger)$ be such that

$$(282) \quad \|e^{\tau \mathcal{L}^\dagger} f\| \leq e^{-1} \|f\|, \quad \forall f \in \mathcal{H}_0.$$

Such a τ exists by the definition of the infimum. For any $t \geq 0$ write $t = n\tau + r$ with $n \in \mathbb{N}_0$ and $r \in [0, \tau)$. The contraction property and the semigroup identity yield

$$(283) \quad \|e^{t \mathcal{L}^\dagger}\| \leq \|e^{r \mathcal{L}^\dagger}\| \|e^{\tau \mathcal{L}^\dagger}\|^n \leq e^{-n}.$$

Since $n \geq t/\tau - 1$, this implies

$$(284) \quad \|e^{t \mathcal{L}^\dagger}\| \leq e e^{-t/\tau}.$$

We may therefore apply Corollary C.2 with $C = e$ and $\nu = \tau^{-1}$ to obtain

$$(285) \quad \tau^{-1} \leq (1 + \log e) s(\mathcal{L}^\dagger) = 2 s(\mathcal{L}^\dagger).$$

Hence $\tau \geq (2s(\mathcal{L}^\dagger))^{-1}$. Letting $\tau \downarrow t_{\text{rel}}(\mathcal{L}^\dagger)$ gives the stated bound. $\square$

C.2. Test function and completion of the proof.

Proof of Theorem 5.4. Set $A := -Q_O|_{\mathcal{H}_{S,0}}$. By Assumption 4.6, A is self-adjoint and $\sigma(A) \subset [\lambda(Q_O), \infty)$. Since $\lambda(Q_O) = \inf \sigma(A)$, the spectral theorem gives a normalized vector in the spectral subspace near the bottom of A . Because $\mathcal{C}_Q \cap \mathcal{H}_{S,0}$ is a core for A , we may approximate that vector in the graph norm closely enough, and then renormalize, so that for every $\varepsilon > 0$ there is $\phi_\varepsilon \in \mathcal{C}_Q \cap \mathcal{H}_{S,0}$ such that

$$(286) \quad \|\phi_\varepsilon\| = 1, \quad \|A\phi_\varepsilon\| \leq \lambda(Q_O) + \varepsilon, \quad \|A^{1/2}\phi_\varepsilon\| \leq \sqrt{\lambda(Q_O) + \varepsilon}.$$

Define the first-order corrector

$$(287) \quad \psi_\varepsilon := \mathcal{R}^+ \mathcal{V}_{\text{skew}} \phi_\varepsilon \in \mathcal{H}_F,$$

and the centered test function $g_{\gamma,\varepsilon} := \phi_\varepsilon + \gamma^{-1} \psi_\varepsilon$. Because ϕ_ε belongs to the common slow core, the standing core convention and Assumption 4.4 make $g_{\gamma,\varepsilon}$ an admissible test vector in $\text{dom}(\mathcal{L}^\dagger) \cap \mathcal{H}_0$; the perturbation estimate below is justified by Assumption 4.5.4. Since $\phi_\varepsilon \in \mathcal{H}_S$ and $\psi_\varepsilon \in \mathcal{H}_F$ we have $\langle \phi_\varepsilon, \psi_\varepsilon \rangle = 0$ and therefore

$$(288) \quad \|g_{\gamma,\varepsilon}\|^2 = \|\phi_\varepsilon\|^2 + \gamma^{-2} \|\psi_\varepsilon\|^2 \geq 1.$$

Using $\mathcal{R}\phi_\varepsilon = 0$ and $\mathcal{R}\mathcal{R}^+ = P_F$ on $\mathcal{H}$, together with centered coupling $\mathcal{V}_{\text{skew}}\phi_\varepsilon \in \mathcal{H}_F$, we obtain

$$(289) \quad \mathcal{R}\psi_\varepsilon = \mathcal{R}\mathcal{R}^+ \mathcal{V}_{\text{skew}}\phi_\varepsilon = P_F \mathcal{V}_{\text{skew}}\phi_\varepsilon = \mathcal{V}_{\text{skew}}\phi_\varepsilon.$$

Consequently,

$$(290) \quad \mathcal{L}^\dagger g_{\gamma,\varepsilon} = (\gamma \mathcal{R} - \mathcal{V}_{\text{skew}} + \eta \mathcal{V}_{\text{pert}}^\dagger)(\phi_\varepsilon + \gamma^{-1} \psi_\varepsilon)$$

$$(291) \quad = \underbrace{\mathcal{R}\psi_\varepsilon - \mathcal{V}_{\text{skew}}\phi_\varepsilon}_{=0} - \gamma^{-1} \mathcal{V}_{\text{skew}}\psi_\varepsilon + \eta \mathcal{V}_{\text{pert}}^\dagger g_{\gamma,\varepsilon}.$$

We decompose the remaining transport term into slow/fast components:

$$(292) \quad \mathcal{V}_{\text{skew}}\psi_\varepsilon = P_S \mathcal{V}_{\text{skew}}\psi_\varepsilon + P_F \mathcal{V}_{\text{skew}}\psi_\varepsilon.$$

For the slow part, apply Assumption 4.4(iii) to ϕ_ε . Since $w_{\phi_\varepsilon} := P_F \mathcal{V}_{\text{skew}} P_S \phi_\varepsilon = \mathcal{V}_{\text{skew}}\phi_\varepsilon$ and $(-\mathcal{R}^+)w_{\phi_\varepsilon} = -\psi_\varepsilon$ on $\mathcal{H}_F$, we have

$$(293) \quad -P_S \mathcal{V}_{\text{skew}}\psi_\varepsilon = P_S \mathcal{V}_{\text{skew}} P_F (-\mathcal{R}^+)w_{\phi_\varepsilon} = Q_O \phi_\varepsilon,$$

and therefore $\|P_S \mathcal{V}_{\text{skew}} \psi_\varepsilon\| = \|A\phi_\varepsilon\| \leq \lambda(Q_O) + \varepsilon$. For the fast part, apply the flux regularity estimate (30) with the slow input ϕ_ε . Using the same notation, we obtain

$$(294) \quad \|P_F \mathcal{V}_{\text{skew}} \psi_\varepsilon\| = \|P_F \mathcal{V}_{\text{skew}} P_F \psi_\varepsilon\| = \|P_F \mathcal{V}_{\text{skew}} P_F (-\mathcal{R}^+) w_{\phi_\varepsilon}\| \leq \kappa_{\text{ell}} \|A\phi_\varepsilon\| + \kappa_{\text{low}} \|A^{1/2} \phi_\varepsilon\|$$

$$(295) \quad \leq \kappa_{\text{ell}} (\lambda(Q_O) + \varepsilon) + \kappa_{\text{low}} \sqrt{\lambda(Q_O) + \varepsilon}.$$

Combining the slow and fast contributions yields

$$(296) \quad \|\mathcal{V}_{\text{skew}} \psi_\varepsilon\| \leq (1 + \kappa_{\text{ell}}) (\lambda(Q_O) + \varepsilon) + \kappa_{\text{low}} \sqrt{\lambda(Q_O) + \varepsilon}.$$

Hence

$$(297) \quad \|\mathcal{L}^\dagger g_{\gamma,\varepsilon}\| \leq \frac{(1 + \kappa_{\text{ell}}) (\lambda(Q_O) + \varepsilon) + \kappa_{\text{low}} \sqrt{\lambda(Q_O) + \varepsilon}}{\gamma} + \eta \|\mathcal{V}_{\text{pert}}^\dagger g_{\gamma,\varepsilon}\|.$$

To bound the perturbation, use Assumption 4.5.4 together with $\mathcal{V}_{\text{pert}}^\dagger = -\mathcal{K}_{\text{pert}} + \mathcal{S}_{\text{pert}}$ to estimate

$$(298) \quad \|\mathcal{V}_{\text{pert}}^\dagger f\| \leq \|\mathcal{K}_{\text{pert}} f\| + \|\mathcal{S}_{\text{pert}} f\|$$

$$(299) \quad \leq (a_1 + a_2) \|(-\mathcal{R})^{1/2} f\| + (b_1 + b_2) \|f\|, \quad f \in \text{dom}((-\mathcal{R})^{1/2}).$$

Since $\phi_\varepsilon \in \ker(\mathcal{R})$ and $g_{\gamma,\varepsilon} = \phi_\varepsilon + \gamma^{-1} \psi_\varepsilon$, we have

$$(300) \quad \|(-\mathcal{R})^{1/2} g_{\gamma,\varepsilon}\| = \gamma^{-1} \|(-\mathcal{R})^{1/2} \psi_\varepsilon\|.$$

Finally, using $\psi_\varepsilon = \mathcal{R}^+ \mathcal{V}_{\text{skew}} \phi_\varepsilon$ and the definition of Q_O ,

$$(301) \quad \begin{aligned} \|(-\mathcal{R})^{1/2} \psi_\varepsilon\|^2 &= \langle (-\mathcal{R}) \psi_\varepsilon, \psi_\varepsilon \rangle \\ &= \langle \mathcal{V}_{\text{skew}} \phi_\varepsilon, (-\mathcal{R}^+) \mathcal{V}_{\text{skew}} \phi_\varepsilon \rangle \\ &= -\langle Q_O \phi_\varepsilon, \phi_\varepsilon \rangle = \langle A\phi_\varepsilon, \phi_\varepsilon \rangle \leq \lambda(Q_O) + \varepsilon. \end{aligned}$$

Therefore $\|(-\mathcal{R})^{1/2} g_{\gamma,\varepsilon}\| \leq \sqrt{\lambda(Q_O) + \varepsilon}/\gamma$, and

$$(302) \quad \|\mathcal{V}_{\text{pert}}^\dagger g_{\gamma,\varepsilon}\| \leq (b_1 + b_2) \|g_{\gamma,\varepsilon}\| + (a_1 + a_2) \frac{\sqrt{\lambda(Q_O) + \varepsilon}}{\gamma}.$$

Dividing by $\|g_{\gamma,\varepsilon}\| \geq 1$ gives

$$(303) \quad \frac{\|\mathcal{L}^\dagger g_{\gamma,\varepsilon}\|}{\|g_{\gamma,\varepsilon}\|} \leq \frac{(1 + \kappa_{\text{ell}}) (\lambda(Q_O) + \varepsilon) + \kappa_{\text{low}} \sqrt{\lambda(Q_O) + \varepsilon}}{\gamma} + \eta (b_1 + b_2) + \eta (a_1 + a_2) \frac{\sqrt{\lambda(Q_O) + \varepsilon}}{\gamma}.$$

By definition of $s(\mathcal{L}^\dagger)$, this implies $s(\mathcal{L}^\dagger) \leq \|\mathcal{L}^\dagger g_{\gamma,\varepsilon}\|/\|g_{\gamma,\varepsilon}\|$. Letting $\varepsilon \downarrow 0$ gives the claimed bound on $s(\mathcal{L}^\dagger)$. Combining with Lemma C.1 yields the stated estimate on ν , and Corollary C.2 gives the improved prefactor under contractivity. $\square$

APPENDIX D. PROOF OF PROPOSITION 6.2

We verify Assumptions 3.1, 4.4, and 4.5.1–4 for the Langevin family

$$(304) \quad \mathcal{V}_{\text{pert}} f = F(q) \cdot \nabla_p f,$$

with $d \geq 2$, $m \in (0, 1]$, $K \geq 0$, a confining potential $U \in C^\infty(\mathbb{R}^d)$ such that $\nabla^2 U \geq -K \text{Id}$ and all derivatives of positive order grow at most polynomially, and a force $F \in C^\infty(\mathbb{R}^d; \mathbb{R}^d)$ whose derivatives of positive order also grow at most polynomially and which satisfies (71). We assume throughout that the position marginal $\mu_q(dq) \propto e^{-\beta U(q)} dq$ has exact spectral gap $m > 0$, in the sense that m is the optimal constant in (67). The qualitative realization input is the standard confining Kramers–Fokker–Planck core theory for the equilibrium part together with the polynomial-coefficient preservation of the Schwartz core by the perturbation terms; the quantitative constants are recorded in terms of $A_\nabla, B_\nabla, A_p, B_p$.

We use the Schwartz core

$$(305) \quad \mathcal{C}_{\text{Lan}} := \mathcal{S}(\mathbb{R}_q^d \times \mathbb{R}_p^d) \subset L^2(\mu),$$

where

$$d\mu(q, p) = Z^{-1} e^{-\beta(\frac{1}{2}|p|^2 + U(q))} dq dp.$$

We use the standard confining Kramers–Fokker–Planck realization for which $\mathcal{C}_{\text{Lan}}$ is a common invariant core for the equilibrium part $\mathcal{V}_{\text{skew}} + \gamma\mathcal{R}$ and its adjoint. Because F and its derivatives have at most polynomial growth, the perturbation operators $\mathcal{V}_{\text{pert}}$, $\mathcal{V}_{\text{pert}}^\dagger$, $\mathcal{K}_{\text{pert}}$, and $\mathcal{S}_{\text{pert}}$ also preserve $\mathcal{C}_{\text{Lan}}$; the same is true for P_S , P_F , and $\mathcal{R}^+$. We therefore use $\mathcal{C}_{\text{Lan}}$ as the common invariant core in the sense of Section 3.1. Moreover, for $\psi \in \mathcal{C}_{\text{Lan}} \cap \mathcal{H}_S = \mathcal{S}(\mathbb{R}^d)$ one has

$$(306) \quad \mathcal{R}^+ \mathcal{V}_{\text{skew}} \psi = -p \cdot \nabla_q \psi \in \mathcal{C}_{\text{Lan}}, \quad Q_O \psi = \beta^{-1} \Delta_q \psi - \nabla_q U \cdot \nabla_q \psi,$$

with $\mathcal{C}_{\text{Lan}} \cap \mathcal{H}_S$ a core for this realization on $L^2(\mu_q)$; see [20, Introduction and §§1–2], [12, §2], and [19, Ch. 5]. The identities above verify Assumption 4.4. The rest of the appendix records the constants in Assumption 4.5 and the slow gap in Assumption 4.6.

D.1. Microscopic spectral gap, centering, and slow gap. The fast dissipator

$$(307) \quad \mathcal{R} = -p \cdot \nabla_p + \beta^{-1} \Delta_p$$

acts only on the Gaussian momentum variable and has spectral gap

$$(308) \quad \lambda_R = 1$$

on $\mathcal{H}_F = \ker(\Pi_p)$. The Hamiltonian part $\mathcal{V}_{\text{skew}} = p \cdot \nabla_q - \nabla_q U \cdot \nabla_p$ is skew-adjoint in $L^2(\mu)$ because the Hamiltonian flow preserves μ . Since $\int p d\mu_p = 0$, one has

$$(309) \quad P_S \mathcal{V}_{\text{skew}} P_S = 0.$$

Also $P_S \mathcal{V}_{\text{pert}} P_S = 0$ because $\nabla_p \phi(q) = 0$ for slow functions; the corresponding adjoint slow compression vanishes by the same Gaussian-centering calculation.

For $\psi = \psi(q)$,

$$(310) \quad w_\psi := P_F \mathcal{V}_{\text{skew}} P_S \psi = p \cdot \nabla_q \psi.$$

Since $\mathcal{R}(p_i) = -p_i$, one has $(-\mathcal{R}^+)w_\psi = w_\psi$, so the conditioning constant is

$$(311) \quad M_R = 1.$$

The effective slow operator is

$$(312) \quad Q_O \psi = \beta^{-1} \Delta_q \psi - \nabla_q U \cdot \nabla_q \psi.$$

By the assumed spectral gap $m > 0$ for μ_q , expressed through the Poincaré inequality (67), one has

$$(313) \quad -\langle \psi, Q_O \psi \rangle_{L^2(\mu_q)} = \beta^{-1} \|\nabla_q \psi\|_{L^2(\mu_q)}^2 \geq m \|\psi - \int \psi d\mu_q\|_{L^2(\mu_q)}^2,$$

so Assumption 4.6 holds with

$$(314) \quad \lambda(Q_O) = m.$$

D.2. Flux regularity (Assumption 4.5.3). Define

$$(315) \quad Z_{ij}(p) := p_i p_j - \beta^{-1} \delta_{ij}.$$

Using $(-\mathcal{R}^+)w_\psi = w_\psi$, we compute

$$(316) \quad \begin{aligned} \mathcal{V}_{\text{skew}} w_\psi &= p \cdot \nabla_q (p \cdot \nabla_q \psi) - \nabla_q U \cdot \nabla_p (p \cdot \nabla_q \psi) \\ &= \sum_{i,j=1}^d p_i p_j \partial_{q_i q_j} \psi - \nabla_q U \cdot \nabla_q \psi. \end{aligned}$$

The p -average of this expression is

$$(317) \quad P_S \mathcal{V}_{\text{skew}} w_\psi = \beta^{-1} \Delta_q \psi - \nabla_q U \cdot \nabla_q \psi = Q_O \psi,$$

so subtracting the slow part gives

$$(318) \quad P_F \mathcal{V}_{\text{skew}} P_F (-\mathcal{R}^+) w_\psi = \sum_{i,j=1}^d (p_i p_j - \beta^{-1} \delta_{ij}) \partial_{q_i q_j} \psi = \sum_{i,j=1}^d Z_{ij}(p) \partial_{q_i q_j} \psi.$$

The Gaussian fourth-moment identity

$$(319) \quad \mathbb{E}_{\mu_p} [Z_{ij} Z_{kl}] = \beta^{-2} (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk})$$

therefore gives

$$(320) \quad \|P_F \mathcal{V}_{\text{skew}} P_F (-\mathcal{R}^+) w_\psi\|_{L^2(\mu)}^2 = 2\beta^{-2} \|\nabla_q^2 \psi\|_{L^2(\mu_q)}^2.$$

To control the Hessian, we use the weighted elliptic estimate obtained from the Bochner identity under the lower Hessian bound. Set $L := -\beta Q_O = -\Delta_q + \beta \nabla_q U \cdot \nabla_q$, the weighted Laplacian on $L^2(\mu_q)$. For every $\psi \in \mathcal{S}(\mathbb{R}^d)$, one has the commutator

$$(321) \quad [\partial_{q_i}, L] = \beta (\partial_{q_i q_k}^2 U) \partial_{q_k},$$

obtained from $[\partial_{q_i}, \nabla_q U \cdot \nabla_q] = (\partial_{q_i q_k}^2 U) \partial_{q_k}$ and $[\partial_{q_i}, -\Delta_q] = 0$. Since L is self-adjoint in $L^2(\mu_q)$ and $\langle g, Lg \rangle_{L^2(\mu_q)} = \|\nabla_q g\|_{L^2(\mu_q)}^2$,

$$(322) \quad \begin{aligned} \|L\psi\|_{L^2(\mu_q)}^2 &= \langle \psi, L^2 \psi \rangle_{L^2(\mu_q)} = \sum_{i=1}^d \langle \partial_{q_i} \psi, \partial_{q_i} (L\psi) \rangle_{L^2(\mu_q)} \\ &= \sum_{i=1}^d \langle \partial_{q_i} \psi, L(\partial_{q_i} \psi) \rangle_{L^2(\mu_q)} + \beta \langle \nabla_q^2 U \nabla_q \psi, \nabla_q \psi \rangle_{L^2(\mu_q)} \\ &= \|\nabla_q^2 \psi\|_{L^2(\mu_q)}^2 + \beta \langle \nabla_q^2 U \nabla_q \psi, \nabla_q \psi \rangle_{L^2(\mu_q)}. \end{aligned}$$

Since $\nabla_q^2 U \geq -K \text{Id}$, the second term is bounded below by $-\beta K \|\nabla_q \psi\|_{L^2(\mu_q)}^2$, so

$$(323) \quad \|\nabla_q^2 \psi\|_{L^2(\mu_q)}^2 \leq \|L\psi\|_{L^2(\mu_q)}^2 + \beta K \|\nabla_q \psi\|_{L^2(\mu_q)}^2.$$

Using $L = -\beta Q_O$ and

$$(324) \quad \beta^{-1} \|\nabla_q \psi\|_{L^2(\mu_q)}^2 = \|(-Q_O)^{1/2} \psi\|_{L^2(\mu_q)}^2,$$

we obtain the weighted elliptic estimate

$$(325) \quad \|\nabla_q^2 \psi\|_{L^2(\mu_q)}^2 \leq \beta^2 \|(-Q_O) \psi\|_{L^2(\mu_q)}^2 + \beta^2 K \|(-Q_O)^{1/2} \psi\|_{L^2(\mu_q)}^2.$$

Combining with (320) yields

$$(326) \quad \|P_F \mathcal{V}_{\text{skew}} P_F (-\mathcal{R}^+) w_\psi\|_{L^2(\mu)}^2 \leq 2 \|(-Q_O) \psi\|_{L^2(\mu_q)}^2 + 2K \|(-Q_O)^{1/2} \psi\|_{L^2(\mu_q)}^2,$$

and therefore

$$(327) \quad \|P_F \mathcal{V}_{\text{skew}} P_F(-\mathcal{R}^+) w_\psi\|_{L^2(\mu)} \leq \sqrt{2} \|(-Q_O)\psi\|_{L^2(\mu_q)} + \sqrt{2K} \|(-Q_O)^{1/2}\psi\|_{L^2(\mu_q)}$$

for every $\psi \in \mathcal{S}(\mathbb{R}^d)$. Since $\mathcal{C}_{\text{Lan}} \cap \mathcal{H}_S = \mathcal{S}(\mathbb{R}^d)$ is a core for Q_O and the right-hand side is continuous in the graph norm of Q_O (using $\|(-Q_O)^{1/2}u\|^2 \leq \|u\| \|(-Q_O)u\|$), the estimate extends by density to every $\psi \in \text{dom}(Q_O)$. Thus Assumption 4.5.3 holds, with $\lambda(Q_O) = m$ and

$$(328) \quad \kappa_{\text{ell}} \leq \sqrt{2}, \quad \kappa_{\text{low}} \leq \sqrt{2K}.$$

D.3. Perturbation bounds (Assumption 4.5.4). The $L^2(\mu)$ -adjoint of $\mathcal{V}_{\text{pert}} = F(q) \cdot \nabla_p$ is

$$(329) \quad \mathcal{V}_{\text{pert}}^\dagger = -F(q) \cdot \nabla_p + \beta F(q) \cdot p,$$

and both $\mathcal{V}_{\text{pert}}$ and $\mathcal{V}_{\text{pert}}^\dagger$ have zero slow compression because a slow input is independent of p and the Gaussian momentum has mean zero. The skew/symmetric decomposition is

$$(330) \quad \mathcal{K}_{\text{pert}} = F(q) \cdot \nabla_p - \frac{\beta}{2} F(q) \cdot p, \quad \mathcal{S}_{\text{pert}} = \frac{\beta}{2} F(q) \cdot p.$$

On $\mathcal{C}_{\text{Lan}}$, the adjoint identity above gives

$$(331) \quad \mathcal{K}_{\text{pert}}^\dagger = -\mathcal{K}_{\text{pert}}, \quad \mathcal{S}_{\text{pert}}^\dagger = \mathcal{S}_{\text{pert}}.$$

Let $f \in \text{dom}((-\mathcal{R})^{1/2})$. Using the relative bounds (71),

$$(332) \quad \begin{aligned} \|\mathcal{K}_{\text{pert}} f\|_{L^2(\mu)} &\leq \|F(q) \cdot \nabla_p f\|_{L^2(\mu)} + \frac{\beta}{2} \|F(q) \cdot p f\|_{L^2(\mu)} \\ &\leq \left(A_\nabla + \frac{\beta}{2} A_p\right) \|(-\mathcal{R})^{1/2} f\|_{L^2(\mu)} + \left(B_\nabla + \frac{\beta}{2} B_p\right) \|f\|_{L^2(\mu)}, \end{aligned}$$

while

$$(333) \quad \|\mathcal{S}_{\text{pert}} f\|_{L^2(\mu)} \leq \frac{\beta}{2} \|F(q) \cdot p f\|_{L^2(\mu)} \leq \frac{\beta}{2} A_p \|(-\mathcal{R})^{1/2} f\|_{L^2(\mu)} + \frac{\beta}{2} B_p \|f\|_{L^2(\mu)}.$$

Therefore Assumption 4.5.4 holds with

$$(334) \quad a_1 = A_\nabla + \frac{\beta}{2} A_p, \quad b_1 = B_\nabla + \frac{\beta}{2} B_p, \quad a_2 = \frac{\beta}{2} A_p, \quad b_2 = \frac{\beta}{2} B_p.$$

The forcing hypothesis is used quantitatively only through these bounds. In the present reference norm the relative bounds are strong: for product test functions they force F to be a bounded multiplier in the position variable. Thus bounded forces give the canonical class covered uniformly by this example, and any unbounded force must be justified directly through (71).

APPENDIX E. PROOF OF PROPOSITION 6.4

We verify Assumptions 3.1, 4.1, 4.4, 4.5, and 4.6 for the gravitactic active-Brownian model from Example 6.2. Work on

$$L^2(\mu_L), \quad d\mu_L(x, \theta) = \frac{dx}{(2\pi L)^2} \frac{d\theta}{2\pi}, \quad x \in \mathbb{T}_L^2,$$

with generator

$$\mathcal{L}f = \gamma \partial_{\theta\theta} f + v_0 \tau(\theta) \cdot \nabla_x f - \eta \sin \theta \partial_\theta f.$$

The common invariant core is

$$\mathcal{C}_{\text{ABP}} := \text{span}\{e^{in \cdot x/L} e^{ik\theta} : n \in \mathbb{Z}^2, k \in \mathbb{Z}\}.$$

It is preserved by $\mathcal{V}_{\text{skew}}$, $\mathcal{V}_{\text{pert}}$, their adjoints, and $\mathcal{R}^+$. We take

$$\mathcal{C}_Q = \text{span}\{e^{in \cdot x/L} : n \in \mathbb{Z}^2\} \subset \mathcal{H}_S.$$

The stationary density is explicit. With $a = \eta/\gamma$ and

$$h_{\gamma, \eta}(x, \theta) = \frac{e^{a \cos \theta}}{I_0(a)}, \quad I_0(a) = \int_{\mathbb{S}^1} e^{a \cos \theta} \frac{d\theta}{2\pi},$$

one has

$$h'_{\gamma,\eta} = -a \sin \theta h_{\gamma,\eta}, \quad h''_{\gamma,\eta} = (-a \cos \theta + a^2 \sin^2 \theta) h_{\gamma,\eta}.$$

Since $h_{\gamma,\eta}$ is independent of x ,

$$\mathcal{L}^\dagger h_{\gamma,\eta} = \gamma h''_{\gamma,\eta} + \eta (\sin \theta h'_{\gamma,\eta} + \cos \theta h_{\gamma,\eta}) = 0.$$

Moreover $\mathcal{L}\mathbf{1} = 0$ and $h_{\gamma,\eta} \in L^2_1(\mu_L)$, so Assumption 4.1 holds for every $\gamma > 0$ and $\eta \geq 0$.

E.1. Microscopic spectral gap, centering, and slow gap. The fast dissipator

$$\mathcal{R} = \partial_{\theta\theta}$$

acts only on the angular variable and satisfies

$$\ker(\mathcal{R}) = \mathcal{H}_S = \{f(x, \theta) = \psi(x)\}, \quad \lambda_R = 1$$

by the Poincaré inequality on $\mathbb{S}^1$. The transport part

$$\mathcal{V}_{\text{skew}} = v_0 \tau(\theta) \cdot \nabla_x$$

is skew-adjoint in $L^2(\mu_L)$ by periodic integration by parts. The centering conditions are immediate:

$$P_S \mathcal{V}_{\text{skew}} P_S = 0, \quad P_S \mathcal{V}_{\text{pert}} P_S = 0,$$

because τ has zero angular average and $\partial_\theta \psi = 0$ on $\mathcal{H}_S$.

For a slow function $\psi = \psi(x)$,

$$w_\psi := P_F \mathcal{V}_{\text{skew}} P_S \psi = v_0 \tau(\theta) \cdot \nabla_x \psi.$$

Since w_ψ is a linear combination of the first angular Fourier modes and $\mathcal{R}(\cos \theta g) = -\cos \theta g$, $\mathcal{R}(\sin \theta g) = -\sin \theta g$, one has

$$(-\mathcal{R}^+) w_\psi = w_\psi.$$

Therefore

$$\|w_\psi\|_{L^2(\mu_L)}^2 = \frac{v_0^2}{2} \|\nabla_x \psi\|_{L^2(dx/(2\pi L)^2)}^2,$$

and the conditioning identity

$$\|w_\psi\|^2 \leq M_R \langle w_\psi, (-\mathcal{R}^+) w_\psi \rangle$$

holds with $M_R = 1$.

Applying Definition 3.4 on $\mathcal{C}_Q$ gives

$$Q_O \psi = -P_S \mathcal{V}_{\text{skew}} \mathcal{R}^+ \mathcal{V}_{\text{skew}} P_S \psi = \frac{v_0^2}{2} \Delta_x \psi.$$

Moreover $\mathcal{R}^+ \mathcal{V}_{\text{skew}} \psi = -v_0 \tau(\theta) \cdot \nabla_x \psi$ lies in $\mathcal{C}_{\text{ABP}}$ for $\psi \in \mathcal{C}_Q$, so the upper-bound test-vector admissibility required in Assumption 4.4 is satisfied. Equivalently,

$$-\langle \psi, Q_O \psi \rangle_{L^2(dx/(2\pi L)^2)} = \frac{v_0^2}{2} \|\nabla_x \psi\|_{L^2(dx/(2\pi L)^2)}^2.$$

The exact Poincaré constant of the flat torus $\mathbb{T}_L^2$ is L^2 , hence

$$\frac{v_0^2}{2} \|\nabla_x \psi\|_{L^2(dx/(2\pi L)^2)}^2 \geq \frac{v_0^2}{2L^2} \left\| \psi - \int \psi \frac{dx}{(2\pi L)^2} \right\|_{L^2(dx/(2\pi L)^2)}^2.$$

Thus Assumption 4.6 holds with

$$\ker(Q_O) = \mathbb{C}\mathbf{1}, \quad \lambda(Q_O) = \frac{v_0^2}{2L^2}.$$

The identities (24)–(26) follow directly from the displayed formulas for w_ψ , $(-\mathcal{R}^+) w_\psi$, and Q_O , so Assumption 4.4 holds on $\mathcal{C}_Q$ and extends by density.

Finally,

$$P_S \mathcal{V}_{\text{pert}} \mathcal{R}^+ \mathcal{V}_{\text{skew}} P_S \psi = -\frac{v_0}{2} \partial_{x_1} \psi, \quad \mathcal{V}_{\text{pert}} P_S = 0.$$

Therefore the full effective generator is

$$\mathcal{L}_O \psi = Q_O \psi + \eta \frac{v_0}{2} \partial_{x_1} \psi,$$

as claimed in (93).

E.2. Flux regularity (Assumption 4.5.3). Since

$$w_\psi(x, \theta) = v_0(\cos \theta) \partial_{x_1} \psi(x) + v_0(\sin \theta) \partial_{x_2} \psi(x),$$

one computes directly that

$$(335) \quad P_F \mathcal{V}_{\text{skew}} P_F(-\mathcal{R}^+) w_\psi = \frac{v_0^2}{2} \cos(2\theta) (\partial_{x_1 x_1} - \partial_{x_2 x_2}) \psi + v_0^2 \sin(2\theta) \partial_{x_1 x_2} \psi.$$

Expanding ψ in the torus Fourier basis,

$$\psi(x) = \sum_{n \in \mathbb{Z}^2} c_n e^{in \cdot x/L},$$

gives

$$\|(-Q_O) \psi\|^2 = \frac{v_0^4}{4} \sum_{n \in \mathbb{Z}^2} \frac{|n|^4}{L^4} |c_n|^2.$$

Since $\cos(2\theta)$ and $\sin(2\theta)$ are orthogonal in $L^2(\mathbb{S}^1)$ with norm $1/\sqrt{2}$, (335) gives

$$(336) \quad \begin{aligned} \|P_F \mathcal{V}_{\text{skew}} P_F(-\mathcal{R}^+) w_\psi\|^2 &= \frac{v_0^4}{8} \|(\partial_{x_1 x_1} - \partial_{x_2 x_2}) \psi\|^2 + \frac{v_0^4}{2} \|\partial_{x_1 x_2} \psi\|^2 \\ &= \frac{v_0^4}{8} \sum_{n \in \mathbb{Z}^2} \frac{(n_1^2 - n_2^2)^2 + 4n_1^2 n_2^2}{L^4} |c_n|^2 \\ &= \frac{v_0^4}{8} \sum_{n \in \mathbb{Z}^2} \frac{|n|^4}{L^4} |c_n|^2 = \frac{1}{2} \|(-Q_O) \psi\|^2. \end{aligned}$$

Thus Assumption 4.5.3 holds in the stronger one-term form with

$$\kappa_{\text{ell}} \leq 1/\sqrt{2}, \quad \kappa_{\text{low}} = 0.$$

E.3. Perturbation bounds (Assumption 4.5.4). The $L^2(\mu_L)$ -adjoint of $\mathcal{V}_{\text{pert}} = -\sin \theta \partial_\theta$ is

$$\mathcal{V}_{\text{pert}}^\dagger = \sin \theta \partial_\theta + \cos \theta.$$

Both $\mathcal{V}_{\text{pert}}$ and $\mathcal{V}_{\text{pert}}^\dagger$ have zero slow compression: the first annihilates slow functions, and the second sends a slow input to $\cos \theta$ times that input, whose angular average is zero. Hence

$$(337) \quad \begin{aligned} \mathcal{K}_{\text{pert}} &= \frac{1}{2} (\mathcal{V}_{\text{pert}} - \mathcal{V}_{\text{pert}}^\dagger) = -\sin \theta \partial_\theta - \frac{1}{2} \cos \theta, \\ \mathcal{S}_{\text{pert}} &= \frac{1}{2} (\mathcal{V}_{\text{pert}} + \mathcal{V}_{\text{pert}}^\dagger) = \frac{1}{2} \cos \theta. \end{aligned}$$

For every $f \in \text{dom}((-\mathcal{R})^{1/2})$,

$$(338) \quad \begin{aligned} \|\mathcal{K}_{\text{pert}} f\|_{L^2(\mu_L)} &\leq \|\partial_\theta f\|_{L^2(\mu_L)} + \frac{1}{2} \|f\|_{L^2(\mu_L)} = \|(-\mathcal{R})^{1/2} f\|_{L^2(\mu_L)} + \frac{1}{2} \|f\|_{L^2(\mu_L)}, \\ \|\mathcal{S}_{\text{pert}} f\|_{L^2(\mu_L)} &\leq \frac{1}{2} \|f\|_{L^2(\mu_L)}. \end{aligned}$$

Thus Assumption 4.5.4 holds with

$$a_1 = 1, \quad b_1 = \frac{1}{2}, \quad a_2 = 0, \quad b_2 = \frac{1}{2}.$$

APPENDIX F. PROOF OF PROPOSITION 6.6

We verify Assumptions 3.1, 4.1, 4.4, 4.5, and 4.6 for the biased circular telegraph/run-and-tumble example from Section 6.3. Write

$$e(\sigma) := \sigma, \quad \sigma \in \{+1, -1\}.$$

With respect to the uniform velocity marginal in μ_{tel} , one has $P_S e = 0$ and $\|e\| = 1$. Every $f \in L^2(\mu_{\text{tel}})$ has the orthogonal decomposition

$$f(\theta, \sigma) = a(\theta) + b(\theta)e(\sigma), \quad a = P_S f, \quad P_F f = be,$$

with

$$\|f\|^2 = \|a\|_{L^2(d\theta/(2\pi))}^2 + \|b\|_{L^2(d\theta/(2\pi))}^2.$$

The common invariant core is

$$\mathcal{C}_{\text{tel}} := \text{span}\{e^{in\theta}, e(\sigma)e^{in\theta} : n \in \mathbb{Z}\},$$

and the slow core is

$$\mathcal{C}_Q := \text{span}\{e^{in\theta} : n \in \mathbb{Z}\}.$$

These cores are preserved by P_S , P_F , $\mathcal{R}^+$, $\mathcal{V}_{\text{skew}}$, $\mathcal{V}_{\text{pert}}$, and the adjoints below.

In the orthonormal velocity basis $\{\mathbf{1}, e\}$,

$$(339) \quad \mathcal{R}(a + be) = -be,$$

$$(340) \quad \mathcal{V}_{\text{pert}}(a + be) = -b,$$

$$(341) \quad \mathcal{V}_{\text{pert}}^\dagger(a + be) = -ae.$$

The formula for $\mathcal{R}$ follows from $\mathcal{R}e = -e$. The formula for $\mathcal{V}_{\text{pert}}$ follows by applying (103) to $a + be$; the adjoint is the transpose in the orthonormal basis.

The stationary density can be written as

$$h_{\gamma, \eta} = 1 - \frac{\eta}{\gamma}e,$$

which is equivalent to (104). It has mean one and is positive for $0 \leq \eta < \gamma$. Since $h_{\gamma, \eta}$ is independent of θ , $\mathcal{V}_{\text{skew}}^\dagger h_{\gamma, \eta} = 0$. Using (339) and (341),

$$\gamma \mathcal{R} h_{\gamma, \eta} + \eta \mathcal{V}_{\text{pert}}^\dagger h_{\gamma, \eta} = \eta e - \eta e = 0.$$

Moreover $\mathcal{L}\mathbf{1} = 0$, so Assumption 4.1 holds.

F.1. Microscopic spectral gap, centering, and slow gap. By (339), $\ker(\mathcal{R}) = \mathcal{H}_S$, $\mathcal{R}$ is self-adjoint and non-positive, and

$$-\langle \mathcal{R}f, f \rangle = \|P_F f\|^2, \quad f \in \mathcal{H}.$$

Thus Assumption 4.5.1 holds with $\lambda_R = 1$, and

$$\mathcal{R}^+(be) = -be, \quad -\mathcal{R}^+(be) = be.$$

The transport operator $\mathcal{V}_{\text{skew}} = ce\partial_\theta$ is skew-adjoint because e is real, independent of θ , and ∂_θ is skew-adjoint on the circle. The centering identities

$$P_S \mathcal{V}_{\text{skew}} P_S = 0, \quad P_S \mathcal{V}_{\text{pert}} P_S = 0$$

follow respectively from $P_S e = 0$ and from $\mathcal{V}_{\text{pert}} \psi = 0$ for slow ψ .

For $\psi = \psi(\theta)$,

$$w_\psi := P_F \mathcal{V}_{\text{skew}} P_S \psi = ce \partial_\theta \psi, \quad (-\mathcal{R}^+)w_\psi = w_\psi.$$

Consequently,

$$\|w_\psi\|^2 = c^2 \|\partial_\theta \psi\|^2, \quad \langle w_\psi, (-\mathcal{R}^+)w_\psi \rangle = c^2 \|\partial_\theta \psi\|^2,$$

so the flux conditioning estimate holds with $M_R = 1$. Moreover,

$$P_S \mathcal{V}_{\text{skew}} P_F (-\mathcal{R}^+) w_\psi = c^2 P_S (e^2) \partial_{\theta\theta} \psi = c^2 \partial_{\theta\theta} \psi.$$

Thus

$$Q_O \psi = c^2 \partial_{\theta\theta} \psi, \quad -\langle \psi, Q_O \psi \rangle = c^2 \|\partial_\theta \psi\|^2,$$

and Assumption 4.4 holds on $\mathcal{C}_Q$ and then by density for the self-adjoint Laplacian realization on the slow space. The Fourier basis gives

$$-Q_O e^{in\theta} = c^2 n^2 e^{in\theta}, \quad n \in \mathbb{Z},$$

so Assumption 4.6 holds with

$$\ker(Q_O) = \mathbb{C}\mathbf{1}, \quad \lambda(Q_O) = c^2.$$

Finally, since $\mathcal{V}_{\text{pert}} P_S = 0$ and

$$\mathcal{R}^+ \mathcal{V}_{\text{skew}} P_S \psi = -c e \partial_\theta \psi,$$

we get from (340)

$$P_S \mathcal{V}_{\text{pert}} \mathcal{R}^+ \mathcal{V}_{\text{skew}} P_S \psi = c \partial_\theta \psi.$$

The full Schur-complement generator is therefore

$$\mathcal{L}_O \psi = -P_S (\mathcal{V}_{\text{skew}} + \eta \mathcal{V}_{\text{pert}}) \mathcal{R}^+ (\mathcal{V}_{\text{skew}} + \eta \mathcal{V}_{\text{pert}}) P_S \psi = c^2 \partial_{\theta\theta} \psi - \eta c \partial_\theta \psi,$$

as claimed in (106).

F.2. Flux regularity (Assumption 4.5.3). Since $(-\mathcal{R}^+) w_\psi = c e \partial_\theta \psi$,

$$P_F \mathcal{V}_{\text{skew}} P_F (-\mathcal{R}^+) w_\psi = c^2 P_F (e^2) \partial_{\theta\theta} \psi = 0.$$

Hence Assumption 4.5.3 holds with

$$\kappa_{\text{ell}} = 0, \quad \kappa_{\text{low}} = 0.$$

F.3. Perturbation bounds (Assumption 4.5.4). In the orthonormal basis $\{\mathbf{1}, e\}$, (340) is the matrix

$$\begin{pmatrix} 0 & -1 \\ 0 & 0 \end{pmatrix},$$

so

$$\mathcal{K}_{\text{pert}} = \frac{1}{2} (\mathcal{V}_{\text{pert}} - \mathcal{V}_{\text{pert}}^\dagger), \quad \mathcal{S}_{\text{pert}} = \frac{1}{2} (\mathcal{V}_{\text{pert}} + \mathcal{V}_{\text{pert}}^\dagger)$$

are respectively skew-adjoint and self-adjoint, and both have operator norm $1/2$. Therefore, for every $f \in \mathcal{H}$,

$$\|\mathcal{K}_{\text{pert}} f\| \leq \frac{1}{2} \|f\|, \quad \|\mathcal{S}_{\text{pert}} f\| \leq \frac{1}{2} \|f\|.$$

Assumption 4.5.4 holds with

$$a_1 = 0, \quad b_1 = \frac{1}{2}, \quad a_2 = 0, \quad b_2 = \frac{1}{2}.$$

APPENDIX G. PROOF OF PROPOSITION 6.8

We verify the assumptions for the two-qubit Lindblad example. The proof is a Pauli-basis calculation in the normalized Hilbert–Schmidt space

$$\mathcal{H} = \mathcal{B}((\mathbb{C}^2)^{\otimes 2}), \quad \langle A, B \rangle = 4^{-1} \operatorname{tr}(A^* B).$$

Let

$$E_{ab} := \sigma_a \otimes \sigma_b, \quad a, b \in \{0, 1, 2, 3\}, \quad \sigma_0 := I.$$

This is an orthonormal basis. We write the slow basis as $E_j := E_{0j}$, $j = 0, 1, 2, 3$.

For the one-spin boundary dissipators, the identities

$$(342) \quad \begin{aligned} \frac{1}{2} \mathcal{D}_{\sigma_+}(I) + \frac{1}{2} \mathcal{D}_{\sigma_-}(I) &= 0, & \frac{1}{2} \mathcal{D}_{\sigma_+}(\sigma_1) + \frac{1}{2} \mathcal{D}_{\sigma_-}(\sigma_1) &= -\sigma_1, \\ \frac{1}{2} \mathcal{D}_{\sigma_+}(\sigma_2) + \frac{1}{2} \mathcal{D}_{\sigma_-}(\sigma_2) &= -\sigma_2, & \frac{1}{2} \mathcal{D}_{\sigma_+}(\sigma_3) + \frac{1}{2} \mathcal{D}_{\sigma_-}(\sigma_3) &= -2\sigma_3, \end{aligned}$$

and

$$(343) \quad \begin{aligned} \frac{1}{2} \mathcal{D}_{\sigma_+}(I) - \frac{1}{2} \mathcal{D}_{\sigma_-}(I) &= 0, & \frac{1}{2} \mathcal{D}_{\sigma_+}(\sigma_1) - \frac{1}{2} \mathcal{D}_{\sigma_-}(\sigma_1) &= 0, \\ \frac{1}{2} \mathcal{D}_{\sigma_+}(\sigma_2) - \frac{1}{2} \mathcal{D}_{\sigma_-}(\sigma_2) &= 0, & \frac{1}{2} \mathcal{D}_{\sigma_+}(\sigma_3) - \frac{1}{2} \mathcal{D}_{\sigma_-}(\sigma_3) &= 2I \end{aligned}$$

are immediate from the Pauli matrices. Tensoring these identities with the second spin gives

$$\mathcal{R}E_{0b} = 0, \quad \mathcal{R}E_{1b} = -E_{1b}, \quad \mathcal{R}E_{2b} = -E_{2b}, \quad \mathcal{R}E_{3b} = -2E_{3b}.$$

Thus $\mathcal{R}$ is self-adjoint and non-positive, $\ker(\mathcal{R}) = \operatorname{span}\{E_{0b} : 0 \leq b \leq 3\}$, and the microscopic gap is $\lambda_R = 1$. The pseudoinverse is

$$\mathcal{R}^+ E_{1b} = -E_{1b}, \quad \mathcal{R}^+ E_{2b} = -E_{2b}, \quad \mathcal{R}^+ E_{3b} = -\frac{1}{2} E_{3b}, \quad \mathcal{R}^+ E_{0b} = 0.$$

Since $\mathcal{V}_{\text{skew}} = i[H_J, \cdot]$ with $H_J = H_J^*$, it is skew-adjoint in the Hilbert–Schmidt inner product. The identities $P_S \mathcal{V}_{\text{skew}} P_S = 0$ and $P_S \mathcal{V}_{\text{pert}} P_S = 0$ follow from the fact that every nonzero action below has a non-identity Pauli matrix on the first spin, except for the perturbation image $E_{3b} \mapsto 2E_{0b}$, which cannot occur from a slow input.

G.1. Schur complement and slow gap. The Hamiltonian commutators needed for slow inputs are

$$(344) \quad \begin{aligned} \mathcal{V}_{\text{skew}} E_1 &= 2JE_{23} - 2J\Delta E_{32}, \\ \mathcal{V}_{\text{skew}} E_2 &= -2JE_{13} + 2J\Delta E_{31}, \\ \mathcal{V}_{\text{skew}} E_3 &= 2JE_{12} - 2JE_{21}. \end{aligned}$$

These vectors lie in $\mathcal{H}_F$, so centering is explicit. Applying $-\mathcal{R}^+$ and then $P_S \mathcal{V}_{\text{skew}} P_F$ gives

$$(345) \quad \begin{aligned} P_S \mathcal{V}_{\text{skew}} P_F (-\mathcal{R}^+) \mathcal{V}_{\text{skew}} E_1 &= -J^2(4 + 2\Delta^2) E_1, \\ P_S \mathcal{V}_{\text{skew}} P_F (-\mathcal{R}^+) \mathcal{V}_{\text{skew}} E_2 &= -J^2(4 + 2\Delta^2) E_2, \\ P_S \mathcal{V}_{\text{skew}} P_F (-\mathcal{R}^+) \mathcal{V}_{\text{skew}} E_3 &= -8J^2 E_3. \end{aligned}$$

This is exactly (117), because $Q_O = -P_S \mathcal{V}_{\text{skew}} \mathcal{R}^+ \mathcal{V}_{\text{skew}} P_S = P_S \mathcal{V}_{\text{skew}} P_F (-\mathcal{R}^+) \mathcal{V}_{\text{skew}} P_S$. Hence Q_O is self-adjoint on the slow space, $\ker(Q_O) = \mathbb{C}\mathbf{1}$, and

$$\lambda(Q_O) = J^2 \min\{4 + 2\Delta^2, 8\}.$$

Assumptions 4.4 and 4.6 follow.

For the perturbative part of the Schur complement, (343) gives $\mathcal{V}_{\text{pert}}E_{3b} = 2E_{0b}$ and $\mathcal{V}_{\text{pert}}E_{1b} = \mathcal{V}_{\text{pert}}E_{2b} = 0$. Combining this with (344) yields

$$(346) \quad -P_S \mathcal{V}_{\text{pert}} \mathcal{R}^+ \mathcal{V}_{\text{skew}} E_1 = -2J\Delta E_2, \quad -P_S \mathcal{V}_{\text{pert}} \mathcal{R}^+ \mathcal{V}_{\text{skew}} E_2 = 2J\Delta E_1, \quad -P_S \mathcal{V}_{\text{pert}} \mathcal{R}^+ \mathcal{V}_{\text{skew}} E_3 = 0.$$

Since $\mathcal{V}_{\text{pert}}P_S = 0$, the other linear cross term vanishes, and (118) follows. This also proves that the effective perturbation is nonzero when $J\Delta\eta \neq 0$.

G.2. Flux regularity. For a slow input $\psi = xE_1 + yE_2 + zE_3$, the flux $w_\psi = \mathcal{V}_{\text{skew}}\psi$ is given by (344). Orthogonality of the Pauli basis gives

$$\|w_\psi\|^2 = J^2(4 + 4\Delta^2)(|x|^2 + |y|^2) + 8J^2|z|^2.$$

Moreover,

$$\langle w_\psi, (-\mathcal{R}^+)w_\psi \rangle = J^2(4 + 2\Delta^2)(|x|^2 + |y|^2) + 8J^2|z|^2.$$

Thus the flux-conditioning estimate holds with

$$M_R = \max \left\{ 1, \frac{4 + 4\Delta^2}{4 + 2\Delta^2} \right\} \leq 2.$$

It remains to bound $P_F \mathcal{V}_{\text{skew}} P_F (-\mathcal{R}^+) w_\psi$. A direct use of the Pauli commutators gives

$$(347) \quad \begin{aligned} P_F \mathcal{V}_{\text{skew}} P_F (-\mathcal{R}^+) \mathcal{V}_{\text{skew}} E_1 &= 6J^2 \Delta E_{10}, \\ P_F \mathcal{V}_{\text{skew}} P_F (-\mathcal{R}^+) \mathcal{V}_{\text{skew}} E_2 &= 6J^2 \Delta E_{20}, \\ P_F \mathcal{V}_{\text{skew}} P_F (-\mathcal{R}^+) \mathcal{V}_{\text{skew}} E_3 &= 8J^2 E_{30}. \end{aligned}$$

The corresponding slow operator has absolute eigenvalues

$$J^2(4 + 2\Delta^2), \quad J^2(4 + 2\Delta^2), \quad 8J^2.$$

Since

$$\frac{6\Delta}{4 + 2\Delta^2} \leq \frac{3\sqrt{2}}{4}$$

for every $\Delta \geq 0$, with equality at $\Delta = \sqrt{2}$, and the E_3 coefficient is exactly 8, we obtain

$$\|P_F \mathcal{V}_{\text{skew}} P_F (-\mathcal{R}^+) w_\psi\| \leq \max \left\{ 1, \frac{6\Delta}{4 + 2\Delta^2} \right\} \|(-Q_O)\psi\|.$$

Hence Assumption 4.5.3 holds with

$$\kappa_{\text{ell}} \leq \max \left\{ 1, \frac{6\Delta}{4 + 2\Delta^2} \right\} \leq \frac{3\sqrt{2}}{4}, \quad \kappa_{\text{low}} = 0.$$

G.3. Perturbation bounds and stationary state. The perturbation has rank four in the Pauli basis:

$$\mathcal{V}_{\text{pert}}E_{3b} = 2E_{0b}, \quad \mathcal{V}_{\text{pert}}E_{ab} = 0 \quad (a \neq 3).$$

Its Hilbert–Schmidt adjoint satisfies

$$\mathcal{V}_{\text{pert}}^\dagger E_{0b} = 2E_{3b}, \quad \mathcal{V}_{\text{pert}}^\dagger E_{ab} = 0 \quad (a \neq 0).$$

Therefore

$$\mathcal{K}_{\text{pert}} = \frac{1}{2}(\mathcal{V}_{\text{pert}} - \mathcal{V}_{\text{pert}}^\dagger), \quad \mathcal{S}_{\text{pert}} = \frac{1}{2}(\mathcal{V}_{\text{pert}} + \mathcal{V}_{\text{pert}}^\dagger)$$

are respectively skew-adjoint and self-adjoint, and both have operator norm one. Thus Assumption 4.5.4 holds with

$$a_1 = 0, \quad b_1 = 1, \quad a_2 = 0, \quad b_2 = 1.$$

Finally, for $0 \leq \eta < \gamma$ the Schrödinger-picture adjoint of (116) is a completely positive trace-preserving finite-dimensional Lindblad generator with boundary rates $(\gamma + \eta)/2$ and $(\gamma - \eta)/2$.

Compactness of the density-matrix simplex gives a stationary density matrix $\rho_{\gamma,\eta}$. Setting $h_{\gamma,\eta} := 4\rho_{\gamma,\eta}$ gives $\langle h_{\gamma,\eta}, \mathbf{1} \rangle = 1$ and $\mathcal{L}^\dagger h_{\gamma,\eta} = 0$, so Assumption 4.1 holds.

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